

PI: Bradshaw, Jessica	Title: The Role of Autonomic Regulation of Attention in the Emergence of ASD	
Received: 11/07/2022	FOA: PA21-201	Council: 05/2023
Competition ID: FORMS-G	FOA Title: Research on Autism Spectrum Disorders (R01 Clinical Trial Optional)	
1 R01 MH132925-01A1	Dual: HD	Accession Number: 4770773
IPF: 1524302	Organization: UNIVERSITY OF SOUTH CAROLINA AT COLUMBIA	
Former Number: 1R01MH132925-01	Department: Psychology	
IRG/SRG: CPDD	AIDS: N	Expedited: N
<u>Subtotal Direct Costs</u> <u>(excludes consortium F&A)</u> Year 1: 499,351 Year 2: 499,731 Year 3: 499,972 Year 4: 499,690 Year 5: 499,690	Animals: N Humans: Y Clinical Trial: N Current HS Code: 48 HESC: N HFT: N	New Investigator: Y Early Stage Investigator: Y
<i>Senior/Key Personnel:</i>	<i>Organization:</i>	<i>Role Category:</i>
Jessica Bradshaw	University of South Carolina	PD/PI
Christian O'Reilly	University of South Carolina	Co-Investigator
Robin Dail	University of South Carolina	Co-Investigator
Jana Iverson	Boston University	Co-Investigator
JOHN RICHARDS	University of South Carolina	Co-Investigator
Jane Roberts	University of South Carolina	Co-Investigator

APPLICATION FOR FEDERAL ASSISTANCE

SF 424 (R&R)

3. DATE RECEIVED BY STATE		State Application Identifier
1. TYPE OF SUBMISSION*		4.a. Federal Identifier MH132925
☐ Pre-application ☒ Application ☐ Changed/Corrected Application		b. Agency Routing Number
2. DATE SUBMITTED	Application Identifier 135800-23-62917	c. Previous Grants.gov Tracking Number
5. APPLICANT INFORMATION UEI*: J22LNTMEDP73		
Legal Name*: UNIVERSITY OF SOUTH CAROLINA AT COLUMBIA		
Department: Sponsored Awards Management		
Division:		
Street1*: UNIVERSITY OF SOUTH CAROLINA AT COLUMBIA		
Street2: 1600 Hampton Street, Suite 414		
City*: COLUMBIA		
County:		
State*: SC: South Carolina		
Province:		
Country*: USA: UNITED STATES		
ZIP / Postal Code*: 292080001		
Person to be contacted on matters involving this application		
Prefix: First Name*: Courtney Middle Name: Last Name*: Swaney Suffix:		
Position/Title: Sponsored Programs Administrator		
Street1*: 1600 Hampton Street, Suite 414		
Street2:		
City*: Columbia		
County: Richland		
State*: SC: South Carolina		
Province:		
Country*: USA: UNITED STATES		
ZIP / Postal Code*: 29208-0001		
Phone Number*: 803-777-2274 Fax Number: Email: cswaney@mailbox.sc.edu		
6. EMPLOYER IDENTIFICATION NUMBER (EIN) or (TIN)*		1576001153A1
7. TYPE OF APPLICANT*		H: Public/State Controlled Institution of Higher Education
Other (Specify):		
Small Business Organization Type ☐ Women Owned ☐ Socially and Economically Disadvantaged		
8. TYPE OF APPLICATION*		If Revision, mark appropriate box(es).
☐ New ☒ Resubmission		☐ A. Increase Award ☐ B. Decrease Award ☐ C. Increase Duration
☐ Renewal ☐ Continuation ☐ Revision		☐ D. Decrease Duration ☐ E. Other (specify) :
Is this application being submitted to other agencies?* ☐ Yes ☒ No What other Agencies?		
9. NAME OF FEDERAL AGENCY* National Institutes of Health		10. CATALOG OF FEDERAL DOMESTIC ASSISTANCE NUMBER TITLE:
11. DESCRIPTIVE TITLE OF APPLICANT'S PROJECT* The Role of Autonomic Regulation of Attention in the Emergence of ASD		
12. PROPOSED PROJECT Start Date* Ending Date* 07/01/2023 06/30/2028		13. CONGRESSIONAL DISTRICTS OF APPLICANT SC-006

SF 424 (R&R) APPLICATION FOR FEDERAL ASSISTANCE**Page 2****14. PROJECT DIRECTOR/PRINCIPAL INVESTIGATOR CONTACT INFORMATION**

Prefix: First Name*: Jessica Middle Name: Last Name*: Bradshaw Suffix:

Position/Title: Associate Professor

Organization Name*: University of South Carolina

Department: Psychology

Division:

Street1*: 1800 Gervais St

Street2:

City*: Columbia

County:

State*: SC: South Carolina

Province:

Country*: USA: UNITED STATES

ZIP / Postal Code*: 292013504

Phone Number*: 9094544652 Fax Number: Email*: jbradshaw@sc.edu

15. ESTIMATED PROJECT FUNDING

a. Total Federal Funds Requested* \$3,745,118.00

b. Total Non-Federal Funds* \$0.00

c. Total Federal & Non-Federal Funds* \$3,745,118.00

d. Estimated Program Income* \$0.00

16. IS APPLICATION SUBJECT TO REVIEW BY STATE EXECUTIVE ORDER 12372 PROCESS?*

a. YES ☐ THIS PREAPPLICATION/APPLICATION WAS MADE AVAILABLE TO THE STATE EXECUTIVE ORDER 12372 PROCESS FOR REVIEW ON:

DATE:

b. NO ☒ PROGRAM IS NOT COVERED BY E.O. 12372; OR

☐ PROGRAM HAS NOT BEEN SELECTED BY STATE FOR REVIEW

17. By signing this application, I certify (1) to the statements contained in the list of certifications* and (2) that the statements herein are true, complete and accurate to the best of my knowledge. I also provide the required assurances * and agree to comply with any resulting terms if I accept an award. I am aware that any false, fictitious, or fraudulent statements or claims may subject me to criminal, civil, or administrative penalties. (U.S. Code, Title 18, Section 1001)

☒ I agree*

* The list of certifications and assurances, or an Internet site where you may obtain this list, is contained in the announcement or agency specific instructions.

18. SFLLL or OTHER EXPLANATORY DOCUMENTATION

File Name:

19. AUTHORIZED REPRESENTATIVE

Prefix: First Name*: Courtney Middle Name: Last Name*: Swaney Suffix:

Position/Title*: Sponsored Programs Administrator

Organization Name*: University of South Carolina

Department: Sponsored Awards Management

Division:

Street1*: 1600 Hampton Street, Suite 414

Street2:

City*: Columbia

County: Richland

State*: SC: South Carolina

Province:

Country*: USA: UNITED STATES

ZIP / Postal Code*: 29208-0001

Phone Number*: 803-777-2274 Fax Number: Email*: cswaney@mail.box.sc.edu

Signature of Authorized Representative*

Angela Wilcox

Date Signed*

11/07/2022

20. PRE-APPLICATION File Name:**21. COVER LETTER ATTACHMENT** File Name: CoverLetter_BradshawR01.pdf

424 R&R and PHS-398 Specific

Table Of Contents

SF 424 R&R Cover Page.....	1
Table of Contents.....	3
Performance Sites.....	4
Research & Related Other Project Information.....	5
Project Summary/Abstract(Description).....	6
Project Narrative.....	7
Facilities & Other Resources.....	8
Equipment.....	19
Research & Related Senior/Key Person.....	20
Research & Related Budget Year - 1.....	50
Research & Related Budget Year - 2.....	53
Research & Related Budget Year - 3.....	56
Research & Related Budget Year - 4.....	59
Research & Related Budget Year - 5.....	62
Budget Justification.....	65
Research & Related Cumulative Budget.....	72
Research & Related Budget - Consortium Budget (Subaward 1).....	74
Total Direct Costs Less Consortium F&A.....	92
PHS398 Cover Page Supplement.....	93
PHS 398 Research Plan.....	95
Introduction to Application.....	96
Specific Aims.....	97
Research Strategy.....	98
PHS Human Subjects and Clinical Trials Information.....	110
Study 1: The Role of Autonomic Regulation of Attention in the Emergence of ASD.....	112
Inclusion Enrollment Reports.....	119
Bibliography & References Cited.....	127
Consortium/Contractual Arrangements.....	135
Letters of Support.....	136
Resource Sharing Plan(s).....	149

Project/Performance Site Location(s)**Project/Performance Site Primary Location**

☐ I am submitting an application as an individual, and not on behalf of a company, state, local or tribal government, academia, or other type of organization.

Organization Name: UNIVERSITY OF SOUTH CAROLINA AT COLUMBIA
UEI: J22LNTMEDP73
Street1*: UNIVERSITY OF SOUTH CAROLINA AT COLUMBIA
Street2: 1600 Hampton Street, Suite 414
City*: COLUMBIA
County:
State*: SC: South Carolina
Province:
Country*: USA: UNITED STATES
Zip / Postal Code*: 292080001
Project/Performance Site Congressional District*: SC-006

Project/Performance Site Location 1

☐ I am submitting an application as an individual, and not on behalf of a company, state, local or tribal government, academia, or other type of organization.

Organization Name: Trustees of Boston University
UEI: THL6A6JLE1S7
Street1*: Sponsored Programs
Street2: 881 Commonwealth Avenue
City*: Boston
County:
State*: MA: Massachusetts
Province:
Country*: USA: UNITED STATES
Zip / Postal Code*: 02215-1300
Project/Performance Site Congressional District*:

Additional Location(s)

File Name:

RESEARCH & RELATED Other Project Information

1. Are Human Subjects Involved?* ☒ Yes ☐ No	
1.a. If YES to Human Subjects Is the Project Exempt from Federal regulations? ☐ Yes ☒ No If YES, check appropriate exemption number: — 1 — 2 — 3 — 4 — 5 — 6 — 7 — 8 If NO, is the IRB review Pending? ☒ Yes ☐ No IRB Approval Date: Human Subject Assurance Number 00000404	
2. Are Vertebrate Animals Used?* ☐ Yes ☒ No	
2.a. If YES to Vertebrate Animals Is the IACUC review Pending? ☐ Yes ☐ No IACUC Approval Date: Animal Welfare Assurance Number	
3. Is proprietary/privileged information included in the application?* ☐ Yes ☒ No	
4.a. Does this project have an actual or potential impact - positive or negative - on the environment?* ☐ Yes ☒ No	
4.b. If yes, please explain: 4.c. If this project has an actual or potential impact on the environment, has an exemption been authorized or an environmental assessment (EA) or environmental impact statement (EIS) been performed? ☐ Yes ☐ No 4.d. If yes, please explain:	
5. Is the research performance site designated, or eligible to be designated, as a historic place?* ☐ Yes ☒ No	
5.a. If yes, please explain:	
6. Does this project involve activities outside the United States or partnership with international collaborators?* ☐ Yes ☒ No	
6.a. If yes, identify countries: 6.b. Optional Explanation:	
7. Project Summary/Abstract*	Filename ProjectSummary_BradshawR01.pdf
8. Project Narrative*	ProjectNarrative_BradshawR01.pdf
9. Bibliography & References Cited	Bibliography_BradshawR01_A1.pdf
10. Facilities & Other Resources	FacilitiesResources_BradshawR01.pdf
11. Equipment	Equipment_BradshawR01.pdf

PROJECT SUMMARY

Disrupted attention is among the earliest-emerging, lifelong features of autism spectrum disorder (ASD) and research suggests a critical point in the neonatal and early infant period at which disrupted attention may be detectable. The timing of this disruption aligns with key maturational shifts in neurodevelopment, but the neurobiological mechanisms associated with disrupted attention in ASD remain elusive. One neurobiological system that regulates attention from early infancy is the autonomic nervous system (ANS). Broad ANS dysfunction is observed in older children already diagnosed with ASD, but whether and when ANS dysfunction contributes to attention abnormalities in ASD remains unknown. The overall goal of this study is to examine how atypical autonomic regulation of attention may be associated with the emergence of ASD symptoms. A key marker of autonomic regulation of attention is heart defined attention. Accordingly, maturation of heart defined attention in the early infant period and the developmental consequences therein for the emergence of interactive behaviors and ASD symptoms will be examined. A critical innovation of this study is leveraging a three-group design in which infants who experience elevated ASD symptoms (infant siblings of children with ASD; ASIBs) will be compared to typically developing (TD) and preterm (PT) control groups. Infants born preterm experience broad ANS dysfunction from birth and comparing ASIBs to PTs will help delineate how attention-specific ANS dysfunction predicts the emergence of ASD-specific symptoms. Aim 1 will compare the emergence of autonomic regulation of attention during a very critical period of neurodevelopment (1-3 months) across ASIBs, PT infants, and TD infants. Aim 2 will quantify the association between heart defined attention and interactive behavior on a moment-to-moment basis at 6, 9, and 12 months, and compare across the three groups. Aim 3 will utilize machine learning techniques to predict ASD symptoms at age 3 years from early autonomic and attentional features (from Aims 1-2). Determining the specificity of autonomic regulation of attention as a key, early emerging feature associated with ASD, has significant translational potential for a cost-effective biomarker. This work will inform developmental models of ASD wherein disrupted autonomic regulation of attention has proximal effects on real-world interactions that may interfere with learning and have cascading effects on long-term outcomes.

PROJECT NARRATIVE

The current study will determine the role of infant autonomic regulation of attention in the unfolding of autism spectrum disorder (ASD). This work will inform developmental models of ASD wherein disrupted autonomic regulation of attention has proximal effects on real-world interactions that may interfere with learning and have cascading effects on long-term outcomes. Findings will lead to new models for earlier intervention that enrich early experiences and enhance learning opportunities from the first months of life, ultimately improving lifelong outcomes for individuals with ASD.

FACILITIES AND OTHER RESOURCES

The University of South Carolina (USC) is ranked by the Carnegie Foundation as being in the highest tier of research institutions (“very high research activity”). The PI is an Associate Professor in Psychology, where there are a large number of supports in place to ensure the success of early stage investigators. In addition, significant resources are available via the PI’s position as an affiliate of the Institute for Mind and Brain and chair of the executive steering committee for the Carolina Autism and Neurodevelopment (CAN) Research Center at USC. The resources to complete this project are substantial and contribute directly to the feasibility of completing the research goals of this proposal. These resources are outlined below regarding Scientific Environment, Recruitment Resources, Institutional Investment, and Facilities (University and Department).

SCIENTIFIC ENVIRONMENT

The Early Social Development Lab, PI: Bradshaw

Dr. Bradshaw’s Early Social Development Lab is located in the **Institute for Mind and Brain (IMB)**, an interdisciplinary institute based in the Psychology Department focused on understanding processes in the human brain, health, and disease. Dr. Bradshaw is an invited faculty affiliate of the IMB. The IMB was created with a \$1 million renovation to a historic 13,675 square foot building on campus. This space houses approximately 12 faculty members from the Department of Psychology, along with their staff, postdocs, students, and equipment. The IMB has several meeting rooms equipped with large LCD TV screens for presentations and training purposes and offers airy, appealing spaces to facilitate discussions and promote collaborations across disciplines. The IMB hosts a monthly colloquium, which attracts internationally renowned psychologists and cognitive scientists. There is also free, easily accessible parking, ideal for families and infants with strollers.



Dr. Bradshaw has five experimental testing rooms within the IMB in addition to a personal office and six offices with 18 total workstations for undergraduate students, graduate students, research staff, postdocs, and research assistant professors. Ample room for data storage is also available. The primary experimental room is equipped with a **Noldus mounted 3-angle camera system** and acoustic treatment. It is adjacent to a control



room that houses the video recording hardware and software. The second experimental testing room includes an **SR Eye Link Eye Tracker** and **six telemetry-based ECG waveform recorders (CamnTech ActiHeart)** outfitted for data collection with infants and toddlers. This equipment is accompanied by eye-tracking and heart-rate data analysis software, including SR Experiment Builder, SR DataViewer, EPrime, and CardioEdit. There is also a family-friendly waiting area that contains comfortable, family-friendly seating, a changing table, and a variety of toys.

In addition, lab space includes tables/chairs for workstations, infant and child-friendly tables/chairs for assessments, clinical assessment kits and protocols, 10 computers, three printers, locked filing cabinets, and bookshelves. Research assistant and student workstations are connected via high-speed local area network (100MB) and interface with the University’s internet connection. Other equipment includes multiple external hard drives, five video cameras with tripods for remote data collection, and clinical assessments and protocols (NICU Network Neurobehavioral Scales (NNNS), Test of Infant Motor Performance, Carey Temperament Scales, Communication and Symbolic Behavior Scales, Preschool Language Scale, Language Use Inventory, Autism Diagnostic Observation Schedule, Bayley Scales of Infant Development, Mullen Scales of Early Learning). Dr. Bradshaw’s personal office is equipped with a personal computer, printer/scanner, locked filing cabinet, bookshelves, and a table with chairs to meet with collaborators and staff. Computer software includes Microsoft Office, SPSS, SAS, and Matlab. Of note, Dr. Bradshaw has significant institutional, discretionary funds to continue building her already well-equipped laboratory space.

As an Associate Professor in the Department of Psychology at the USC in Columbia, the PI has an extremely supportive scientific environment in which to conduct research. One of the primary resources is a strong and extensive network of collaborators and colleagues within the department and across the state of South Carolina. In addition, Dr. Bradshaw has developed strong collaborations with all co-investigators on this R01 proposal. Dr.

Jane Roberts directs the Neurodevelopmental Disorders Lab and leads a successful NIH-funded line of research focused on understanding the biological mechanisms that underlie cognitive and behavioral functioning in children with neurodevelopmental disorders such as autism, fragile X syndrome, and ADHD. Dr. John Richards is an internationally recognized expert in infant development of visual attention and cortical source localization. Drs. Roberts and Richards are both mentors on the PI's K23 Career Development Award and their offices and labs are housed in the same building as Dr. Bradshaw, allowing significant opportunity for collaboration.

Secure Data Storage. Keyed entry is required to enter the PI's lab. Three large, five-drawer locked cabinets are available to store assessment protocols in a dedicated administrative office that requires additional lock and key access. The lab also has 10 TB of available space on a secure networked database that only research personnel can access via password. This space will be available to store and back-up video recordings so that no hard copies of the videos (e.g., SD cards) will be created or stored. Data are also routinely backed up on a secure network.

Undergraduate and graduate paid and unpaid research assistantships are also a significant source of scientific support for research programs at the USC. Significant university research infrastructure (described in more detail below) provides several mechanisms by which undergraduate and graduate-level students can contribute to research. Relevant to the current proposal, Dr. Bradshaw has recruited four clinical and school psychology doctoral students and over 20 undergraduate research assistants who are currently working in her lab. The doctoral students are highly clinically trained (ADOS-2, NNNS research reliability) and contribute significantly to ongoing research operations. Graduate and undergraduate students will contribute to behavioral coding of attention and the heart-defined attention processing pipeline, as well as to the development of co-authored manuscripts and conference presentations.

The Neurodevelopmental Disorders Lab, PI: Roberts

Dr. Roberts (Co-I), a Carolina Distinguished Professor in the Department of Psychology in the College of Arts and Sciences at the USC, is the Director of the Neurodevelopmental Disorders (NDD) Lab. NDD Lab grant personnel currently consists of a research assistant professor, a program coordinator, two postdoctoral fellows, three graduate students, two research specialists, and numerous undergraduate students led by Dr. Roberts. The NDD Lab is co-located with Dr. Bradshaw's lab at the IMB, and within this renovated space, the NDD Lab has five offices, each requiring keyed entry, as well as dedicated undergraduate workspace and access to small- and large-group spaces for meetings and trainings. Dr. Roberts' IMB office is equipped with a personal computer and a table/chairs suitable for private meetings and is adjacent to Dr. Bradshaw's, facilitating their bi-monthly meetings. Dr. Roberts also has a secondary office at Barnwell College equipped with a personal computer, printer, phone, and table/chairs for small meetings. At the IMB, all NDD Lab staff has access to high-speed internet access, two phones, four LaserJet printers, a fax/scanner, and desktop computers equipped with software such as Microsoft Office, E-Prime, SPSS, SAS, Noldus Observer XT (behavior coding software), CardioEdit/CardioBatch (heart rate editing and analysis software), Matlab, R, and Datavyu. Furthermore, the NDD Lab has an assessment room on the lower level of the IMB equipped with a table/chairs for assessments, along with comfortable chairs, a sofa, a child's table set, storage cabinets, and digital audio/video recording equipment (e.g., Panasonic HC-V770 Full HD Camcorder). Dr. Roberts' lab contains four laptop computers dedicated to behavioral and heart activity data acquisition. Three mobile SR EyeLink 1000 Plus eye-tracking systems, along with the corresponding software (i.e., SR Experiment Builder, SR DataViewer, and E-Prime) and a remote arm to accommodate infant participants, are available. The NDD Lab also has four Actiheart heart activity monitors, allowing for non-invasive, portable, and compact recording of ECG waveforms. A range of electronic equipment is available for mobile testing, including six video cameras, six tripods, and several external hard drives. A variety of clinical assessment kits and protocols, including two complete Bayley-4 testing kits, two complete ADOS-2 testing kits, and three AOSI testing kits, are also available and stored in the NDD Lab's equipment room.

The Infant Lab, PI: Richards

Dr. Richards' Infant Laboratory is a 1,875 square foot lab space located in the same building as the PI and Co-I Roberts – Institute for Mind and Brain (IMB). The IMB contains testing rooms, control rooms, IT rooms, equipment rooms for research, and additional space for offices. This space is easily accessible to parents bringing their infants in for studies. The Infant EEG laboratory is an 800-sq-ft space that includes a reception area and EEG testing suite, both on the lower level of the IMB. The reception area is a 236-sq-ft room where participants and their parents will fill out consent forms and any pre-screening forms. The 562-sq-ft EEG testing suite is located directly across the hall from the reception area. Equipment in this suite includes the high-density EEG recording system, a control room equipped with computers, HD-EEG amplifiers, EGI Geodesic

Photogrammetry system for registration of electrode placement on the scalp, and a large room with a sink and supplies for net preparation and storage.

Computers and Data Storage. Dr. Richards' Infant Lab has the following resources: (a) laboratory microcomputers: two computers with data acquisition and experimental control devices for generating experimental stimuli and collecting data (Dell Precision Workstation 530, dual 1.7Ghz Xeon processors, 18G storage, 1M memory); (b) microcomputer: a Mac-Mini system for interfacing with 128-channel physiological data acquisition system made by "Electrical Geodesics Incorporated" to do high-density EEG recording; (c) local network for laboratory computers for sharing data, data acquisition, data analysis, access to internet via TCP/IP protocol, including 1G switch connection within laboratories, and 10GHz switch connection from server storage to McCausland Brain Imaging Center; (d) university-wide consulting support for statistical packages, hardware, mainframe, and minicomputer work; (e) virtual computers and Linux Clusters: Dr. Richards has rack-mounted computer systems housing virtual machines (VM's) for data storage, data analysis, and other processing. The Research Computing Infrastructure group at the USC administers a Linux cluster with about 230 nodes and 7000 CPUs. Dr. Richards has priority access to 18 CPU nodes and two GPU nodes, in addition to shared access to all the nodes in the cluster. The cluster has attached storage in the petabyte range.

Research Staff: The Infant Development Lab has a full-time research assistant with 10 years of experience in the lab. The research assistant manages the lab, recruits participants, collects data, and assists in data analysis.

The O'Reilly Lab at the Artificial Intelligence Institute (AIISC), PI: O'Reilly

Co-I, Christian O'Reilly, is an Assistant Professor in Computer Science and Engineering and the Artificial Intelligence Institute (AIISC). To bolster collaboration and interactions across students and faculty, faculty of the AIISC share their space and infrastructure in common "laboratories" rather than having separate physical space for their laboratories. As such, the equipment available for Co-I Dr. O'Reilly are as listed for the AIISC. The Co-I has a Mac OS computer networked to the USC system for word-processing, imaging/graphics, bibliographical and statistical software, plus wireless internet access to an extensive collection of electronic databases and journals, via the entire USC Library system. This computer is set up for complex LFP analyses using Python and its rich scientific (Numpy, Scipy, Scikit-learn), statistical (StatsModels, Pandas, XArray, rpy2, Pyro), graphic (Matplotlib, Seaborn), AI/ML (PyTorch, Keras, TensorFlow), neuroimaging (NiBabel, MNE-Python), and computational neuroscience (Brian2, LFPy) software stack. For more intensive applications, the Co-I has a 24/7 access to high-performance computing (HPC) and associate technical support in the AIISC. This HPC infrastructure includes:

- Multiple GPU clusters for deep learning and GPU intensive computations: (a) eight A100 NVidia "Multi-Instance GPU" (usable as 56 GPUs) with 1TB of RAM; Performance: five petaFLOPS AI, 10 petaOPS INT8; (b) Nvidia V100 GPU with 32GB memory and 6,000 GPU cores for compute-intense calculations and three M-6000 GPUs each with 24 GB memory and 3,072 GPU cores for computational research. Reserved budget is available for the procurement (if and when needed) of an additional dedicated node on this cluster for data-intensive analyses being performed by members of Dr. O'Reilly's Laboratory.
- High-Performance Computing (HPC) cluster, Hyperion, which includes 6,760 compute cores that include powerful Graphical Processing Unit (GPU) and Big Data nodes connected by a very high-speed 100 Gigabit/second network.
- The OpenPOWER server powered by an IBM Power-8 CPU supporting up to 160 concurrent threads. Additionally, high-end GPUs communicate with the CPU over high-speed NV-LINK connections that significantly increases throughput for GPU-intensive application.

Dail Research Lab, PI: Dail

Dr. Dail (Co-I) maintains a dedicated research lab (Dail Research Lab) on the 5th floor of the College of Nursing. The lab is equipped with three computer stations, each with dual monitors. The research coordinator space for Dail's present R01 study (1R01NR017872-01A1) with active funding (09/26/2018-06/30/2023) is located in the research lab with computer and dual monitors, equipment storage, and an instrument testing area. The lab also has conferencing space with video projection, microphone, and speakers to use for study meetings and education for research personnel in Dr. Dail's and colleagues' studies. Dr. Dail has research funds from start-up funds and industry studies to facilitate education of doctoral, masters, and undergraduate students conducting research projects with Dr. Dail. Several students are working on Dr. Dail's research studies and the

lab provides a rich environment to experience all aspects of study setup and methods, instrument testing, study team meetings, and data analysis.

RECRUITMENT RESOURCES

Collaborative relationships with local community partners have been put in place to support participant recruitment and establish a mutual relationship between the local infant, autism, and preterm/early intervention community and Dr. Bradshaw's Early Social Development Lab. The South Carolina Department of Disabilities and Special Needs (DDSN) has an established relationship with USC researchers and has served as a significant resource for recruitment of infant siblings of children with ASD. The DDSN Associate State Director has consistently pledged support for all of the PI's NIH and Foundation projects and will disseminate information about this research project to their database of over 1600 local families with a young child with ASD. Notably, our DDSN partnership is active and strong. Our most recently mailing with this organization occurred in March 2022. Additionally, lead practitioner of one of the largest pediatric practices in the area, Dr. Deborah Greenhouse, is exceedingly supportive of this research and will actively recruit participants in her practice. The PRISMA Developmental Pediatrics clinic is one of the premier diagnostic clinics in the area and lead psychologist, Dr. Emily Lowell, has pledged her support for continued recruitment efforts. She personally refers all eligible participants to our lab. Similarly, Dr. Laura Carpenter, psychologist at Developmental Pediatrics at the Medical University of South Carolina (MUSC) and investigator on the nation's largest genetic study of autism (SPARK), has pledged her support of this project through direct referrals. Dr. Katie Franke is the Director of the Unumb Center for Neurodevelopment Psychological Clinic, serving over 400 individuals with ASD in the Columbia metropolitan area. The Unumb Center offers a variety of assessment and treatment services for children with ASD and is closely connected to the local autism community. Dr. Franke is enthusiastic about the scientific aims of this research and will dedicate significant resources to assure Dr. Bradshaw meets her recruitment targets.

The CAN Center Research Registry is a database of contact information of families with children with ASD and children who are considered typically developing. The database is supported by the CAN Center at USC and is accessible to faculty who are conducting research in the area of autism and neurodevelopment. Families in this database have expressed interest in being contacted about research studies and receive periodic updates about new study openings within CAN. This database was established in April 2021 and **currently includes over 1,100 families**, with approximately 4-5 families being added each week. Local pediatricians, OBGYN providers, and other community partners have been instrumental in supporting the growth of this database and continue to serve as helpful referral sources.

The USC Infants and Children Research Registry was created to support recruitment efforts for studies across USC more broadly and includes research faculty from various departments. Similar to the CAN Center Research Registry, this registry includes the contact information for families with infants and children who have expressed an interest in participating in developmental research. The information in this registry is available upon request to researchers at the USC who have an approved research protocol by the USC Institutional Review Board for the use of human subjects in research. Funds for this database come from various sources such as individual grants, department support, college or university funds, and direct grants. This database maintains a strong online presence through the registry's active website and by continually advertising the registry across social media and other online platforms. Additional funds for both databases will also be used to attend and recruit at state conferences, strengthen relationships with providers, and further establish a presence in the local community (e.g. attending the State Fair, setting up at local markets, and organizing outreach events).

PRISMA Health Richland's Children's Hospital Neonatal Intensive Care Unit (NICU). Dr. Victor Iskersky, head neonatologist for PRISMA Health, has pledged significant support for this project (see *Letters of Support*). The Birthplace at PRISMA Health Richland features a 70-bed level III neonatal intensive care unit which is designed for care of infants of all gestational ages. This unit is the Midlands regional perinatal center, caring for infants from across the 16-county Midlands Perinatal Region. PRISMA Health is the largest and most comprehensive integrated health care system in the South Carolina Midlands region and serves 15 counties in South Carolina. The system is made up of PRISMA Health Baptist (413 beds), PRISMA Parkridge (76 beds), PRISMA Health Children's Hospital (163 beds at PRISMA Health Richland), PRISMA Heart Hospital (124 beds at PRISMA Health Richland), Palmetto Health Richland (649 beds), and PRISMA Health Tuomey (301 beds). This not-for-profit system is locally owned and consists of six Joint Commission-accredited acute-care hospitals.



PRISMA Health is supported by a network of physicians and advanced practice providers in the PRISMA Health-USC Medical Group and the PRISMA Health Quality Collaborative, as well as two 501(c)(3) foundations. Annually, PRISMA Health provides care for more than one million patients and has collectively over 7,000 infant births. In addition to its seven acute care hospitals, PRISMA Health has an expansive physician network and dozens of affiliated clinics and specialty care practices. PRISMA Health is a teaching hospital with 24 residency and fellowship programs affiliated with the USC School of Medicine. The mission of the **PRISMA Health Research Division** is to improve the health and quality of life of all the individuals served by providing patients with a supportive and safe environment to participate in cutting edge research; researchers with a framework of quality, compliant, and ethical services for the development, coordination, and oversight of their research; and sponsors the assurance that PRISMA Health is a partner in research. The Research Division is composed of a clinical trials division, institutional review board (IRB) administration, research compliance division, and resident research division. The Research Division hosts bi-monthly meetings for those interested in clinical research, including investigators, research staff professionals, and IRB members. PRISMA Health Research Division has full accreditation by the Association for the Accreditation of Human Research Protection Programs, Inc. The Clinical Trials department provides project management and centralized budget, planning, and contract processes. The department provides seamless scheduling, case report form completion, and submissions to the local IRB. The department also provides services ranging from multi-center study coordination to assistance with setting up data management systems and patient recruitment. The Research Compliance department coordinates research contracting and subaward review, establishes appropriate research billing and coding mechanisms, provides biostatistical support for protocol development and data analysis, offers grant support services, administers periodic audits to those conducting research, and administers the PRISMA Health Administrative Review (PHARR) process. This process provides a standard method for research compliance issues to be assessed. Items to be reviewed include conflict of interests, scientific validity of projects, research/subaward contracts, and financial feasibility.

Medical University of South Carolina (MUSC) Children's Hospital. MUSC Neonatologist, Dr. Heidi Steflik is a Co-I on Dr. Dail's current R01 (Dr. Dail is a Co-I on this proposal). Dr. Steflik has committed significant support for recruitment within the MUSC Children's Hospital and NICU. The 178-bed MUSC Children's Hospital is ranked in the top 20 hospitals for children's heart programs and in the top 50 for pediatric gastroenterology programs in *U.S. News and World Report 2012* and also nationally ranked for its specialty services by *Child Magazine*, and by *Best Doctors in America* (235 faculty listed). MUSC Children's Hospital **neonatal services** are located in Charleston, SC. Neonatal services contains 38 neonatal intensive care unit (NICU) beds, and 28 neonatal intermediary care beds. It is the only NICU covering an 8-county region of South Carolina (Beaufort, Berkeley, Charleston, Colleton, Dorchester, Georgetown, Hampton, and Jasper). The Children's Hospital's mission is to offer the best possible health care to the children of South Carolina. The **Department of Pediatrics' Division of Neonatology** has a strong commitment to the nutrition of neonates, including preterm infants. Members of the Division of Neonatology have published numerous articles and performed studies on lactation and the composition of breast milk. Active nutrition studies in the Division of Neonatology include two funded by NIH, one funded by Thrasher Research Fund, and one funded by industry. Two prior nutrition studies were completed within the last two years, one funded by NIH and one funded by Thrasher Research Fund.

INSTITUTIONAL INVESTMENT

The University, College of Arts and Sciences, and Department of Psychology have demonstrated **a great deal of investment in the PI's success** through a variety of resources spanning resources, career enrichment programs, formalized department mentorship, logistical support, and financial support. Each new faculty member in the department is provided a formal mentor to assist with starting their research program and demystifying administrative processes within the department. In regard to financial support, the startup package for Dr. Bradshaw was quite generous and the **PI recently received an institutional funding package that more than quadrupled the size of her original startup**. These funds represent significant institutional investment that can support the science of the proposed project. This investment will allow the PI to support graduate students, purchase equipment, and travel to scientific meetings to develop collaborations. In addition, the Department of Psychology is very supportive of protected research time for all faculty, especially Early Stage Investigators. There are a number of departmental **Research Incentives** designed to decrease teaching and increase support for grant-funded faculty. Investigators with grant funding in our department routinely teach only 1-2 classes *per year*. Thus, the PI will have significant protected research time. The Psychology Department has significant

administrative resources that assist with finance management, travel planning, grants pre- and post-award. The PI has unlimited access to these resources to assist with the proposed project.

USC Psychology Students in the Undergraduate Program and Clinical-Community Doctoral Program.

Another valuable resource to the PI through the department is the availability of undergraduate and graduate students. With nearly 1,000 undergraduate psychology majors at the USC, a large number of students seek out research experiences with faculty holding interests common to theirs. The PI routinely receives dozens of applications (>40) after each advertisement for research assistants. The PI and lab staff currently oversee ~20 undergraduate students who assist with data collection, data management, behavioral coding, literature reviews, and general administrative tasks. With regard to graduate students, the department provides a full stipend (\$18,250 academic year) and covers tuition (\$13,650) for every graduate student in the program. In addition, the department prioritizes Early Stage Investigators for selection of students and funding packages. This results in very high-qualified candidates and the ability for faculty to be selective in admission selections to enroll the highest qualified applicants with interests lined up to respective faculty. The department uses a mentorship model and has a primary goal of preparing students in research and only accepts Ph.D. level students. The PI currently has four students funded under an **NIH T32 training program**.

Commitment to Supporting Early Stage Investigators by the Office of the Vice President for Research.

This office at the USC plays a primary role in strategic development of the research infrastructure of the University in order to help build the institution's existing research base and provide support and assistance for faculty and students conducting research throughout the University. There is a particular emphasis on Early Stage Investigators. The office supports faculty in securing funding for their research and scholarly activities. The Office enhances the teaching and learning environment at the University by integrating research into the educational experience of undergraduate and graduate students, ensuring the integrity of research, providing assurance to governmental and private funding agencies and to the public that USC research is conducted in accordance with the law and the highest ethical standards, and promoting interdisciplinary research as an important component of economic development in the Columbia region and across the state of South Carolina. Specifically, the office offers:

- Grantsmanship and research coursework for faculty, such as the "R01's and Beyond" and "National Institutes of Health Peer Review" series.
- Several research support offices including: Sponsored Awards Management (pre-award), Office of Research Compliance, Information and Technology Data Management, and Office of Undergraduate Research (centralized resource for engaging undergraduates in research)
- Research Cyberinfrastructure works with researchers to improve project performance and secure computing resources.
- Advanced Support Programs for Innovative Research Excellence (ASPIRE) Funding Program allows faculty the opportunity to compete for funding to begin a promising new research endeavor for later external funding, build up a collaborative, interdisciplinary research project, and/or invest in the university's research infrastructure. Awards are given annually and range from \$15,000- \$100,000.
- Breakthrough Faculty Awards recognize relatively early-career assistant and associate professors who demonstrate considerable contributions to their field in terms of research and scholarly activity while at USC. This recognition can contribute to recipients' career advancement and retention and helps the university attract the best and brightest young faculty to join USC.
- Governor's Awards for Excellence in Science is an internal selection process to nominate a candidate for the Governor's Awards for Excellence in Science, helping to ensure that USC is as competitive as possible in this prestigious statewide awards program.
- The Research Initiative for Summer Engagement, or RISE, program supports summer research for faculty members at USC. This program employs a competitive application process to provide money for summer salary, research supplies, travel related to research, and undergraduate student support to bolster scholarship throughout USC.
- Support for Minority Advancement in Research Training aims to enhance the training of under-represented minority students by providing funds for USC faculty to mentor minority undergraduate students through summer research projects.

The University also supports several centers and initiatives dedicated to supporting the careers of faculty at USC. The Center for Teaching Excellence provides a variety of resources for teaching, including workshops and consultations. The Center hosts the New Faculty Academy, a cohort-based program for new faculty at USC. This program offers monthly breakfast seminars where the PI has opportunities to engage with peers and discuss issues relevant to early career scientists. USC is also an institutional member of the **National Center for Faculty**

Development and Diversity, which provides monthly writing “boot camps,” online support networks, writing clubs, webinars, and forums aimed at promoting faculty development. Finally, Faculty Tuition Assistance is available to full-time faculty members, and the PI is eligible to enroll in one 3-credit course per semester with the tuition fee waived.

COLLEGE FACILITIES & RESOURCES

The College of Arts and Sciences

The PI's home department of Psychology is housed within the College of Arts and Sciences. The Psychology department is located in the Barnwell and Hamilton buildings which are five stories and 110,000 square feet. The graduate psychology programs offer both the Master of Arts and the Doctor of Philosophy degree in one of four emphases: experimental psychology, quantitative psychology, clinical-community psychology, and school psychology. In addition to the four graduate degree programs, the Department of Psychology has developed research focus areas in Developmental Cognitive Neuroscience, Prevention Science (youth, family, school, and community domains), and Child and Family Outcomes. Students in any of the graduate degree programs can interface with any of the research areas. The Department of Psychology Information Technology Services (IT Services) provides complete file and print services, internet connectivity, electronic mail, webmail, and web publishing support to the Early Social Development and Intervention Lab. The IT staff maintains all computer, web, and network services and supports a standard suite of software, as well as advanced statistical software, database management, web publishing, and print publishing.

The Psychological Services Center (PSC) is a service, training, and research facility of the Psychology Department at the USC. The PSC is a nonprofit facility established in 1965 in order to provide assessment and treatment services to the USC community and the Columbia Metropolitan area. This facility has 4,700 square feet of professional office space centrally located in downtown Columbia. The space has a large two-level parking garage, ample handicapped parking spaces, is clearly visible from the street, and is located on a city bus line to provide convenient access. Thus, the facility serves as a natural focal point for face-to-face interactions between the Department of Psychology and the local community. The space features the PSC, a Psychology Research Suite, and shared facilities for these two operations. The PSC is a 2,100 square foot outpatient mental health clinic staffed by the faculty and graduate students of the Psychology Department. In addition to serving as an outpatient mental health clinic, the PSC is used for clinical research projects. The PSC is a source of patient populations for research and also provides opportunities to evaluate assessment and intervention practices conducted within the clinic. The PSC features eight meeting rooms (~130 sq. ft. each) for individuals, couples, or families receiving services through the center, a conference room for group meetings (~230 sq. ft.), a reception area (~380 sq. ft.), and three offices for core PSC staff (~145 sq. ft. each). The Psychology Research Suite is a 750 square foot facility adjacent to the PSC which features two meeting rooms for individuals or small groups (~130 sq. ft.), as well as a conference room (~200 sq. ft.) and reception area (~300 sq. ft.). Shared facilities include a large conference room (~600 sq. ft.), an information technology and computing lab (~240 sq. ft.), and a flexible use multi-purpose room (~850 sq. ft.). All meeting and conference rooms in the center are equipped with desktop Pentium 4 Personal Computers linked to the College of Arts and Sciences computer services. The computing lab has an additional four Pentium 4 computers, plus photocopying, scanning, and fax equipment. In addition, all conference and meeting rooms have built-in digital audio-video equipment (Axis Communications 211A network cameras with microphones) linked and secured via a dedicated server with audio-video recording software to support observational studies of behavior. There is extra capacity built into the recording system so that it can accommodate additional cameras as needed for special projects. All of the Ethernet-based equipment for the facility is connected directly to the University backbone via a T1 line.

The College of Engineering and Computing

Co-I Christian O'Reilly's home college is the College of Engineering and Computing (CEC). Based at USC's main campus in Columbia, CEC is ranked by both *U.S. News and World Report* and the National Research Council. The first Engineering classes were taught at the USC in 1848, and engineering degrees have been offered since 1894. The CEC was established in 1909 and currently has an enrollment of more than 3300 students, including 423 full-time graduate students and 2545 full-time undergraduates. The College currently employs 124 tenure, research, and clinical track faculty members. The CEC is located in the Swearingen Engineering Center, which is one of the largest engineering facilities in the Southeast. The USC College of Engineering and Computing's mission is to become a leader in technology innovation, engineering and computing education, and entrepreneurship. The CEC provides support to the Artificial Intelligence Institute of South Carolina.

Artificial Intelligence Institute (AIISC) is a university-wide AI Institute where Dr. O'Reilly is located. The AIISC is a major new initiative at USC. It is located on the top floor of the Science & Technology Building on main campus. Since its inception in 2019, AIISC has grown to around 35 faculty and graduate students in Fall 2021. This growth continues as the institute has embarked on hiring five additional core AI and five new interdisciplinary faculty members at all ranks. These faculty are complemented by existing faculty collaborators in Computer Science and Engineering department, as well as other collaborating centers and faculty members engaged in interdisciplinary research. Faculty at AIISC conduct research on a broad array of AI topics, providing a rich environment supportive of the proposed work. Further, the AIISC has identified the intersection of neuroscience and AI as a priority target for the development of the institute and, in a joint effort with the CAN Center of Excellence, has recently hired two new faculty (including Dr. O'Reilly) in that area.

The College of Nursing

Dr. Robin Dail (Co-I) is the Associate Dean of Faculty Affairs and Professor in the College of Nursing. The College of Nursing is located on USC's Columbia campus, housed in the Williams-Brice building, a six-story facility that has been remodeled over time to maintain state-of-the-art technology to support the College's mission. The College is located next to the College of Liberal Arts and is in close proximity to other colleges on the main campus and the library. The College of Nursing is one of five health-related colleges including pharmacy, social work, public health, and medicine. The faculty has a philosophical commitment to collaborate with other health disciplines. This commitment has been fulfilled through interdisciplinary collaboration in health planning, grant writing, delivering services, and developing educational programs. The College of Nursing offers both graduate and undergraduate programs. Graduate programs include two doctoral programs, the Doctor of Nursing Practice (DNP) degree and the Doctor of Philosophy in Nursing Science degree, and a Master of Science in Nursing (MSN) degree. The College offers a four-year undergraduate program that leads to the Bachelor of Science in Nursing (BSN) degree with an RN/BSN option. A combined BSN/MSN Education Plan is available for both BSN and RN/BSN.

College of Nursing Research Resources. The College of Nursing has significant infrastructure to support research operations. There are four video conferencing rooms, three Polycom rooms and one setup for a variety of web conferencing protocols. This technology affords the ability to offer a wide range of video and data collaboration functions. Web collaboration is possible through the use of Adobe Connect and Gotomeeting which can provide unlimited on-demand web meetings. Faculty and staff can create virtual conference rooms from any web-connected location. The College also provides the Technology Resource Center (TRC) for faculty and student use. The TRC provides 52 general use computer workstations and provides AV and portable computer resources to the faculty. The TRC is equipped with a laser printer, nursing computer applications, web access, statistical software packages, including SAS (9.4) and SPSS, and access to RedCap database. The College of Nursing Office of Research (OR) is under the leadership of an Associate Dean for Research and staffed with two program coordinators, a biostatistician, and graduate assistants. The OR assists faculty with research proposal development and submission including budget preparation. The OR also provides assistance to faculty with dissemination activities such as abstract, manuscript, and poster preparation and provides information on funding opportunities. A full-time doctorally prepared statistician and part-time graduate assistants augment the research capabilities of faculty and graduate students who are conducting research, preparing proposals, and/or developing manuscripts. Faculty have access to retrieval systems containing national census and health data within the University computer system. A variety of packaged computer software (statistical, graphics, word processing, and spreadsheets) are available for faculty use. The campus also provides access to several supercomputer cores for major processing applications.

UNIVERSITY RESOURCES

The University of South Carolina in Columbia is the state's flagship research university and is ranked by the Carnegie Foundation as being in the highest tier of research institutions ("very high research activity") in the United States. It is located in Columbia, the capital city and geographical center of the state, and is adjacent to the state government complex and the downtown business and cultural districts. The Greater Columbia area has a population of more than 600,000 and is easily accessible by public transportation and interstate highways. The University enrolls 16,000 undergraduate students and 8,400 graduate and professional students. It offers more than 350 major degree programs, including baccalaureate degrees in 126 areas, master's degrees in 172 areas, doctoral degrees in 74 areas, and professional doctorates in business, education, engineering, law, medicine, music, pharmacy, and public health. More than two dozen programs are nationally and internationally ranked.

Carolina Autism and Neurodevelopment (CAN) Research Center at USC. The CAN Center is an exciting

new resource at USC. In 2019, Co-I Jane Roberts was awarded \$4.8 million by the USC Office of the Provost to establish the CAN Center of Excellence through 2024. The initiative has three primary aims: (1) Increase USC's research capacity in autism and neurodevelopmental disorders through new faculty hires strategically designed to fill critical gaps in research and teaching; (2) Build a collaborative research center through new opportunities to enhance multidisciplinary approaches and cultivate innovative collaborations; (3) Advance USC's neuroscience education by developing a multi-departmental Neuroscience major that engages faculty across USC's broad neuroscience community. As a result of this initiative, USC has hired four new autism-focused faculty, including one of the Co-Is on this project (O'Reilly). In addition, the investigative team is heavily involved in CAN with access to significant infrastructure including pilot grant funds, the neurodevelopment research database, interdisciplinary faculty with expertise in neurodevelopmental disorders, quarterly seminar series, and structured meetings to foster collaboration.

The Research Consortium of Children and Families (RCCF) is a multidisciplinary USC endeavor drawing together faculty from behavioral/social sciences and closely related disciplines who conduct grant-funded research related to children/youth. Dr. Bradshaw is an invited member of RCCF, directed by Carolina Distinguished Professor Dr. Ron Prinz. The unique and innovative feature of the RCCF is that it draws together faculty from disparate units, without creating a structural entity such as a center or institute, to provide a vehicle for collaboration that builds on existing areas of funding success, nurtures research capacities of promising junior faculty, attracts talented new faculty, and creates a multidisciplinary climate that is responsive to societal priorities. Among the goals of the RCCF is to create and promote a scientific network among child/family researchers at USC to facilitate state-of-the-art research and multi-disciplinary collaboration. RCCF employs a full-time Research Manager, June Greenlaw, dedicated to pre-award grant services, including pre-submission, internal and external peer review, and budget consultation. Annual retreats are organized for all faculty affiliates that allow for discussions with other University scientists with overlapping interests.

The Maternal Child Research Interest Group. The Maternal Child Research Interest Group (MCRIG) is a group of faculty within the College of Nursing at the USC meeting together for research collaboration, education, and networking in the area of maternal, child, and infant research. The mission of the MCRIG is to improve health and well-being for families in South Carolina, the United States, and globally through research and innovation, leadership, and excellence by the members of the MCRIG. The MCRIG members will accomplish their endeavors with respect to diversity, inclusivity, commitment, caring, integrity, respect, and professionalism, which are the core values of the USC College of Nursing. MCRIG members will be held to the highest standard of ethical scholarship in all research and development endeavors. The MCRIG has been formed with the goal to become a USC College of Nursing Research Center, with this application for research center designation in progress. The MCRIG personnel include faculty, staff, graduate and undergraduate student research assistants (GRAS/UGRs), and people not affiliated with USC. The MCRIG Director, Core Faculty, and Staff are employees of the USC College of Nursing. Associate MCRIG Faculty are USC faculty who have appointments at Colleges or Schools other than the College of Nursing. Affiliate MCRIG Faculty are people who contribute to the MCRIG but who are not USC employees or students. The MCRIG sponsors faculty speakers across the nation to present their research each month to the MCRIG. These efforts are designed to expand group research knowledge and expertise and to foster research collaboration amongst group members. **Dr. Robin Dail (Co-I), Professor & Associate Dean for Faculty Affairs, is the director of the MCRIG. Dr. Bradshaw (PI) is an Associate Faculty member in this group.**

USC Honors College. Additionally, the USC is home to the South Carolina Honors College, which has been ranked the nation's top public honors college since 2012. The Honors College strongly encourages student research participation and provides funds for undergraduates to engage in research (up to \$10,000 per student). This is an incredible resource for the PI, whose lab model incorporates undergraduate research.

South Carolina Autism and Neurodevelopmental Disorders (SCAND) Consortium is a network of professionals in various fields who are united by their mission to understand neurodevelopmental and autism disorders to provide support for families affected by these disorders throughout the state of South Carolina. The mission of SCAND is to drive research excellence in South Carolina for clinical, translational, and mechanistic studies aimed at understanding and treating neurodevelopmental and autism spectrum disorders through a state-wide, multi-disciplinary research consortium. SCAND holds an annual research symposium, bringing in internationally recognized experts in ASD, such as Matthew State (UCSF) and Marsha Mailick (U Wisconsin-Madison). The first annual SCAND Symposium was held in 2017, with 110 people in attendance from 14 institutions and organizations, including state legislators, throughout South Carolina. Dr. Bradshaw is an affiliate of this consortium.

The SmartState Center for Childhood Neurotherapeutics, supported by the Office of the Vice President for Research, is an initiative that has attracted more than \$1.4 billion in investments. The SmartState Center for Childhood Neurotherapeutics, led by endowed chairs Dr. Jeffery Twiss and Dr. Manuel Casanova, supports research on the prevention of brain damage in premature infants and curing infant brain diseases through cellular engineering. Dr. Twiss, the Associate Dean for Research, co-founded the SCAND Consortium.

The ACORN Center. The Advancing Chronic Care Outcomes through Research & iNnovation (ACORN Center) is one of South Carolina's SmartState Centers. The SmartState program was established by the state's General Assembly in 2002 with \$180 million of non-tax revenue generated from the South Carolina Education Lottery. SmartState research centers (1) expand the state's knowledge base, (2) create public-private partnerships, (3) support startup firms, and (4) help retain highly talented workers to actively support ongoing development of the knowledge economy. The ACORN Center focuses on innovative, multidisciplinary research to improve health and quality of life among people with chronic conditions. The Center is located within the College of Nursing on the USC Columbia campus and is directed by Dr. Cynthia Corbett, Professor and Endowed Chair. The Center also has a project coordinator, six additional full-time faculty members, and several undergraduate and graduate research assistants. The Center has funds to support pilot and feasibility projects, three offices with workspace, filing cabinet storage, and technology such as digital recorders, iPads, and laptops available for research, as well as space for students that are in active research collaboration with ACORN faculty.

McCausland Center for Brain Imaging, with its new Siemens Magnetom Trio (3 Tesla), Magnetic Resonance Imaging system, will provide USC faculty, as well as collaborative and externally funded researchers, the ability to perform high-resolution, total body imaging studies in Columbia, South Carolina. The McCausland Center was established with funds from the Department of Defense, Center of Economic Excellence, and South Carolina Education Lottery funds, as well as \$1.5 million gifts from both the Palmetto Richland and Greenville Hospital systems, and \$1.75 million in matching funds from Peter J. McCausland, a '71 graduate of the USC.

Statistical Consulting Laboratory (Stat Lab) provides statistical services to projects with government agencies, private industry, and members of the University community working on grants. The Stat Lab also offers free and fee-for-service statistical consulting services for faculty, staff, and students of USC.

University libraries house over 7,000,000 processed items including 2,639,171 volumes, 4,075,000 units in microform, and 17,405 current periodicals. Thomas Cooper Library contains all of the University's library collections in Columbia except those located in the South Carolina Library, Law Library, Mathematics Library, Music Library, Business Library, and Medical Library. Thomas Cooper Library seats approximately 2,500 readers. Included in the seating are more than 900 private locked facilities for faculty and graduate students involved in research. Special areas in the library include the Student Computer Labs, the Science Library, Special Collections, and the Map Library. Access to all library collections is obtained through the USCAN/NOTIS Online Card Catalog. CD ROM stations are available for user searching of multiple databases. New databases are added regularly, and information on them is available in the Reference Department. Medline Search is also available from all libraries. The USC Library system offers access to literature from international sources through the online computer-assisted reference department. The computerized reference service provides bibliographic citations, statistics, and international news from recent literature on a wide range of subjects. This service is particularly strong in the natural and social sciences. Also included are government activities, publications, and grant sources.

Office of Information Technology oversees centralized and distributed computing and telecommunications services for academic, research, and administrative use to meet the needs of USC faculty, staff, and students. The Office of Information Technology provides the campus community with computing, voice, and data communications, networking, data security, video transport, information technology training, web services, customer support, desktop and server support, installation and maintenance of IT infrastructure, policies and procedures assistance, PC labs, software licensing and distribution, IT planning, applications development and support, and operational systems. The Columbia campus is covered by wireless service.

Office of Sponsored Awards Management (SAM) provides support to faculty and staff for the development of research grant applications and other external funding requests. SAM is responsible for the initial review of all proposals to ensure that they conform to regulations and requirements of the State, University, and sponsor. Upon receipt of an external award, SAM initiates the processes that establish an account with the University and works with the investigator(s), University administrators, and the sponsoring organization to ensure that the project complies with all grant or contractual agreements. SAM also serves as an interface between investigators and the accounting departments at the University.

Contract and Grant Accounting at the University ensures proper fiscal management and reporting of contract and grant funds. Contract and Grant Accounting works closely with SAM to coordinate pre- and post-

award management of grant proposals and grants, and to facilitate access by the Principal Investigators to their grant accounts.

The Office of Research Compliance, a division of the SAM, monitors the human subjects review process through the University's IRB. The IRB has an outstanding record of compliance with federal regulations and of providing service to faculty involved in research. All proposal submissions are managed electronically through the USCeRA system.

USC's Research Foundation is an element in the University's strategy for research growth, supporting faculty efforts and helping faculty with their research goals. The primary objectives of the Research Foundation are to administer proposals and awards more efficiently, assist faculty more effectively in competing for external funding, and maximize the University's returns on its research efforts. The foundation provides greater flexibility and competitiveness in implementing research grants and contracts, building infrastructure, and developing intellectual property management responsibilities.

University Computer Services include an IBM 2066-OX2 Enterprise Server and associated storage arrays, libraries, tape drives, and optical scanners. The University's Computer Services Division provides a full range of software, training, and technical support to faculty researchers. The Gambrell Computer Lab, located in the lower level of Gambrell Hall, is the main computing facility of the College of Arts & Sciences. The facility contains four labs, consisting of a mix of PC and Macintosh computers. In addition, there are a number of peripheral devices available (scanning, printing, video importing) to students in the College. The Research Computing Center supports and manages applications on multiple hardware platforms for the advanced computing technology needs for research, academic, and administrative activities within the College of Arts and Sciences.

EQUIPMENT

No major equipment will be required for this proposal.

RESEARCH & RELATED Senior/Key Person Profile (Expanded)

PROFILE - Project Director/Principal Investigator				
Prefix:	First Name*: Jessica	Middle Name	Last Name*: Bradshaw	Suffix:
Position/Title*:	Associate Professor			
Organization Name*:	University of South Carolina			
Department:	Psychology			
Division:				
Street1*:	1800 Gervais St			
Street2:				
City*:	Columbia			
County:				
State*:	SC: South Carolina			
Province:				
Country*:	USA: UNITED STATES			
Zip / Postal Code*:	292013504			
Phone Number*: 9094544652	Fax Number:			
E-Mail*: jbradshaw@sc.edu				
Credential, e.g., agency login: ucsbbradshaw				
Project Role*: PD/PI			Other Project Role Category:	
Degree Type: PHD,MA,BS			Degree Year: 2015,2012,2003	
Attach Biographical Sketch*:	File Name:	Biosketch_Bradshaw_BradshawR01.pdf		
Attach Current & Pending Support:	File Name:			

PROFILE - Senior/Key Person				
Prefix:	First Name*: Christian	Middle Name	Last Name*: O'Reilly	Suffix:
Position/Title*:	Assistant Professor			
Organization Name*:	University of South Carolina			
Department:	Computer Science and Engineering			
Division:				
Street1*:	1112 Greene St			
Street2:				
City*:	Columbia			
County:				
State*:	SC: South Carolina			
Province:				
Country*:	USA: UNITED STATES			
Zip / Postal Code*:	292010000			
Phone Number*: 8036266667	Fax Number:			
E-Mail*: christian.oreilly@sc.edu				
Credential, e.g., agency login: coreilly				
Project Role*: Co-Investigator			Other Project Role Category:	
Degree Type: PhD			Degree Year: 2012	
Attach Biographical Sketch*:	File Name:	Biosketch_O'Reilly_BradshawR01.pdf		
Attach Current & Pending Support: File Name:				

PROFILE - Senior/Key Person				
Prefix:	First Name*: Robin	Middle Name Britt	Last Name*: Dail	Suffix:
Position/Title*:	Associate Dean for Faculty Affairs			
Organization Name*:	University of South Carolina			
Department:	Nursing			
Division:				
Street1*:	1601 Greene Street			
Street2:				
City*:	Columbia			
County:				
State*:	SC: South Carolina			
Province:				
Country*:	USA: UNITED STATES			
Zip / Postal Code*:	292080000			
Phone Number*: 803-576-6187	Fax Number:			
E-Mail*: rdail@mailbox.sc.edu				
Credential, e.g., agency login: rkobel				
Project Role*: Co-Investigator			Other Project Role Category:	
Degree Type: PHD,MSN			Degree Year: 2006,2000	
Attach Biographical Sketch*:	File Name:	Biosketch_Dail_BradshawR01.pdf		
Attach Current & Pending Support: File Name:				

PROFILE - Senior/Key Person				
Prefix:	First Name*: Jana	Middle Name M	Last Name*: Iverson	Suffix:
Position/Title*:	Professor			
Organization Name*:	Boston University			
Department:				
Division:				
Street1*:	Boston University			
Street2:	635 Commonwealth Ave.			
City*:	Boston			
County:				
State*:	MA: Massachusetts			
Province:				
Country*:	USA: UNITED STATES			
Zip / Postal Code*:	022150000			
Phone Number*: 617-353-2713	Fax Number:			
E-Mail*: iversonj@bu.edu				
Credential, e.g., agency login: jiverson				
Project Role*: Co-Investigator			Other Project Role Category:	
Degree Type: PHD,MA,AB			Degree Year: 1996,1994,1991	
Attach Biographical Sketch*:	File Name:	Biosketch_Iverson_BradshawR01.pdf		
Attach Current & Pending Support: File Name:				

PROFILE - Senior/Key Person				
Prefix:	First Name*: JOHN	Middle Name E	Last Name*: RICHARDS	Suffix:
Position/Title*:	Professor			
Organization Name*:	University of South Carolina			
Department:	Psychology			
Division:				
Street1*:	UNIVERSITY OF SOUTH CAROLINA			
Street2:	DEPT OF PSYCHOLOGY			
City*:	COLUMBIA			
County:				
State*:	SC: South Carolina			
Province:				
Country*:	USA: UNITED STATES			
Zip / Postal Code*:	292080000			
Phone Number*: (803) 777-2079	Fax Number: (803) 777-9558			
E-Mail*: RICHARDS-JOHN@SC.EDU				
Credential, e.g., agency login: jrichards				
Project Role*: Co-Investigator			Other Project Role Category:	
Degree Type: PHD,MA,BA			Degree Year: 1982,1978,1976	
Attach Biographical Sketch*:	File Name:	Biosketch_Richards_BradshawR01.pdf		
Attach Current & Pending Support: File Name:				

PROFILE - Senior/Key Person				
Prefix:	First Name*: Jane	Middle Name E	Last Name*: Roberts	Suffix:
Position/Title*:	Professor			
Organization Name*:	University of South Carolina			
Department:	Psychology			
Division:				
Street1*:	Barnwell College			
Street2:	1512 Pendleton St			
City*:	Columbia			
County:				
State*:	SC: South Carolina			
Province:				
Country*:	USA: UNITED STATES			
Zip / Postal Code*:	292080000			
Phone Number*: 803-777-7000		Fax Number:		
E-Mail*: jane.roberts@sc.edu				
Credential, e.g., agency login: janeroberts				
Project Role*: Co-Investigator		Other Project Role Category:		
Degree Type: PHD		Degree Year: 1998		
Attach Biographical Sketch*:	File Name:	Biosketch_Roberts_BradshawR01.pdf		
Attach Current & Pending Support:	File Name:			

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Jessica Bradshaw

eRA COMMONS USER NAME: ucsbbradshaw

POSITION TITLE: Associate Professor, Department of Psychology, University of South Carolina

EDUCATION/TRAINING

INSTITUTION AND LOCATION	DEGREE	Completion Date MM/YYYY	FIELD OF STUDY
University of California, San Diego	B.S.	06/2007	Cognitive Science
University of California, Santa Barbara	Ph.D.	09/2015	Clinical Psychology
Emory University School of Medicine, Department of Pediatrics	Postdoctoral	08/2018	Clinical Psychology and Social Neuroscience

A. Personal Statement

My long-term research interests involve methods for 1) quantifying neonatal and infant behavior, 2) identifying diverse developmental pathways that lead to atypical development of social communication and the emergence of autism spectrum disorder (ASD), and 3) translating these basic findings to actionable early detection and intervention strategies. I use a variety of experimental methodologies (electrocardiogram (ECG), behavioral coding, eye tracking) to evaluate attentional and neurobiological abnormalities and phenotypic variation in the context of early detection and intervention for infants who are at an elevated likelihood for developing ASD (infant siblings of children with ASD; ASIBs). I am particularly interested incorporating unique control groups in the study of atypical development. In this application, we propose a control group of infants born preterm (PT) as a model of broad autonomic nervous system (ANS) dysfunction. This will be contrasted to ASIBs who we hypothesize to have ANS dysfunction specific to the regulation of attention.

I have spent much of the last 14 years engaged in prospective, longitudinal research of infant siblings of children with ASD and contributing to NIH awards (PIs: Kasia Chawarska, Yale; Ami Klin, Yale, Emory). I currently have a K23 award (mentors: Jane Roberts, John Richards) and R21 (R21DC017252, 2019-2022), both utilizing prospective, longitudinal designs of ASIBs and PTs. These awards have contributed to my expertise in: 1) recruitment of ASIB and PT infants in South Carolina, 2) collection and analysis of the measures proposed in this study (heart activity, heart-defined attention, naturalistic interactive behavior), 3) clinical training of staff and clinical care of participants, and 4) leadership and management of largescale, longitudinal studies of infants. The focus of the proposed study on very early, mechanistic underpinnings of disrupted attention in ASD, represents a programmatic extension from my K23 and R21 and I have the scientific expertise and grant management experience to lead the proposed project. Importantly, Jane Roberts (Co-I), John Richards (Co-I), Robin Dail (Co-I), and Jana Iverson (Co-I) are senior investigators and full professors who have extensive experience leading largescale projects. I have active collaborations with all Co-Is through awarded grants or published manuscripts (see below). Their role on the proposed project includes scientific expertise as well as assistance and guidance with largescale grant management.

In addition to scientific and methodological expertise in attention and neonatal neurobehavior, I am a licensed clinical psychologist with expertise in infants and children with neurodevelopmental disorders and ASD. I am a certified trainer of the NICU Network Neurobehavioral Scale (NNNS) and research reliable ADOS examiner. I have extensive training and years of experience in all developmental measures to be used in this study.

In sum, my experience in study and grant management, scientific and clinical expertise in neonatal development and infants with ASD, and the experience and expertise of my interdisciplinary, diverse investigative team (spanning senior to early-stage investigators), provide an excellent foundation for successful achievement of the proposed aims that will inform new etiological theories of ASD.

Ongoing and recently completed projects that I would like to highlight include:

K23 MH120476

Bradshaw (PI)

Roberts (Mentor), Richards (Mentor)

7/19/2019-6/30/2024

Heart Defined Attention in Infancy: Predicting Social Communication and ASD Symptomology

R21 DC017252

Bradshaw (PI)

12/1/19-11/30/22

Neurobehavioral Determinants of Social Communication and Language Impairments in at-risk Infants

R01 HD099295

Feinberg, Hock (MPIs) Role: Co-Investigator

6/17/2020- 2/25/2025

Test of an Innovative, Scalable Support Program for Parents with a Young Child Recently Diagnosed with Autism Spectrum Disorder

Carolina Autism and Neurodevelopment Research Center Pilot Award

Bradshaw/Dail (MPIs)

7/1/21-7/1/22

Examining Neonatal Thermal Gradients and Heart Activity as Predictive Risk Factors for Autism Spectrum Disorder in Premature Infants

Carolina Autism and Neurodevelopment Research Center Pilot Award

O'Reilly (PI), Role: Co-Investigator

6/1/22-6/1/23

Model-Driven Analysis of Autonomic Control in ASD Using ECG

Citations:

1. **Bradshaw**, J., Schwichtenberg, A. J., & Iverson, J. M. (2022). Capturing the complexity of autism: Applying a developmental cascades framework. *Child Development Perspectives*.
2. Roberts, J. E., **Bradshaw**, J., Will, E., Hogan, A. L., McQuillin, S., & Hills, K. (2020). Emergence and rate of autism in fragile X syndrome across the first years of life. *Development and psychopathology*, 32(4), 1335-1352.
3. **Bradshaw**, J., Vinyard, A., Platt, E., Abney, D. (2022). Infant Attention and Physiological Arousal During Dyadic Interactions as Predictors of Social Communication. XXIII ICIS Biennial Congress: Developmental Cascades. 2022 Jul 7-10; Ottawa, ON Canada.

B. Positions, Scientific Appointments, and Honors

Positions and Employment

2007-2009	Research Assistant, Yale Early Social Cognition Lab, Yale Child Study Center, New Haven, CT
2009-2010	Lab Manager, Yale Early Social Cognition Lab, Yale Child Study Center, New Haven, CT
2010-2014	Graduate Student Researcher, Koegel Autism Center, University of California, Santa Barbara
2014-2015	Clinical Psychology Intern, Rady Children's Hospital, University of California, San Diego
2015-2018	Postdoctoral Research Fellow, Marcus Autism Center, Emory University School of Medicine, Atlanta, GA
2018-	Assistant Professor, Department of Psychology, University of South Carolina, Columbia, SC
2022-	Associate Professor, Department of Psychology, University of South Carolina, Columbia, SC

Other Experiences and Professional Memberships

2010 -	Member, International Society for Autism Research
2012 -	Adhoc Reviewer, Journal of Autism and Developmental Disorders
2014-2015	Chief Intern, UCSD/VA Psychology Internship Training Program, San Diego, CA
2016 -	Adhoc Reviewer, Autism

- 2016- International Society for Infant Studies
- 2016- Licensed Psychologist, State of Georgia, PSY004065
- 2019- Adhoc Reviewer, Journal of Child Psychology and Psychiatry, Development and Psychopathology, Journal of Neurodevelopmental Disabilities, Autism Research, International Journal of Developmental Disabilities
- 2020 - Member, Society for Developmental Psychobiology
- 2020 - Member, Society for Research on Child Development
- 2021 - Member, Journal of Autism and Neurodevelopmental Disabilities Editorial Board
- 2021 - Member, International Society for Developmental Psychobiology
- 2021 - Member, International Congress for Infant Studies Communications Committee
- 2020 - Member, Baby Siblings Research Consortium
- 2021 Early Career Reviewer, National Institute of Health, Child Psychopathology Dev. Disabilities Study Section
- 2022- Elected Board Member, International Congress for Infant Studies

Honors

- 2006 Phi Beta Kappa
- 2007 Cognitive Science Honors Program, UCSD
- 2007 BS awarded *cum laude* with *Distinction in Cognitive Science*, University of California, San Diego
- 2011-2012 Dean's Fellowship, UCSB
- 2011-2012 Autism Science Foundation Predoctoral Fellowship
- 2012 Travel Award, Graduate Division, UCSB
- 2013 Outstanding Research Award, UCSB
- 2013-2014 Graduate Research Mentorship Program, Graduate Division, UCSB
- 2017 American Speech-Language-Hearing Association Lessons for Success Participant
- 2017 Institute of Education Sciences Summer Research Training Institute Participant
- 2018 International Society for Autism Research – Travel Award
- 2021 University of South Carolina Breakthrough Star Award

C. Contributions to Science

My Contributions to Science are organized into five distinct categories: 1) The Role of Neonatal Attention and Neurobehavior in the Emergence of Social Communication and ASD, 2) Early Identification of ASD in Infants and Toddlers, 3) Early Intervention for Infants and Toddlers, 4) Parent-Implemented and Parent-Oriented Interventions for Children with ASD, and 5) Attention and Early Social Cognition in ASD.

1. The Role of Neonatal Attention and Neurobehavior in the Emergence of Social Communication and

ASD: I am pursuing an area of autism research that, to my knowledge, is largely unexplored: the role of neonatal neurobehavior in the pathogenesis of ASD and related social and communication impairments. I have begun to map early infant experience from birth to 6 months across domains of visual attention, neurological integrity, and motor development for those at elevated and low likelihood of ASD. My research in this area suggests that very early biological and behavioral factors, including preterm birth, motor skills, ANS function, and attention, may be associated with developmental outcome, including social communication and ASD^{a-d}. Recent analysis of heart-defined attention in 3-4-month-old infants show potential value for this measure as a predictor later social-communication skills^c. Outside of ASD, I have contributed to research on developmental outcomes for infants born preterm^d.

- a. **Bradshaw, J., Shi, D., Federico, A., Klaiman, C., & Saulnier, C. (2022).** The Pull-to-Sit Task: Examining Infant Postural Development in Autism Spectrum Disorder. *The Journal of Pediatrics*, (PMID: 36202237)
- b. **Bradshaw, J., Shi, D., Hendrix, C., Saulnier, C., Klaiman, C. (2022).** Neonatal Neurobehavior in Infants with Autism Spectrum Disorder. *Developmental Medicine and Child Neurology*, (PMID: 34713902)
- c. **Bradshaw, J., Abney, D. (2021).** Infant Physiological Activity and the Early Emergence of Social Communication. *Developmental Psychobiology*, 63(7), e22145. (PMID: 34674237)
- d. **Bills, S., Johnston, J., Shi, D., Bradshaw, J. (2020).** Social-environmental moderators of neurodevelopmental outcomes in youth born preterm: A systematic review. *Child Neuropsychology*, 27(3), 351-370. (PMID: 33342364).

2. Early Identification of ASD in Infants and Toddlers: My contributions in early identification and of ASD have thus far focused on evaluating early developmental profiles of infants between 9-12 months who are later diagnosed with ASD as well as early behavioral trajectories that distinguish typically developing infants from those at elevated genetic likelihood for ASD.

- a. **Bradshaw, J.**, McCracken, C., Lewis, M., Bran, N., Delehanty, A., Day, T., Klaiman, C., Saulnier, C., Klin, A., Wetherby, A. (2021). Early Social Communication Development in Infants with Autism Spectrum Disorder. *Child Development*.
- b. Pileggi, M., Brane, N., **Bradshaw, J.**, Delehanty, A., Day, T., McCracken, C., Stapel-Wax, J., Wetherby, A. (2021). Early Observation of Red Flags in 12-month-old Infants Later Diagnosed with Autism Spectrum Disorder. *American Journal of Speech-Language Pathology*. (PMID: 33989505)
- c. **Bradshaw, J.**, Klin, A., Evans, L., Klaiman, C., Saulnier, C., McCracken, C. (2019). Development of Attention from Birth to 5 Months in Infants at Risk for Autism Spectrum Disorder. *Development and Psychopathology*, 32(2) 491-501. (PMID: 31012398).
- d. **Bradshaw, J.**, Gillespie, S., Klaiman, C., Klin, A., & Saulnier, C. (2018). Early emergence of discrepancy in adaptive behavior and cognitive skills in toddlers with autism spectrum disorder. *Autism*, 23(6), 1485-1496. (PMID: 30525959).

3. Early Intervention for Infants and Toddlers: A large part of my research has also focused on developing and evaluating early intervention strategies for infants, toddlers, and children^c, as well as examining optimal methods for conducting randomized clinical trials^a. I have investigated strategies for improving verbal and preverbal skills for infants and toddlers^d and the use of novel methods for improving social attention^b.

- a. **Bradshaw, J.**, Trumbull, A., Stapel-Wax, J., Gillespie, S., George, N., Saulnier, C., Klaiman, C., Woods, J., Call, N., Klin, A., Wetherby, A. (2020). Factors Associated with Enrollment into a Clinical Trial of Caregiver-Implemented Intervention for Infants at Risk for Autism Spectrum Disorder. *Autism*. (PMID: 32594763)
- b. Wang, Q., Wall, C. A., Barney, E. C., **Bradshaw, J.**, Macari, S. L., Chawarska, K., & Shic, F. (2019). Promoting Social Attention in 3-Year-Olds with ASD through Gaze-Contingent Eye Tracking. *Autism Research*, 13(10), 61-73. (PMID: 31468735)
- c. **Bradshaw, J.**, Steiner, A. M., Gengoux, G., & Koegel, L. K. (2015). Feasibility and Effectiveness of Very Early Intervention for Infants At-Risk for Autism Spectrum Disorder: A Systematic Review. *Journal of Autism and Developmental Disorders*. (PMID: 25218848).
- d. Koegel, R. L., **Bradshaw, J. L.**, Ashbaugh, K., & Koegel, L. K. (2014). Improving Question-Asking Initiations in Young Children with Autism Using Pivotal Response Treatment. *Journal of Autism and Developmental Disorders*, 44(4), 816-827. (PMID: 24014174).

4. Parent-Implemented and Parent-Oriented Interventions for Children with ASD: Effective interventions for infants and children with ASD require close involvement of parents and caregivers^a. My work in this area has focused on evaluating the effectiveness of parent-implemented interventions^b, including one of the first studies to focus behavioral intervention for 6-12-month-old infants^d. This work documented preliminary effectiveness of increasing social engagement in the first year of life. In an effort to individualize treatment strategies, I also aim to understand predictors of treatment response for parent-implemented interventions^c.

- a. **Bradshaw, J.**, Wolfe, K., Hock, R., & Scopano, L. (2022). Advances in Supporting Parents in Interventions for Autism Spectrum Disorder. *Pediatric Clinics*. (PMID: 35934491).
- b. **Bradshaw, J.** Koegel, L., Koegel, R. (2017). Improving Functional Language and Social Motivation with a Parent-Mediated Intervention for Toddlers with Symptoms of Autism Spectrum Disorder. *Journal of Autism and Developmental Disorders*, 47(8), 2443-24458. (PMID: 28536956)
- c. **Bradshaw, J.**, Bearss, K., McCracken, C., Smith, T., Johnson, C., Lecavalier, L., Swiezy, N., Scahill, L. (2017). Parent Education for Young Children with Autism and Disruptive Behavior: Response to Active Control Treatment. *Journal of Clinical Child and Adolescent Psychology*, 47(sup1), S445-S455. (PMID 29048234).
- d. Koegel, L., Singh, A., Koegel, R., Hollingsworth, J., & **Bradshaw, J.** (2013). Assessing and Improving Early Social Engagement in Infants. *Journal of Positive Behavior Intervention*, 16(2), 69-80. (PMID: 25313271).

5. Attention and Social Cognition in ASD: I have used eye-tracking methodology and basic experimental paradigms to understand differences in habituation, recognition, attentional disengagement, and patterns of activity monitoring in ASD. I have also applied these methodologies to evaluation of treatment response^b.

- a. Kaliukhovich, D., Manyakov, N., Bangerter, S., Skalkin, A., Goodwin, M., Dawson, G., Hendren, R., Leventhal, B., Hudac, C., **Bradshaw, J.**, Shic, F., Pandina, G. (2020). Social Attention to Activities in Children and Adults with Autism Spectrum Disorder: Effects of Context and Age. *Molecular Autism*, 11(1), 1-14. (PMID: 33076994)
- b. **Bradshaw, J.**, Shic, F., Holden, A., Horowitz, E., Barret, A., German, T., Vernon, T. (2019). The Use of Eye Tracking as a Biomarker of Treatment Outcome in a Pilot Randomized Clinical Trial for Young Children with Autism. *Autism Research*, 12(5), 779-793. (PMID: 30891960).
- c. **Bradshaw, J.**, Shic, F., & Chawarska, K. (2011). Brief report: Face-specific Recognition Deficits in Young Children with Autism Spectrum Disorders. *Journal of Autism and Developmental Disorders*, 41(10), 1429-1435 (PMID: 21373955).
- d. Shic, F., **Bradshaw, J.**, Scassellati, B., Chawarska, K. (2011). Limited Activity Monitoring in Toddlers with Autism Spectrum Disorders. *Brain Research*, 1380, 246-254. (PMID: 21129365).

Complete list of published works on MyBibliography:

<https://www.ncbi.nlm.nih.gov/myncbi/jessica.bradshaw.1/bibliography/public/>

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. DO NOT EXCEED FIVE PAGES.

NAME: O'Reilly, Christian

eRA COMMONS USER NAME (credential, e.g., agency login): COREILLY

POSITION TITLE: Assistant Professor

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE (if applicable)	END DATE MM/YYYY	FIELD OF STUDY
Polytechnique Montreal, Montreal, Quebec	BENG	12/2007	Electrical Engineering
Polytechnique Montreal, Montreal, Quebec	MENG	05/2011	Biomedical Engineering
Polytechnique Montreal, Montreal, Quebec	PHD	05/2012	Biomedical Engineering
Université de Montréal, Montreal, Quebec	Postdoctoral Fellow	05/2014	Postdoctoral researcher in biomedical sciences – analysis of sleep EEG
McGill University, Montreal, Quebec	Postdoctoral Fellow	09/2015	Brain imaging in sleep and autism
École Polytechnique Fédérale de Lausanne, Geneva, Geneva	Postdoctoral Fellow	11/2018	Modeling of the thalamocortical system

A. Personal Statement

My curriculum demonstrates that I can bring a significant contribution to this project. Although I am an early-career researcher, I showed a high level of productivity with currently 61 peer-reviewed publications (including 39 papers in international journals), over 1800 citations, and an h-index of 22 (Google Scholar). I gathered a diversified experience in modeling biological systems, analyzing time-series from physiological recordings such as human movement kinematics, electroencephalogram, polysomnogram (including ECG), magnetoencephalogram, patch-clamp recordings, and implanted electrodes (local field potentials). This experience demonstrates that I master the techniques required to process signals such as ECG, as proposed in this project. I also worked on diagnostic applications (e.g., for brain stroke, ADHD) based on pattern recognition and machine learning techniques. I gained further expertise in this field by designing automatic classifiers (e.g., for sleep spindles) and biomarkers (e.g., for early diagnosis of ASD using EEG and ECG recordings). Aside from developing significant expertise in signal processing, statistical analysis, and machine learning in the biomedical field, I also specialized in ASD. Further, apart from my professional experience, I can count on a well-established network of local (e.g., at the Artificial Intelligence Institute of South Carolina) and international collaborators. For example, my active involvement Quebec 1000 Families (Q1K) project – a world-class cohort on ASD and related neurodevelopmental disorders led by renowned Canadian institutions like McGill's Azrieli Centre for Autism Research – allows me to draw support from world-renown experts in ASD. I have been collaborating with Dr. Bradshaw (PI) since my arrival to USC, meeting 1-2x/month. Our work consists of analysis of ECG data to be used as biomarkers, resulting in the preliminary data presented in this proposal. In addition, she is a Co-PI on my pilot grant from the Carolina Autism and Neurodevelopment (CAN) Center at USC in which I will use Dr. Bradshaw's ECG data to create a computational model of the heart to inform use of heart activity as biomarkers. Given our successful collaborations thus far, my interest in generating ASD biomarkers, and my signal processing and computational skills, I am well equipped to serve as a Co-Investigator for this project and contribute to successful achievement of the specific aims.

- a. **O'Reilly C**, Elsabbagh M. Intracranial recordings reveal ubiquitous in-phase and in-antiphase functional connectivity between homotopic brain regions in humans. *Journal of Neuroscience Research*. 2020 November 15; 99(3):887-897. DOI: 10.1002/jnr.24748
- b. **O'Reilly C**, Lewis J, Elsabbagh M. Is functional brain connectivity atypical in autism? A systematic review of EEG and MEG studies. *PLOS ONE*. 2017; 12(5):e0175870-. DOI: 10.1371/journal.pone.0175870
- c. **O'Reilly C**, Nielsen T. Automatic sleep spindle detection: benchmarking with fine temporal resolution using open science tools. *Front Hum Neurosci*. 2015;9:353. PubMed Central PMCID: PMC4478395.
- d. **O'Reilly C**, Godbout J, Carrier J, Lina JM. Combining time-frequency and spatial information for the detection of sleep spindles. *Frontiers in human neuroscience*. 2015; February 19. DOI: 10.3389/fnhum.2015.00070

B. Positions, Scientific Appointments and Honors

Positions and Scientific Appointments

2021 -	Assistant Professor, University of South Carolina, Artificial Intelligence Institute, Department of Computer Science and Engineering, Columbia, SC
2020 - 2021	Research Associate, McGill University, Montreal Neurological Institute-Hospital, Transforming Autism Care Consortium, Montreal
2016 - 2017	Editor, Sleep Spindles & Cortical Up States, Budapest
2015 - 2015	Visiting scholar, Wellcome Center for Human Neuroimaging, UCL, London
2013 - 2016	Guest editor, <i>Frontiers in Human Neuroscience</i> , Lausanne
2011 -	Senior Member, Institute of Electrical and Electronics Engineers, New York, NY

Honors

2013 - 2015	Postdoctoral fellowship, Natural Sciences and Engineering Research Council of Canada
2010 - 2012	Doctoral fellowship, Natural Sciences and Engineering Research Council of Canada
2005 - 2006	Bourses UPIR - Unité de participation et d'initiation à la recherche, Polytechnique Montreal
2013	Postdoctoral fellowship (declined), Fonds de recherche Nature et technologies du Québec
2012	Recommended for best thesis award, Polytechnique Montreal
2011	Best student paper - Conference of the International Graphonomics Society, Maryland University
2011	Best paper, Association des étudiants des cycles supérieurs de Polytechnique
2010	Doctoral fellowship (declined), Fonds de recherche Nature et technologies
2006	Natural Sciences and Engineering Research Council of Canada, Undergraduate Student Research Awards

C. Contribution to Science

1. **EEG functional connectivity in ASD:** Following the publication of a very well-cited systematic review paper on EEG and magnetoencephalography functional connectivity in autism, I studied how atypical neural connectivity develops in infants. To improve EEG analyses for neurodevelopmental cohorts, I prepared and made available to the community a set of 13 realistic head models for infants between two weeks and 24 months of age. These templates can now be used to perform source reconstruction in infants with common neuroimaging software. Further, we are currently proofing a manuscript using these templates to study functional connectivity between EEG cortical sources in infants at high familial risk for autism. Aside from corroborating generalized underconnectivity in high-risk infants and showing an increased severity for the subsets of infants that were later diagnosed with ASD, it also showed different trajectories for males and females in how functional connectivity correlates with Autism Diagnostic Observation Schedule (ADOS) severity scores. Such differences in trajectories might partly explain why females are more resilient to ASD than males.

- a. **O'Reilly C**, Larson E, Richards JE, Elsabbagh M. Structural templates for imaging EEG cortical sources in infants. *NeuroImage*. 2021; 227:117682. DOI: 10.1016/j.neuroimage.2020.117682

- Page 31

study, I highlighted the large bias introduced by testing detectors on different datasets and expert scorings, i.e., the effect of the dataset and the expert scoring was typically larger than the difference in performance between different detection approaches. These results further supported the importance of providing a common open-access database and expert scorings to benchmark contributions from various research laboratories using the same gold standard. Further, in a two-part paper series, I developed and tested a novel method to assess how sleep spindles propagate across the scalp. This approach revealed a systematic pattern across subjects, with slower propagation from posterior to anterior and from central to lateral regions. These patterns are potentially indicating a systematic topology in the thalamocortical generation of sleep spindles which might provide key insights into rhythmogenesis in healthy thalamocortical interactions as well as in diseased conditions like epilepsy.

- a. **O'Reilly C**, Godin I, Montplaisir J, Nielsen T. REM sleep behaviour disorder is associated with lower fast and higher slow sleep spindle densities. *Journal of Sleep Research*. 2015 December; 24(6):593-601. DOI: 10.1111/jsr.12309
 - b. **O'Reilly C**, Nielsen T. Automatic sleep spindle detection: benchmarking with fine temporal resolution using open science tools. *Front Hum Neurosci*. 2015;9:353. PubMed Central PMCID: PMC4478395.
 - c. **O'Reilly C**, Gosselin N, Carrier J, Nielsen T. Montreal Archive of Sleep Studies: an open-access resource for instrument benchmarking and exploratory research. *J Sleep Res*. 2014 Dec;23(6):628-635. PubMed PMID: 24909981.
 - d. **O'Reilly C**, Nielsen T. Assessing EEG sleep spindle propagation. Part 2: experimental characterization. *J Neurosci Methods*. 2014 Jan 15;221:215-27. PubMed PMID: 23999173.
5. **Human movement modeling:** As a Ph.D. student, I used a lognormal decomposition to model the kinematics of human movements. In a highly cited paper, I explained how complex movements like those involved in handwriting signatures can be decomposed into lognormal components to efficiently and accurately model fine motor control. I then used this approach to study correlates of the neuromuscular system with physiological processes like aging and clinical conditions like the main risk factor for brain stroke. Further, I demonstrated how such an approach can be used to developed diagnostic tools based on movement kinematics.
- a. **O'Reilly C**, Plamondon R, Lebrun LH. Linking brain stroke risk factors to human movement features for the development of preventive tools. *Front Aging Neurosci*. 2014;6:150. PubMed Central PMCID: PMC4085479.
 - b. Plamondon R, **O'Reilly C**, Rémi C, Duval T. The lognormal handwriter: learning, performing, and declining. *Frontiers in Psychology*. 2013; 4:945. DOI: 10.3389/fpsyg.2013.00945
 - c. **O'Reilly C**, Plamondon R. Impact of the principal stroke risk factors on human movements. *Human Movement Science*. 2011; 30(4):792-806. DOI: 10.1016/j.humov.2010.07.010
 - d. **O'Reilly C**, Plamondon R. Development of a Sigma–Lognormal representation for on-line signatures. *Pattern Recognition*. 2009 December; 42(12):3324-3337. DOI: 10.1016/j.patcog.2008.10.017

Complete list of published works on MyBibliography:

https://www.ncbi.nlm.nih.gov/myncbi/1Pk_o3EaJVZ5f/bibliography/public/

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Robin B. (Knobel) Dail PhD, RN, FAAN

eRA COMMONS USER NAME (credential, e.g., agency login): RKNOBEL

POSITION TITLE: Professor & Associate Dean for Faculty Affairs, College of Nursing

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE (if applicable)	Completion Date MM/YYYY	FIELD OF STUDY
University of Nevada, Las Vegas, NV	ADN	12/1981	Nursing
East Carolina University, Greenville, NC	NNP	08/1994	Neonatal Nurse Practitioner
East Carolina University, Greenville, NC	MSN	12/2000	Nursing
University of North Carolina at Chapel Hill, Chapel Hill, NC	PhD	08/2006	Nursing

A. Personal Statement

I am delighted to serve as co-investigator with Dr. Bradshaw and team to examine infants' maturation of autonomic regulation of attention during the infant period and influence of autonomic regulation of attention on interactive behavior across with their first year of life to move the science forward to identify early risk indicators of autism spectral disorder (ASD). I am ideally suited to offer expertise on recruitment, enrollment, and evaluation of preterm infants, especially as it relates to autonomic dysregulation and their surrounding medical complexities due to prematurity.

As a nurse scientist with a program of research to reduce morbidity and mortality in preterm infants, I am an ideal co-investigator and senior researcher to support Dr. Bradshaw, an early investigator, on this proposal. Currently, I am funded by NIH/NINR (1R01NR017872) to lead a 5-site observational study of body temperature and heart rate in 440 preterm infants to predict infection including NEC.¹⁰ This study is an extension of my program of research examining thermal gradients, or the central-peripheral temperature difference (CPTd) as a marker of autonomic nervous system function. Current evidence indicates that when the CPTd is >2°C, it is an indicator of autonomic and cardiovascular stress related to infection. Our five-year study examines this marker and is comparing the CPTd model to heart rate characteristic scores from a HeRO monitor, a well known score using heart rate variability in predicting infection. Our prior studies and those of others, have also found CPTd <0°C is a marker for autonomic dysregulation due to stress, stimulation, and immaturity. With this knowledge, Dr. Bradshaw and I began collaborations to leverage the data from my current R01 from our local NICU participants, in a university funded pilot study, to follow these infants until 12 months for emerging signs of ASD. Dr. Bradshaw and I are MPIs on this pilot study (2021-2022) and have a unique, interdisciplinary collaboration with her expertise in psychology and neurodevelopment and my expertise with preterm infants and autonomic dysregulation. This pilot study is ongoing, with 25 infants enrolled to date and follow up is in progress on all infants. Our interim analyses was submitted to Council for the Advancement of Nursing science (CANS) state of the science conference for presentation. Dr. Bradshaw and I meet regularly with our research team and our offices/research labs are within walking distance on the UofSC Campus.

Ongoing and recently completed projects that I would like to highlight include:

The Gerber Foundation PI: Everhart

01/01/2022-12/31-2024 Role: Co-I

Evaluation of a blood warming device for packed red blood cell transfusions to decrease hypothermia in very preterm infants

UofSC, Carolina Autism and Neurodevelopment Research Center Pilot Award MPI: Dail & Bradshaw

07/01/2021-07/01/2022 Role: Co-PI

Examining Neonatal Thermal Gradients and Heart Activity as Predictive Risk Factors for Autism Spectrum Disorder in Premature Infants

NIH/NINR: 1R01NR017872 PI: Dail

09/26/2018-11/30/2023 Role: PI

Central & peripheral body temperature in VLBW preterm infants during the neonatal period: Relationship to neonatal infection and necrotizing enterocolitis

B. Positions, Scientific Appointments, and Honors

Academic Appointments:

2006	Assistant Professor, University of North Carolina, Chapel Hill, NC
2008-2014	Assistant Professor, Duke University School of Nursing, Durham, NC
2014-2017	Associate Professor, Duke University School of Nursing, Durham, NC
2017-present	Consulting Associate Professor, Duke University School of Nursing
2017-present	Associate Dean for Faculty Affairs & Health Sciences Endowed Professor University of South Carolina, College of Nursing, Columbia, SC

Hospital Positions Held:

1981-1983	Registered Nurse, Neonatal, Sunrise Hospital, Las Vegas, NV
1984-1986	Registered Nurse, Neonatal, Landstuhl 2 nd General Hospital, Germany
1986-1988	Registered Nurse, Maternity, Williamsburg Community Hospital, Williamsburg, VA
1987-1988	Registered Nurse, Coronary Care Unit, Riverside Hospital, Newport News, VA
1988-1992	Registered Nurse, Neonatal, Medical College of VA, Richmond, VA
1993	Registered Nurse, Neonatal, New Hanover Medical Center, Wilmington, NC
1994-2006	Neonatal Nurse Practitioner, Children's Hospital of Eastern NC, Greenville, NC
2008-2009	Neonatal Nurse Practitioner, Duke University Hospital

Honors:

2000	Induction into Sigma Theta Tau
2002	University of North Carolina Graduate Transportation Grant
2001-2002	T32 Trainee Predoctoral Fellowship, Interventions to Prevent and Manage Chronic Illness
2003	Glaxo Scholar
2004	Induction into Alpha Epsilon Lambda
2004-2005	Academy of Neonatal Nurses Graduate Nursing Scholarship
2004-2005	March of Dimes Scholarship for Nursing Graduate Students
2010-2013	Robert Wood Johnson Foundation Nurse Faculty Scholar
2010	Inducted into the American Physiological Society
2013	University of North Carolina, Chapel Hill School of Nursing Distinguished Alumna of the Year Award
2014	Fellow of the American Academy of Nursing
2016-2017	Amy V. Cockcroft Nursing Leadership Fellow
2019	Best Student Research Mentor, University of South Carolina College of Nursing
2019	Senior Investigator Award, University of South Carolina, College of Nursing
2019	University of Nevada, Las Vegas, 2019 School of Nursing Alumna Award

2022	Sigma International Nurse Researcher Hall of Fame Inductee
2022	Senior Investigator Award, University of South Carolina, College of Nursing
2022	Student Research Mentor Award, University of South Carolina, College of Nursing

Special International Responsibilities:

1996	Participation in Pediatric Heart Surgeries, Managua, Nicaragua
2002	Research Practice and Possibilities Inaugural Postgraduate Nursing Conference/International Doctoral Student Research Seminar, Melbourne, AU
2005	International Scientific Advisory Panel, Royal Canadian Nursing Association
2011	Medical Research Council, UK, Grant Review
2011	NHS Greater Glasgow & Clyde Research Endowment Fund Committee Review
2015	Health Research Board Reviewer, Ireland

Special National Responsibilities:

1998-present	Neonatal Resuscitation Program Instructor
2001-2003	Advisory Board Member, Academy of Neonatal Nurses
2012	AWHONN Research Advisory Panel, for grant review
2009-2012	American Nurses Foundation- Grant Review
2013	American Nurses Foundation, Vice-Chair for scientific grant review
2014	American Nurses Foundation, Chair for scientific grant review
2012-2013	Neonatal Network, Column Editor: Neonatal Lab Values
2012-2013	Council for Advancement of Nursing Science (CANS) Grant Review
2017-2019	Council for the Advancement of Nursing, Program Committee Member
2019-2022	Council for the Advancement of Nursing, Councilor of Programs (elected)
2019	NIH: NRCS Study Section, Temporary Member
2020	NIH: HSOD Study Section, Temporary Member
2020	NIH: ZRGI HDM-H Study Section, Temporary Member
2021-2025	NIH: ICSC Study Section Permanent Member

C. Contributions to Science

1. Using polyurethane wrap to reduce heat loss for premature infants in the delivery room. Hypothermia is a major problem after birth in premature infants and has not improved over the last 50 years of research and interventions. Our team conducted a chart review to establish the extent of hypothermia after birth at a southeastern neonatal intensive care unit (NICU) which led to a study designed to test an intervention in 100 infants to decrease heat loss in the delivery room for this vulnerable population. Intervention infants were placed in polyurethane bags, up to their necks and while still wet with amniotic fluid, to envelop the infant and retain body heat. We compared rectal temperatures at NICU admission between wrapped infants and control infants who received standard thermal care in the delivery room. We attempted to increase delivery room temperature for all infants; however, this procedure was not followed and we analyzed delivery room temperature as a covariate. **Central findings:** We found infants who were wrapped in polyurethane were warmer and not hypothermic (< 36.5C) upon NICU admission compared to infants who received standard thermal care (1). Infants who were in warmer delivery rooms had higher admission temperatures regardless of wrap versus standard care. We also surveyed for thermal practices around the nation and published those data (2) in an effort to change practice. **Influence on science and health:** This RCT contributed to the AAP and AHA changing delivery room resuscitation to include plastic wrap on infants less than 29 weeks gestation after birth in National Resuscitation Standards in 2006. WHO also now recommends to increase delivery room temperatures to approximately 80°F in addition to wrapping infants in plastic wrap.

1. Knobel, R.B., Wimmer, J., & Holbert, D. (2005). Heat loss prevention for preterm infants in the delivery room. *Journal of Perinatology*, 25, 304-308.

2. Knobel, R.B., Vohra, S., & Lehmann, C. (2005). Heat loss prevention in the delivery room for preterm infants: A national survey of Newborn Intensive Care Units. *Journal of Perinatology*. 25, 514-518.

2. Thermoregulation in the first 12 hours of life in extremely low birth weight premature infants. This study (NIH/NINR: 1F31 NR09143-01), measured abdominal and skin temperature and clinical context

variables in 10 extremely low birth weight infants over their first 12 hours of life to examine hypothermia and determine if they could exhibit peripheral vasoconstriction in response to hypothermia (decreasing their foot temperature below their abdominal temperature by 2° C to send blood to their core for protection). **Central Findings:** Only one infant was able to exhibit a peripheral vasoconstriction response and more importantly, 7 of 10 infants kept the majority of their foot temperatures higher than their abdominal ones, or a central - peripheral temperature difference (CPTd) <0°C for much of the 12 hours of study (1). This finding indicates an abnormal thermal pattern and is representative of altered vasomotor tone, or dysregulation of the autonomic nervous system. Findings from this study also indicated that infants' heart rates were within normal parameters if their central body temperature was maintained between 36.8°-37° C (2). **Influence on science and health:** This abnormal temperature pattern was reported in the literature very little and had never been studied in depth. Some researchers have advocated for dual temperature measures for standard of care in premature infants and this evidence established that dual temperature monitoring is important and can be indicative of an abnormal temperature pattern, requiring further study. Very premature infants have increased morbidity and mortality due to immature systems and the need for intensive care. They also have inefficient heat production, making them susceptible to hypothermia. If they are unable to control vasomotor tone, it leaves their central core unprotected and susceptible to injury from low perfusion. Our findings of the minimum standard control temperature of 36.8° C for premature infants housed in incubators has become routine in many neonatal units and is frequently cited in standards of care.

3. Knobel, R.B., Holditch-Davis, D., Schwartz, T., & Wimmer, J. (2009). Extremely low birth weight preterm infants lack vasomotor response in relationship to cold body temperatures at birth. *Journal of Perinatology*, 29, 814-21. PMID: PMC2787712
4. Knobel, R.B., Holditch-Davis, D., & Schwartz, T.A. (2010). Optimal body temperature in transitional ELBW infants using heart rate and temperature as indicators. *Journal of Obstetrics, Gynecological & Neonatal Nursing*, 39, 3-14. PMID: PMC3864588
5. Knobel, R. & Holditch-Davis, D. (2007). Thermoregulation and heat loss prevention after birth and during neonatal-intensive-care unit stabilization of ELBW Infants. *Journal of Obstetrics, Gynecological & Neonatal Nursing*, 36, 280-7.

3. Central and peripheral temperature in the first two weeks of life in premature infants. As a next step, I led an interdisciplinary team to examine central and peripheral temperature in 30 preterm infants, less than 1200 grams at birth, for longitudinal patterns of abnormal thermal patterns (CPTd<0°C) in relationship to morbidity (1). This study was funded by NIH/NINR: 1R15NR012157-01 and Robert Wood Johnson Foundation #68041 to examine body temperature in very preterm infants over their first 2 weeks of life. This study examined each infant using a within case, observational design and mixed methods. Cases were analyzed to generate hypotheses for future study. **Central Findings:** All infants had periods of time with CPTd <0°C throughout their first two weeks of life. We also found that infants with a higher percentage of time with CPTd >2°C cold were associated with having an infection during the first month of life, being born to a mother who smoked during pregnancy, being African American and being male (3). All associations, except for infection, were not statistically significant because this was a small sample for an exploratory case study design; however, these associations need to be further studied. We also found that increased temperature in the foot was significantly correlated with increased perfusion in the foot (2). This finding indicates that when CPTd <0°C, there is also increased blood flow in the foot, leaving the central body vulnerable to low blood flow and ischemia. **Influence on science and health:** This study gave evidence that continuous monitoring of central and peripheral body temperature might be used as a biomarker to alert clinicians that the infant is in an abnormal circulatory state and that they should suspect possible infection.

6. Knobel, R. B., Levy, J., Katz, L., Guenther, B., & Holditch-Davis, D. (2013). A pilot study to examine maturation of body temperature control in preterm infants. *Journal of Obstetrics, Gynecological & Neonatal Nursing*, 42(5), 562-74. PMID: PMC3783951
7. Knobel-Dail, R. B., Tanaka, D. T., Holditch-Davis, D., White, J. T. (2016). Perfusion index in very low birth weight premature infants during their first two weeks of life. *Biological Research for Nursing*, E. pub ahead of print. PMID: 27352610. <http://dx.doi.org/10.1177/1099800416656914> PMID: PMC5942507
8. Knobel-Dail, R. B., Sloane, R., Holditch-Davis, D. & Tanaka, D. T. (2017). Negative temperature differential in preterm infants less than 29 weeks gestational age: Associations with infection and maternal

smoking. *Nursing Research*, 66 (6), 442-453. PMID: 29095375. DOI: 10.1097/NNR.0000000000000250
PMC5679714

4. Dual point body temperature measurement with servo-control in premature infants. This study examined the usability of measuring central and peripheral body temperature and the CPTd using a research software and the Giraffe Omnibed with a within case design and mixed methods. **Central findings:** Nurses liked having dual temperature display with the CPTd and found that this display made them pay more attention to the premature infants' thermal stability. Nurses wanted larger font and alarms associated with abnormal thermal gradients. Two case study results revealed an associate between abnormally large thermal gradients and the onset of infection. **Influence on science and health:** Monitoring abdominal and foot temperature and their difference was feasible under servo-control and augmented nurses' abilities to care for their infants' thermal stability.

9. Dail, R. B., Tavakoli, A. S., Moseley, S. L., Bergstedt, V. B., Kuehn, D., & Chang, W. (2021) Examining the use of dual point temperature display in servo-control with early preterm infants. *Advances in Neonatal Care*. 21 (6). PMID: 33783382 DOI: 10.1097/ANC.0000000000000827

5. Using thermal gradients to predict infection in premature infants. This study is examining infants' longitudinal CPTds in relationship to the onset of infection in 440 preterm infants at five NICUs in North and South Carolina. Each site has a site PI and a research nurse. Infant quantitative data are collected prospectively with temperature dataloggers and HeRO monitors for their first 28 days of life. Data are sent to University of South Carolina electronically and a RedCap database is used to collect clinical outcome data. Currently (May 2022) there are 250 infants who have completed data collection. (see Ongoing Research Support)

10. Dail, R. B., Everhart, K., Hardin, J., Chang, W., Kuehn, D., Iskersky, V. Fisher, K. & Murphy, H. J. (2020). Predicting Infection in Very Preterm Infants: A Study Protocol. *Nursing Research*. doi: 10.1097/NNR.0000000000000483. [Epub ahead of print] PubMed PMID: 33060417;

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Jana M. Iverson

eRA COMMONS USER NAME (credential, e.g., agency login): JIVERSON

POSITION TITLE: Dudley Allen Sargent Professor of Pediatric Rehabilitation and Associate Dean for Research

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE (if applicable)	Completion Date MM/YYYY	FIELD OF STUDY
Bryn Mawr College, Bryn Mawr, PA	AB	05/1991	Psychology
The University of Chicago, Chicago, IL	MA	06/1994	Psychology
The University of Chicago, Chicago, IL	PhD	12/1996	Psychology
Indiana University, Bloomington, IN	Postdoctoral	08/1999	Developmental Psychology

A. Personal Statement

The proposed research will examine autonomic regulation of attention in the neonatal period and the developmental consequences for the emergence of key exploratory and interactive behaviors and ASD outcomes. I will support the proposed work by providing expertise in behavioral video coding and analysis and in recording and analyzing infant object exploration and social interaction in naturalistic and structured settings. All of my work to date has been in developmental psychology with a particular emphasis on infancy. I have extensive experience as a researcher in motor and communicative and language development (100+ publications). A major focus of my work (supported by NIH since 2002) has been early motor and communicative development in infants with an older sibling with autism. In three projects (R01 HD41677, HD054979, HD073255), I have overseen the collection and management of three long-term longitudinal datasets (infants followed for 2-3 years) with intensive observation schedules and little to no participant attrition. All of this work has involved the development and application of detailed coding systems examining production of infant motor and communicative behavior. I first met Dr. Bradshaw in 2016 where she was a participant (and I was a mentor) at the American Speech-Language-Hear Association (ASHA) Lessons for Success program. Lessons for Success is designed to provide intensive training to promising early career scientists in the development and management of a successful research program. Since then, we have published one manuscript on developmental cascades and ASD (Bradshaw et. Al, 2022⁴). We are also collaborating on a study investigating the impact of motor development on language through the Baby Siblings Consortium Motor Workgroup. Our successful work together, along with my expertise, lays an important foundation for my contribution as a Co-Investigator for the proposed study.

1. **Iverson, J.M.** (2018). Early motor and communicative development in infants with an older sibling with autism spectrum disorder. *Journal of Speech, Language, and Hearing Research*, 61, 2673-2684. PMCID: PMC6693573

2. West, K.L., Northrup, J.B., Leezenbaum, N.B., & **Iverson, J.M.** (2019). The relation between walking and language in infant siblings of children with autism spectrum disorder. *Child Development*, 90, e356-e372. PMCID: PMC5913008

3. **Iverson, J.M.** (2021). Developmental variability and developmental cascades: Lessons from infant motor and language development. *Current Directions in Psychological Science*, 30, 228-235. PMID: PMC8240753
4. Bradshaw, J., Schwichtenberg, A. J., & **Iverson, J.M.** (2022). Capturing the complexity of autism: Applying a developmental cascades framework. *Child Development Perspectives*, 16, 18-26. PMID: in progress

B. Positions, Scientific Appointments, and Honors

Positions and Employment

1999 – 2003	Assistant Professor, Department of Psychological Sciences, University of Missouri-Columbia, Columbia, MO
2003 – 2006	Assistant Professor, Department of Psychology, University of Pittsburgh, Pittsburgh, PA
2005 – Present	Secondary appointment, Department of Linguistics, Pittsburgh, PA
2006 – 2012	Associate Professor, Department of Psychology, University of Pittsburgh, Pittsburgh, PA
2006 – 2022	Secondary appointment, Department of Communication Sciences and Disorders, Pittsburgh, PA
2012 – 2022	Professor, Department of Psychology, University of Pittsburgh, Pittsburgh, PA
2022 – Present	Dudley Allen Sargent Professor of Pediatric Rehabilitation, Department of Physical Therapy, Boston University, Boston, MA
2022 – Present	Associate Dean for Research, College of Health & Rehabilitation Sciences: Sargent College, Boston University, Boston, MA

Other Experience and Professional Memberships

1992 – Present	Member, Society for Research in Child Development
1998 – Present	Member, International Society for Infant Studies
2001	Visiting Professor, Institute of Psychology, National Research Council (CNR), Rome, Italy
2005 – Present	Editorial Board, <i>Language Learning and Development</i>
2006	Visiting Professor, University of Rome “La Sapienza”
2007 – Present	Member, International Society for Autism Research
2010 – Present	Visiting Professor, Department of Human Physiology, Università Campus Bio-Medico, Rome
2011 – Present	Editorial Board, <i>Journal of Child Language</i>
2012 – 2018	Editorial Board, <i>Infancy</i>
2012 – 2016	Permanent Member, NIH CPDD Study Section
2015 – 2019	Member, NIH ZRG1 F16-L20
2017	Faculty Mentor, American Speech and Hearing Association <i>Lessons for Success</i> Workshop,
2018 – Present	Mentor, American Speech and Hearing Association <i>Mentoring Academic Research Careers</i> Program
2018 – Present	Advisory Committee, American Speech and Hearing Association <i>Lessons for Success</i> Workshop, 2018
2019 – Present	Editorial Board, <i>Logopedia e Comunicazione</i>

Honors

1991	Rotary Foundation Graduate Scholarship
1993	National Science Foundation Graduate Fellowship
1995	John Dewey Prize Lectureship
1995	American Association of University Women Dissertation Fellowship
2003	AAAS Women’s International Science Collaboration Travel Award
2003	Italian National Research Council (CNR) International Collaborator Travel Award
2007	University of Pittsburgh Chancellor’s Distinguished Research Award
2018	University of Pittsburgh Chancellor’s Distinguished Teaching Award
2019	Fellow, Association for Psychological Science

C. Contributions to Science

1. Over the past 15 years, my students and I have followed three cohorts of infants at heightened risk (HR) for ASD and are currently following a fourth, using a prospective, longitudinal design with frequent observations. These are children who have an older brother or sister already diagnosed with ASD. For these later-born siblings, the ASD recurrence risk is 18.7%. In addition, as we now know, HR infants are also at risk for a variety of other delays, most especially in motor development and in language and

communication. The risk for clinically significant language delay at 36 months among those with no ASD diagnosis is about 20-25%. Our work has demonstrated, among other things, that substantial proportions of HR infants exhibit early motor and communication delays, that these early communication delays shape the input to which HR infants are exposed, and that early gross motor abilities are predictive of the emergence of communication and language milestones.

- a. Jarvis, E.N., West, K.L., & **Iverson, J.M.** (2020). Object exploration during the transition to sitting: A study of infants at heightened risk for autism spectrum disorder. *Infancy*, 25, 640-657. PMID: PMC7460267
- b. West, K.L., Roemer, E.J., Northrup, J.B., & Iverson, J.M. (2020). Profiles of action-gesture development in infants with an older sibling with ASD. *Journal of Speech, Language, and Hearing Research*, 63, 1195-1211. PMID: PMC7242984
- c. **Iverson, J.M.**, Shic, F., Wall, C.A., Chawarska, K., Curtin, S., Estes, A., Gardner, J.M., Hutman, T., Landa, R.J., Levin, A.R., Libertus, K., Messinger, D.S., Nelson, C.A., Ozonoff, A., Sacrey, L.-A. R., Sheperd, K., Stone, W.L., Tager-Flusberg, H.B., Wolff, J.A., Yirmiya, N., & Young, G.S. (2019). Early motor abilities in infants at heightened vs. low risk for autism spectrum disorder: A Baby Siblings Research Consortium (BSRC) study. *Journal of Abnormal Psychology*, 128, 69-80. PMID: PMC6338079
- d. Leezenbaum, N.B. & **Iverson, J.M.** (2019). Trajectories of posture development in infants with vs. without familial risk for ASD. *Journal of Autism and Developmental Disorders*, 49, 3257-3277. PMID: in progress

2. I began my program of research with a series of studies focused on a set of questions concerning gesture and the gesture-speech link in children's early communication. These questions have a rich history in developmental research, but ways in which gesture use changes and potentially supports the transition to complex speech had not been fully described at the time I began working in this area. In my earliest publications, we found that gesture allows children to communicate meanings for which they do not yet have words. This raised the possibility that gesture may facilitate certain aspects of language acquisition by providing an expressive medium through which children improve communicative skills prior to manifesting these skills in speech. In a subsequent longitudinal study, we found that aspects of children's vocabularies could be predicted from their prior use of gesture, and that the age at which children began to talk in two-word phrases was highly related ($r = .92$) to the age at which they had begun to use gesture-word combinations conveying two different but related pieces of information. In follow up work designed to examine the nature of gesture in maternal input to very young children, we found that maternal gesture production was sensitive to the developmental level of the listener; and the nature of the gestural input that children receive relates both to their use of gesture and to the course of their early lexical development. I served as the primary investigator or co-investigator on all of these studies.

- a. **Iverson, J.M.** & Goldin-Meadow, S. (2005). Gesture paves the way for language development. *Psychological Science*, 16, 367-371.
- b. **Iverson, J.M.**, Capirci, O., Longobardi, E., & Caselli, M.C. (1999). Gesturing in mother-child interactions. *Cognitive Development*, 14, 57-75.
- c. Capirci, O., **Iverson, J.M.**, Pizzuto, E., & Volterra, V. (1996). Communicative gestures during the transition to two-word speech. *Journal of Child Language*, 23, 645-673.
- d. **Iverson, J.M.**, Capirci, O., & Caselli, M.C. (1994). From communication to language in two modalities. *Cognitive Development*, 9, 23-43.

3. Several studies have illustrated that gesture and speech form a tightly integrated communicative system. However, the question of whether gesture-speech links result from observing the gestures of others, or whether their integration exists even in the absence of visual experience remained unaddressed. Thus, I studied the gestures and speech of children who were congenitally blind. Surprisingly, I discovered not only that blind children gesture in accompaniment to speech, they do so even when they are speaking to another person they know to be blind. Neither visual-gestural input nor a sighted interlocutor, in other words, is a necessary precondition for gestural expression.

- a. **Iverson, J.M.** & Goldin-Meadow, S. (2001). The resilience of gesture in talk: Gesture in blind speakers and listeners. *Developmental Science*, 4, 416-422.

- b. **Iverson, J.M.** (1999). How to get to the cafeteria: Gesture and speech in blind and sighted children's spatial descriptions. *Developmental Psychology*, 35, 1132-1142.
- c. **Iverson, J.M.** & Goldin-Meadow, S. (1998). Why people gesture when they speak. *Nature*, 396, 228.
- d. **Iverson, J.M.** & Goldin-Meadow, S. (1997). What's communication got to do with it? Gesture in congenitally blind children. *Developmental Psychology*, 33, 453-467.

4. The fact that the gesture-speech link emerges in blind children even in the absence of visual experience suggests that this link reflects more general connections between language and motor activity. This provided the impetus for theoretical and empirical work designed to address the developmental origins of the gesture-speech link in coordinations between vocalization and limb movement in early infancy. A key assumption of the theoretical model is that development of the gesture-speech link is an instance of the more general problem of motor coordination. The model generates specific predictions about the conditions under which vocal and motor behaviors are likely to be coordinated and how developmental change in one system may impact its ability to influence activity in the complementary system. Three of these predictions were tested and confirmed in a cross-sectional study of temporally coordinated bouts of vocal and rhythmic limb activity in 6- to 9-month-old infants. I subsequently extended these cross-sectional findings in a longitudinal study of 30 infants whose vocalizations, rhythmic limb movements, gestures, and words were observed biweekly from 2 to 19 months.
- a. **Iverson, J.M.**, Hall, A.J., & Nickel, L., & Wozniak, R.H. (2007). The relationship between onset of reduplicated babble and laterality biases in infant rhythmic arm movements. *Brain and Language*, 101, 198-207. PMCID: PMC2034511
 - b. **Iverson, J.M.** & Fagan, M.K. (2004). Infant vocal-motor coordination: Precursor to the gesture-speech system? *Child Development*, 75, 1053-1066.
 - c. **Iverson, J.M.** & Thelen, E. (1999). Hand, mouth, and brain: The dynamic emergence of speech and gesture. *Journal of Consciousness Studies*, 6, 19-40.

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: John E. Richards

eRA COMMONS USER NAME (credential, e.g., agency login): jrichards

POSITION TITLE: Emeritus Carolina Distinguished Professor, Research Professor

EDUCATION/TRAINING (*Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.*)

INSTITUTION AND LOCATION	DEGREE (if applicable)	Completion Date MM/YYYY	FIELD OF STUDY
Biola College, La Mirada, CA	B.A.	May,1976	Psychology, Humanities
California State University, Fullerton, CA	M.A.	Dec 1978	Psychology
University of California, Los Angeles CA	Ph.D.	May. 1983	Psychology

A. Personal Statement

I have expertise in infant attention, including laboratory observation measures, heart rate measures, behavioral coding, and brain measurements using EEG and MRI. I work primarily with infants in the first year (e.g., 3 to 12 months) and study the relation between attention development and brain development. I have a relatively well known theoretical model about the development of sustained attention in infants. I have extensive experiments with ERP source analysis at several ages (infants, children, adults). I have expertise in the structural neurodevelopment of infants and children.

I can contribute to the current proposal in several ways. First, I can assist in the collection and analysis of heart rate data. My research collects ECG on every participant, and has tools to identify R-waves for computing heart rate. Second, I have a well-known theoretical model that links heart rate to attention phases. We have a set of algorithms for identifying periods of sustained attention, and inattention (attention termination, pre-stimulus orienting, post-sustained-attention inattention). We have a relatively user-friendly MATLAB algorithm that does that. Third, I have several publications that link heart rate and sustained attention in ongoing viewing situations with infants and young children.

Ongoing and recently completed projects that I would like to highlight include:

R01 HD19894-31-35 Richards (PI) 8/01/17 – 5/31/2023, Development of sustained attention in infants

Citations

Xie, W., Mallin, B.A., & Richards, J.E. (2017). Development of infant sustained attention and its relation to EEG oscillations: An EEG and cortical source analysis study. *Developmental Science*, 21; e12562.

<https://doi.org/10.1111/desc.12562>

Conte, S., Richards, J.E., Guy, M.W., Xie, W., & Roberts, J.E. (2020). Face-sensitive brain responses in the first year of life. *Neuroimage*

Guy, M.W., Richards, J.E., Tonnsen, B., & Roberts, J.E. (2017). Neural correlates of face processing in etiologically-distinct 12-month-old infants at high-risk of autism spectrum disorder. *Developmental Cognitive Neuroscience*, 29:61-71. doi: 10.1016/j.dcn.2017.03.002. Epub 2017 Mar 16.

Conte, S., & Richards, J.E. (2021). The influence of the head model conductor on the source localization of auditory evoked potentials. *Brain Topography*, DOI: 10.1007/s10548-021-00871-z

B. Positions, Scientific Appointments, and Honors

Carolina Distinguished Professor Emeritus and Research Professor, January 2021 to present
Carolina Distinguished Professor, August 2003 to December 2020
Professor, August, 1992 to July, 2003
Associate Professor, August, 1987, to July, 1992.
Assistant Professor, August, 1982, to July, 1987.

Honors

Fellow, Society for Psychophysiological Research, May 2022 to present
Carolina Distinguished Professor Emeritus, University of South Carolina, January 2021 - present
Fellow, American Association for Advancement in Science, 2010-...
Fellow, American Psychological Society, January 2008-...
MERIT Award, National Institute of Child Health and Human Development, National Institutes of Health, March, 2007 to February, 2018
Carolina Distinguished Professor, University of South Carolina, August, 2003-2020
Fellow, Division 3, American Psychological Association. January, 2003-...
Fellow, Division 7, American Psychological Association. January, 2000 - ...
Research Scientist Development Award, National Institute of Mental Health, National Institutes of Health, May 1992 to April, 1997.
Russell Award for Research in Humanities and Social Sciences, University of South Carolina. 1991.
New Investigator Research Award, National Institute of Child Health and Human Development, National Institutes of Health. August, 1985 to July, 1988.

C. Contributions to Science

1—I have a long-standing and continuing expertise in the development of sustained attention in young infants. Using psychophysiological measures and innovative behavioral manipulations, I have shown that fixation in young infants towards interesting visual stimuli can be distinguished into stimulus orienting, sustained attention, and attention termination. These attention phases show different developmental pathways; i.e., early onset of stimulus orienting, and extended development of sustained attention throughout the first year of postnatal life. This distinction has been applied across a wide range of stimuli and testing situations, such as to simple stimuli, interesting video stimuli, and television and movie programs.

Richards, J.E. (2010). The development of attention to simple and complex visual stimuli in infants: Behavioral and psychophysiological measures. *Developmental Review*, 30, 203-219.

Mallin, B.A., & Richards, J.E. (2012). Peripheral stimulus localization by infants of moving stimuli on complex backgrounds. *Infancy*, 17, 692-714.

Xie, W., & Richards, J. E. (2016). Effects of interstimulus intervals on behavioral, heart rate, and event-related potential indices of infant engagement and sustained attention. *Psychophysiology*, doi:10.1111/psyp.12670

Tonnsen, B., Richards, J.E., & Roberts, J.E. (2018). Heart Rate-Defined Sustained Attention in Infants at Risk for Autism. *Journal of Neurodevelopmental Disorders*, 10:7, doi: 10.1186/s11689-018-9224-2

2—I have integrated my study of attention with studies of responses to stimuli using EEG and ERP. The EEG is a direct measure of brain activity. I have shown that a commonly studied ERP component in infant participants, the Nc, is primarily related to attention, rather than short-term recognition memory or other long-term recognition processes. To study the brain sources of the ERP during attention, I developed new methods for using cortical source analysis of ERP with infant participants. This includes the use of structural MRI with individual infants who then participate in

psychophysiological experiments. From the structural MRIs a realistic model of the head is made, and then realistic finite-element-method meshes can be used to do source analysis. One strategy of mine is to do experiments with adult participants to sharpen the cortical source tools, and then to apply these to infant participants

Buzell, G.A., Richards, J.E., White, L.K., Pine, D.S., & Fox, N.A. (2017). Development of the error-monitoring system from ages 9-35: unique insight provided by MRI-constrained source localization of EEG. *Neuroimage*. Doi: 10.1016/j.neuroimage.2017.05.045

Xie, W., Mallin, B.A., & Richards, J.E. (2018). Development of brain functional connectivity and its relation to infant sustained attention in the first year of life. *Developmental Science*, e12703. <https://doi.org/10.1111/desc.12703>

Lunghi, M., Piccardi, E.S., Richards, J.E., & Simion, F. (2019). The neural correlates of orienting to walking direction in 6-Month-old Infants: an ERP study. *Developmental Science* 22(6). ARTN e12811 10.1111/desc.12811

Sullivan, E.F., Xie, W., Conte, S., Richards, J.E., Shama, T., Haque, R., Petri, W.A., & Nelson, C.A. (2022). Neural correlates of inhibitory control and associations with cognitive outcomes in Bangladeshi children exposed to early adversities. *Developmental Science*

3—A recent area of my research is the relation between brain development and infants processing of faces. We have identified the neural source of several components of the face-sensitive ERP responses in infants from 3 to 12 months of age. We are currently parsing the aspects of face processing that show development in the first year with the goal of determining the onset of adult-like face-configuration responding. This work has been extended to neurodevelopmental disorders groups, such as Fragile X Syndrome infants and infants with a sibling with autism spectrum disorder diagnosis.

Conte, S., Richards, J.E., Guy, M.W., Xie, W., & Roberts, J.E. (2020). Face-sensitive brain responses in the first year of life. *Neuroimage*

Guy, M. W., Richards, J. E., Hogan, A. L., & Roberts, J. E. (2021). Neural Correlates of Infant Face Processing and Later Emerging Autism Symptoms in Fragile X Syndrome [Original Research]. *Frontiers in Psychiatry*, 12(2019). <https://doi.org/10.3389/fpsyt.2021.716642>

Fu, X., & Richards, J.E. (2022). Evaluating Head Models for Cortical Source Localization of the Face-Sensitive N290 Component in Infants. *Brain Topography*, doi: 10.1007/s10548-022-00899-9

Guy, M.W., Richards, J.E., & Roberts, J.E. (2022). Head model selection and cortical source analysis of the N290 ERP component in infants at high risk for autism. *Brain Sciences*

4--In the process of developing realistic head models for infants, I realized that many neuroimaging methods use a standard MRI template to do media segmentation (WM, GM, CSF) and stereotaxic atlas identification (e.g., macroanatomical identification with Talairach maps). However, the brain composition of infants is significantly different than that of adults, so I began a set of studies to develop neuroanatomical MRI templates for infants. As a result of this impetus, I now have a “Neurodevelopmental MRI Database” that provides average MRI templates, head media priors, and stereotaxic atlases in small age increments across the lifespan (e.g., 1.5 increments in first year; 3 months in second year, yearly through age 5; every six months from age 5 to 20-24; 5 year increments to age 89 years This database is publically available and is has been accessed by over 150 labs both in the US and internationally.

Sanchez, C.E., Richards, J.E., & Almli, C.R. (2011). Neurodevelopmental MRI brain templates for children from 2 weeks to 4 years of age. *Developmental Psychobiology*. Doi:10. 1-02/dev20579. PMC3184192

Richards, J.E., & Xie, W. (2015). Brains for all the ages: Structural neurodevelopment in infants and children from a life-span perspective. In J. Benson (ed), *Advances in Child Development and Behavior* (Vol 7).

Fillmore, P.T., Richards, J.E., Phillips-Meek, M.C., Cryer, A., & Stevens, M. (2015) Stereotaxic MRI brain atlases for infants from 3 to 12 months. *Developmental Neuroscience*. DOI:10.1159/000438749.

Richards, J.E., Sanchez, C., Phillips-Meek, M., & Xie, W. (2015) A database of age-appropriate average MRI templates. *Neuroimage*, doi:10.1016/j.neuroimage.2015.04.055

5—I have used the “Neurodevelopmental MRI Database” to develop tools to be used for neuroimaging with infants and children. I have structural MRIs from over 12000 participants, including nearly 200 infants from age of 3 to 12 months. I have used these MRIs and the segmenting from the database to develop new tools that can be used with infant neuroimaging. For example, I developed techniques that determine scalp positions on the infants head and the underlying cortical areas at those scalp locations. This may be used to identify the neural sources that are likely generating the optical measurements used in “NIRS” (near-infrared spectroscopy). Additionally, I have created average electrode positions for adults, adolescents, children, and infants. I have used these with collaborators to study infant attention and cognition with neuroimaging tools.

Bulgarelli, C., de Klerk, C.C.J.M, Richards, J.E., Southgate, V., Hamilton, A., & Blasi, A. (2020). The developmental trajectory of fronto-temporoparietal connectivity as a proxy of the default mode network: a longitudinal fNIRS investigation. *Human Brain Mapping*.

Fu, X., & Richards, J.E. (2021). Age-related changes in Diffuse Optical Tomography sensitivity profiles in infancy. *PLOS One*

Fu, X. & Richards, J.E. (2021). Investigating developmental changes in scalp-to-cortex correspondence using diffuse optical tomography sensitivity in infancy. *Neurophotonics*. **8**(3), 035003, doi: 10.1117/1.NPh.8.3.035003.

Fu, X., & Richards, J.E. (2022) Age-related changes in diffuse optical tomography sensitivity profiles from childhood to adulthood, *Journal of Biomedical Optics*, 27(8), 083004 (2022), doi: 10.1117/1.JBO.27.8.083004.

Complete List of Published Works in My NCBI Collections

Full bibliography: <http://www.ncbi.nlm.nih.gov/myncbi/browse/collection/40569197/>

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Jane E. Roberts, Ph.D.

eRA COMMONS USER NAME (credential, e.g., agency login): JANEROBERTS

POSITION TITLE: Carolina Distinguished Professor, Department of Psychology and Associate Dean for Natural Sciences, College of Arts and Sciences, University of South Carolina

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE (if applicable)	Completion Date MM/YYYY	FIELD OF STUDY
Asbury College	B.A.	05/1986	Psychology
University of Connecticut, Storrs	Ed.S.	12/1991	School Psychology
University of North Carolina, Chapel Hill	Ph.D.	05/1998	Early Childhood Special Ed.
University of North Carolina, Chapel Hill	Post-Doc	12/1999	Developmental Psychophysiology

A. Personal Statement

My training and expertise in the phenotypic characterization of infants and young children with autism spectrum disorder (ASD), fragile X syndrome (FXS), the *FMR1* premutation (FXpm), and Down syndrome (DS) align well with my role as Co-Investigator on this proposal. My program of research utilizes a biobehavioral framework employing a longitudinal design to address questions associated with the emergence, stability, and predictors of maladaptive outcomes for infants and young children with neurodevelopmental disorders, with a specific focus on the emergence of autism in high-risk infants using a multi-modal biomarker approach.

With over 100 publications and multiple completed and current R01's (>\$20m as PI) focused on the phenotype of ASD, FXS, FXpm, DS, and infant siblings of children with ASD, I have expertise across multiple content areas including anxiety, physiological arousal, intellectual disabilities, cognitive neuroscience, and temperament. In addition, I have launched multiple cross-disciplinary research initiatives, including co-founding a statewide South Carolina Autism and Neurodevelopmental Disorders (SCAND) initiative that has had four successful research meetings (100+ in attendance) and serving as co-director of a university-focused center of excellence, the Carolina Autism and Neurodevelopment (CAN) Research Center, that promotes and supports research on autism and other neurodevelopmental disorders. I have strong, existing, collaborative relationship with Dr. Bradshaw (PI), having worked with her over the last four years across multiple grants, including (1) serving as the primary mentor on her K23; (2) having Dr. Bradshaw as a Co-I on my R01 supplement; and (3) serving a Co-I with Dr. Bradshaw on a CAN Pilot Award (PI: O'Reilly). Dr. Bradshaw and I have also published together and collaborated extensively on CAN initiatives bolstering outreach endeavors within the autism community. This R01 proposal is a direct extension of her K23 and her idea to include infants born premature as a unique control group is quite innovative. My previous and ongoing R01's focus on the emergence of autism and anxiety symptomatology, as well as autonomic functioning, use three-group research designs, as is proposed in this application. As such, I have the scientific knowledge and professional experience to contribute to this project, which is a natural extension of my program of research comparatively examining the mechanistic underpinnings of behavior in infants and children with neurodevelopmental disorders, and I look forward to contributing my expertise to this proposal.

Ongoing projects that I would like to highlight include:

R01 HD106652

Roberts (PI)

5/1/2022 – 4/30/2027

Autonomic and Sensory Dysfunctions in FMR1 Conditions: Development, Mechanisms and Consequences

R01 MH90194

Roberts (PI)

4/1/2016 - 2/28/2023

Emergence and Stability of Autism in Fragile X Syndrome

R01 MH107573

Roberts (PI)

9/1/2016 - 8/31/2023

Emergence, Stability and Predictors of Anxiety in Fragile X Syndrome

R01 HD024356

Abbeduto (PI), Role: Co-Investigator

7/15/2018 –6/30/2023

Language Development in Fragile X Syndrome

Un-numbered award

Roberts & Twiss (mPIs)

6/2019 - 6/2024

UofSC Autism and Neurodevelopmental Disorders Center of Excellence

B. Positions, Scientific Appointments, and Honors

Positions and Scientific Appointments

2022-present	Associate Dean for Natural Sciences, College of Arts and Sciences, University of South Carolina
2020-present	Carolina Distinguished Professor, University of South Carolina
2018-2022	Chair, Department of Psychology, University of South Carolina
2016-present	Co-director of South Carolina Autism and Neurodevelopmental Disorders Consortium
2014-present	Adjunct Professor, Biomedical & Psychological Sciences, Monash University
2013-present	Professor, Department of Psychology, University of South Carolina
2013-2016	Program Director, School Psychology, Department of Psychology, University of South Carolina
2012-present	Faculty Affiliate, Institute for Mind and Brain, University of South Carolina
2008-2013	Associate Professor, Department of Psychology, University of South Carolina
2006-2008	Scientist, Frank Porter Graham Child Development Institute, Chapel Hill, NC
1999-2006	Investigator, Frank Porter Graham Child Development Institute, Chapel Hill, NC
1998-1999	Post-Doctoral Fellow, Frank Porter Graham Child Development Center, Chapel Hill, NC
1989-1995	School Psychologist, Palm Beach County School System, West Palm Beach, FL

Honors

2020	Carolina Distinguished Professor, University of South Carolina
2011	USC Rising Star, Faculty Research Award

Other Experience and Professional Memberships/Grant Reviews/Editorial Roles

2015-2020	Baby Siblings Research Consortium, Invited Member
2015-2019	National Institute of Health, Child Psychopathology Dev. Disabilities Study Section
2014	Medical Research Council, United Kingdom, Grant Reviewer
2013	Brain Canada: Axrieli Neurodevelopmental Research Program, Grant Reviewer
2012-2013	Associate Editor, American Journal of Intellectual and Developmental Disabilities
2011	USC Rising Star, Faculty Research Award

C. Contributions to Science

My program of research focuses on understanding the profiles and predictors of developmental outcomes in young children with neurodevelopmental disorders with an emphasis on fragile X syndrome (FXS). Fragile X syndrome is a single gene trinucleotide (CGG) repeat disorder located on the *FMR1* gene (Xq27.3). In most individuals with FXS, the gene is hypermethylated resulting in reduced *FMR1* protein (FMRP) causing abnormal brain development and function. Traditional categorization includes the full mutation (>200 CGG repeats), occurring in approximately 1:2500-3600 persons, and premutation (50-200 CGG repeats), occurring in 1:120 females. FXS is the leading known single-gene cause of autism and the number one inherited cause of intellectual disabilities. My program of research focuses on four primary areas:

1. Early developmental trajectory of infants and young children with FXS, *FMR1* premutation, ASD, and Down syndrome
2. Co-morbidity of autism and anxiety and their interface in young children with *FMR1* conditions
3. Psychiatric profile of adult women with the *FMR1* premutation and its association with their child's early development
4. Role of biomarkers to advance our understanding of the underlying mechanisms associated with deleterious outcomes across the spectrum of impairment in FX-associated conditions

(1) Early Developmental Trajectories in *FMR1* conditions, ASD, and Down syndrome.

My initial research involved contributions to a series of studies on the behavioral phenotype of fragile X syndrome during early childhood. Findings from our series of studies on refinement of the infant phenotype of fragile X syndrome represent some of the earliest and most extensive work published on the early development of children with fragile X syndrome. This work has contributed to the theoretical and empirical basis of early development in fragile X to reflect the emergence of aberrant development earlier than initially purported and the specificity of the developmental profiles from other high-risk populations which has clear implications for the initiation and direction of treatment.

1. **Roberts, J. E.**, Crawford, H., Hogan, A. L., Fairchild, A., Tonnsen, B., Brewe, A., O'Conner, S., Roberts, D. A., & Abbeduto, L. (2019). Social Avoidance Emerges in Infancy and Persists into Adulthood in Fragile X Syndrome. *Journal of Autism and Developmental Disorders*, 49(9), 3753–3766. PMID: PMC6698894
2. **Roberts, J. E.**, Crawford, H., Will, E. A., Hogan, A. L., McQuillin, S., Tonnsen, B. L., O'Conner, S., Roberts, D. A., & Brewe, A. M. (2019). Infant social avoidance predicts autism but not anxiety in fragile X syndrome. *Frontiers in Psychiatry*, 10, 199. PMID: PMC6514151
3. Will, E. A., Caravella, K. E., Hahn, L. J., Fidler, D. J., & **Roberts, J. E.** (2018). Adaptive behavior in infants and toddlers with Down syndrome and fragile X syndrome. *American Journal of Medical Genetics Part B Neuropsychiatric Genetics*, 177(3), 358–368. PMID: PMC7294771
4. **Roberts, J. E.**, McCary, L. M., Shinkareva, S. V., & Bailey, D. B. (2016). Infant development in fragile X syndrome: cross-syndrome comparisons. *Journal of Autism and Developmental Disabilities*, 46(6), 2088–2099. PMID: PMC4860352

(2) Co-Morbidity of Autism Spectrum Disorder and Anxiety in *FMR1* conditions.

Intellectual impairment and social deficits are hallmark features of FXS with a high association of co-morbid conditions in FXS. Anxiety (~80%) and autism (~60%) represent two of the most prevalent and debilitating co-morbid conditions. In our own work, we identified important within-group distinctions with social avoidance, anxiety, and ASD sharing overlapping features, yet, with potentially distinct boundaries with regard to initial versus sustained social responsivity and putative biological mechanisms that differentiated boys with FXS and co-morbid ASD from those with FXS only (Roberts et al., 2007). Overall, this work has contributed to theoretical models of the presence of autism and anxiety in FXS by recognizing the importance of developmental surveillance and the value of integrating multiple behavioral and physiological factors.

1. Black, C. J., Hogan, A. L., Smith, K. D., & **Roberts, J. E.** (2021). Early behavioral and physiological markers of social anxiety in infants with fragile X syndrome. *Journal of Neurodevelopmental Disorders*, 13(1), 1–9. PMID: PMC7980359
2. Smith, K., Hogan, A. L., Will, E., & **Roberts, J. E.** (2021). Attention Bias and Prodromal Anxiety Symptoms in Toddlers With Fragile X Syndrome and Down Syndrome. *American Journal on Intellectual and Developmental Disabilities*, 126(2), 167–181. PMID: PMC8108623

3. **Roberts, J. E.**, Bradshaw, J., Will, E., Hogan, A. L., McQuillin, S., & Hills, K. (2020). Emergence and rate of autism in fragile x syndrome across the first years of life. *Development and Psychopathology*, 32(4), 1335–1352. PMCID: PMC8080303
4. Wall, C. A., Hogan, A. L., Will, E. A., McQuillin, S., Kelleher, B. L., & **Roberts, J. E.** (2019). Early negative affect in males and females with fragile X syndrome: Implications for anxiety and autism. *Journal of Developmental Disorders*, 11(1), 1–11. PMCID: PMC6744625

(3) Psychiatric Profile in Women with the *FMR1* premutation.

Given the high prevalence of the *FMR1* premutation in women (1:150), research has accelerated to characterize the phenotypic profile of these individuals. Our work and others' indicate that persons with the premutation display a range of clinical features that have challenged previously held assumptions that these individuals were largely asymptomatic.

1. Klusek, J., Moser, C., Schmidt, J., Abbeduto, L., & **Roberts, J. E.** (2019). A novel eye-tracking paradigm for indexing social avoidance-related behavior in fragile X syndrome. *American Journal of Medical Genetics, Part B: Neuropsychiatric Genetics*, 183(1), 5–16. PMCID: PMC6898737
2. Klusek, J., LaFauci, G., Adayev, T., Brown, W. T., Tassone, F., & **Roberts, J. E.** (2017). Reduced vagal tone in women with the *FMR1* premutation is associated with *FMR1* mRNA but not depression or anxiety. *Journal of Neurodevelopmental Disorders*, 9(1), 1–16. PMCID: PMC5414146
3. **Roberts, J. E.**, Tonnsen, B. L., McCary, L. M., Ford, A. L., Golden, R. N., & Bailey, D. B. (2016). Trajectory and predictors of depression and anxiety disorders in mothers with the *FMR1* premutation. *Biological Psychiatry*, 79(10), 850–857. PMCID: PMC4733592
4. Klusek, J., McGrath, S. E., Abbeduto, L., & **Roberts, J. E.** (2016). Pragmatic language features of mothers with the *FMR1* premutation are associated with the language outcomes of adolescents and young adults with fragile X syndrome. *Journal of Speech Language Hearing Research*, 59(1), 49–61. PMCID: PMC4867932

(4) The Mechanistic Role of Biomarkers in Fragile X.

My lab has integrated physiological markers to investigate mechanistic factors associated with atypical behaviors or risk status in young children with FXS and ASD. These biomarkers include genetic factors (e.g., FMRP, methylation status, mRNA), heart activity, salivary cortisol, and event-related-potentials. This work has contributed to translational work examining treatment in FXS with initial results indicating that stimulant medication positively impacts attention, academic performance, and regulates heart activity in young males with fragile X syndrome.

1. Guy, M. W., Richards, J. E., & **Roberts, J. E.** (2022). Cortical Source Analysis of the Face Sensitive N290 ERP Component in Infants at High Risk for Autism. *Brain Sciences*, 12(9), 1129. PMCID: PMC5042801
2. Hogan, A. L., Hunt, E., Smith, K., Black, C., Bangert, K., Klusek, J., & **Roberts, J.** (2021). Trajectories of Heart Activity Across Infancy to Early Childhood Differentially Predict Autism and Anxiety Symptoms in Fragile X Syndrome. *Frontiers in Psychiatry*, 12, 727559. PMCID: PMC8526850
3. Conte, S., Richards, J. E., Guy, M. W., Xie, W., & **Roberts, J. E.** (2020). Face-sensitive brain responses in the first year of life. *NeuroImage*, 211(1), 116602. PMCID: PMC7085434
4. Guy, M. W., Richards, J. E., Hogan A. L., & **Roberts, J. E.** (2021). Neural Correlates of Infant Face Processing and Later Emerging Autism Symptoms in Fragile X Syndrome. *Frontiers in Psychiatry*, 12. PMCID: PMC8651978

Complete List of Published Work in MyBibliography:

<http://www.ncbi.nlm.nih.gov/sites/myncbi/jane.roberts.1/bibliography/43271569/public/?sort=date&direction=ascending>

PHS 398 Cover Page Supplement

OMB Number: 0925-0001

Expiration Date: 09/30/2024

1. Vertebrate Animals Section

Are vertebrate animals euthanized? ☐ Yes ☐ No

If "Yes" to euthanasia

Is the method consistent with American Veterinary Medical Association (AVMA) guidelines?

☐ Yes ☐ No

If "No" to AVMA guidelines, describe method and provide scientific justification

.....

2. *Program Income Section

*Is program income anticipated during the periods for which the grant support is requested?

☐ Yes ☒ No

If you checked "yes" above (indicating that program income is anticipated), then use the format below to reflect the amount and source(s). Otherwise, leave this section blank.

*Budget Period	*Anticipated Amount (\$)	*Source(s)
----------------	--------------------------	------------

PHS 398 Cover Page Supplement

3. Human Embryonic Stem Cells Section

*Does the proposed project involve human embryonic stem cells? ☐ Yes ☒ No

If the proposed project involves human embryonic stem cells, list below the registration number of the specific cell line(s) from the following list: http://grants.nih.gov/stem_cells/registry/current.htm. Or, if a specific stem cell line cannot be referenced at this time, check the box indicating that one from the registry will be used:

☐ Specific stem cell line cannot be referenced at this time. One from the registry will be used.

Cell Line(s) (Example: 0004):

4. Human Fetal Tissue Section

*Does the proposed project involve human fetal tissue obtained from elective abortions? ☐ Yes ☒ No

If "yes" then provide the HFT Compliance Assurance

If "yes" then provide the HFT Sample IRB Consent Form

5. Inventions and Patents Section (Renewal applications)

*Inventions and Patents: ☐ Yes ☒ No

If the answer is "Yes" then please answer the following:

*Previously Reported: ☐ Yes ☐ No

6. Change of Investigator/Change of Institution Section

☐ Change of Project Director/Principal Investigator

Name of former Project Director/Principal Investigator

Prefix:

*First Name:

Middle Name:

*Last Name:

Suffix:

☐ Change of Grantee Institution

*Name of former institution:

PHS 398 Research Plan

OMB Number: 0925-0001

Expiration Date: 09/30/2024

Introduction	
1. Introduction to Application (for Resubmission and Revision applications)	Introduction_R01Resubmission_Bradshaw.pdf
Research Plan Section	
2. Specific Aims	SpecificAims_BradshawR01_A1.pdf
3. Research Strategy*	ResearchStrategy_BradshawR01_A1.pdf
4. Progress Report Publication List	
Other Research Plan Section	
5. Vertebrate Animals	
6. Select Agent Research	
7. Multiple PD/PI Leadership Plan	
8. Consortium/Contractual Arrangements	Bradshaw_Consortium_R01_A1.pdf
9. Letters of Support	LOS_BradshawR01.pdf
10. Resource Sharing Plan(s)	ResouceSharingPlan_BradshawR01.pdf
11. Authentication of Key Biological and/or Chemical Resources	
Appendix	
12. Appendix	

INTRODUCTION

We are grateful to the Reviewers for a thorough evaluation and are encouraged by their enthusiasm for the project, described as “highly significant and innovative” with a “strong analytic plan” and “compelling preliminary data,” led by a “highly productive” PI with “established expertise.” We have endeavored to address Reviewer critiques in their entirety, throughout all aspects of the planned study, as described below.

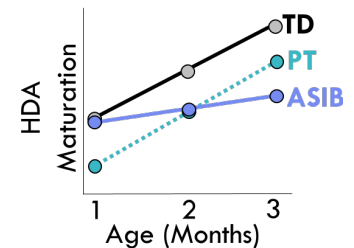
1. **Baseline Measure of Autonomic Function.** We agree that it is critical to measure baseline ANS function to fully evaluate the specificity of attention-related ANS markers. We now include a 5-min baseline recording of heart activity at each visit to extract RSA as a measure of baseline ANS function.
2. **Discussion of Sympathetic and Parasympathetic Nervous System.** A full discussion of ANS component systems and their relation to our research design and heart-defined attention (i.e., parasympathetic influence drives HR responses during attention), along with the importance of baseline ANS integrity in our research design is now included in *Significance*.
3. **Clarify Scientific Basis, Gap to be filled, and Model of ANS to ASD.** Current evidence that forms the scientific basis of this study and specific gaps to be filled is now included in *Significance*. We have also expanded the explanation of our model of ASD wherein initial global ANS function supporting neonatal arousal and reactivity is intact at birth, but subsequent ANS dysregulation of attention emerges over time.
4. **High Rate of ASD in PTs and Proposed Model of ANS Dysfunction and ASD.** Our model describes attention-specific ANS dysfunction as an early indicator of ASD. The rate of ASD in PTs is higher than that of the general population, and we hypothesize that the subset of PTs who develop ASD will have initial global ANS dysfunction (like most PTs) *and* subsequent, persistent ANS-dysregulated attention. This is in contrast to PTs *without* ASD wherein the initial ANS dysfunction resolves over time and does *not* have longstanding effects on attention. Mapping trajectories of heart-defined attention and baseline ANS function, and including PT infants with ASD in all analyses, will help delineate this pathway and clarify the association between broad vs. specific ANS dysfunction and ASD.
5. **Home vs. Lab Visits.** Our current data (N>450 infants, 1wk- 3yrs) evidence that 99% of families prefer in-home visits and only 4/450+ families request “in-lab” in lieu of “in-home”. One ongoing study of 4-8-month-olds requires in-home visits and no parents have declined enrollment due to this requirement. To make all visits comparable (i.e., data collection in a familiar, home environment), we will conduct *all* visits in home.
6. **Masked Examiners.** We anticipate that many siblings will be at school or daycare during the home visits, limiting examiner contact with siblings and reducing the likelihood of unmasking. However, we acknowledge the moderate-to-high likelihood of unintentional unmasking during some study visits. As such, examiners will record when they were unmasked for analytic consideration. In addition, masked RAs will process data (ECG editing, video coding) and masked psychologists can score clinical measures via video.
7. **PT vs. ASIB Preliminary Data** supporting our hypotheses, is now presented (*Fig 4a*).
8. **Inclusion of Only PTs with Siblings.** We agree with the potential confound of PTs without siblings and will therefore only include PTs *with* siblings. This is feasible as ~65% of PTs in our NICU have siblings.
9. **Equal Sampling of Gestational Age Bins for PTs.** Our data (and e.g., Schlatterer et al., 2022) show that the vast majority of infants <32w have dysregulated ANS and that prematurity-related morbidities have a greater impact on ANS than GA at birth. While we view it as a strength of this study to have some variability of ANS dysfunction within our PT group, we agree with the concern about the possible lack of ANS dysfunction in later-born PT infants. We have lowered the birth GA of the PT group and will recruit 3 even groups (n=25/group) of PTs born with GA of: 24-26w; 27-29w; 30-32w. We acknowledge that this sample is not large enough to answer complex questions of the broad range of ANS dysfunction in this population. However, no studies of this kind, that link ANS dysfunction to developmental trajectories/ASD outcomes, exist and our plan will ensure a PT sample with some degree of ANS immaturity, providing critical data for future investigations. Feasibility evidence: our NICU admits ~75 neonates per bin per year.
10. **Attrition.** Attrition rates reported include the same age range and longitudinal design as proposed here.
11. **Excluding PTs and FTs with ASD Limits Generalizability.** We anticipate a small number of PTs and full-term non-ASD siblings with ASD. However, we agree that developmental trajectories of PTs with ASD will be a major contribution to science. Therefore, we will include PTs and FTs with ASD in all analyses.
12. **Full Investigative Team Meetings** will occur 3x/year at key points in the project to troubleshoot issues and think through larger project questions from multiple perspectives.
13. **Human Subjects Comment.** Delays and referrals for ASIBs, PTs, and TDs will be determined using annual ASQ-3 screeners, parental concerns questionnaires, expert clinical judgement (*Protection of HS*).
14. **Budget Comment** has been addressed. Annual direct costs less subaward F&A are all below \$500,000 (see “Total Direct Costs Less Consortium F&A”).

SPECIFIC AIMS

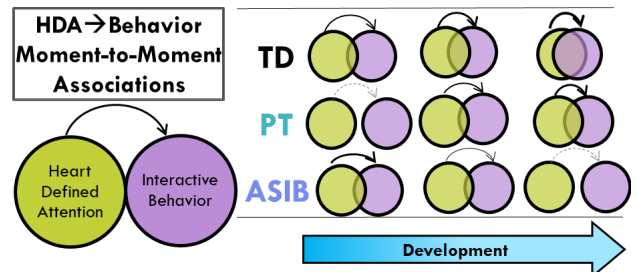
Atypical attention is among the earliest-emerging, lifelong features of autism spectrum disorder (ASD) and a particularly compelling candidate for an ASD biomarker¹⁻⁴. Our work, and that of others, suggests that disrupted attention may be detectable and developmentally predictive in early infancy⁵⁻⁸, yet the biological underpinnings of this disruption remain unknown. The autonomic nervous system (ANS) is an early-emerging bioregulatory system critical for supporting neonatal arousal and infant attention⁹⁻¹¹. Early disruptions in ANS's regulation of attention ("autonomic regulation of attention") may have cascading consequences for interaction, learning, and development^{8,9,11}. Broad ANS dysfunction is observed in older children already diagnosed with ASD¹², but there remains a tremendous gap in knowledge regarding **whether** and **when** ANS dysfunction contributes to attention abnormalities and how infant autonomic regulation of attention predicts the emergence of ASD symptoms.

The current study fills this gap by examining autonomic regulation of attention in the neonatal period and the developmental consequences therein for the emergence of ASD symptoms in infant siblings of children with ASD (**ASIBs**), infants born preterm (**PTs**), and typically developing (**TD**) infants. A critical innovation of this study is leveraging a comparison group of PTs who, intriguingly, experience global ANS dysfunction from birth^{9,13-15}. While PTs show slightly elevated rates of ASD (~7%) compared to the general population (~2%)¹⁶, the vast majority of PTs do *not* develop ASD. In this study, **inclusion of PT infants as a model for global ANS dysfunction** will help investigate our hypothesis that early and broad ANS dysfunction (not specific to attention) *does not* lead to ASD-specific symptoms. This developmental process (broad ANS dysfunction → non-ASD specific outcomes) will be contrasted to **ASIBs** who experience significantly higher rates of ASD (~20%)¹⁷. Contrasting **PTs** to **ASIBs** will help delineate how *attention-specific* ANS dysfunction predicts the emergence of *ASD-specific* symptoms. Thus, our innovative **three-group research design** will compare **ASIBs** to two control groups (**PTs** and **TD** infants) to determine the specificity of autonomic regulation of attention as a key, early emerging feature associated with ASD, with incredible translational potential for a cost-effective biomarker.

Aim 1. Compare the maturation of autonomic regulation of attention in the early infant period (1-3m) across ASIBs, PTs, and TDs. Attention in the first months of life is supported by flexible modulation of arousal and alertness, regulated by the ANS¹⁸⁻²⁰. Autonomic regulation allows for attention to salient features of the environment and inhibition of responses to irrelevant (internal or external) stimuli. ANS regulation of attention is vital for learning and can be measured using **heart-defined attention (HDA)** (i.e., coordination of heart rate changes within moments of visual attention^{21,22}). Maturation of ANS and attention systems across the first months of life lead to increasing coordination of heart rate with attention. Our pilot work with ASIBs shows differences in HDA quantity (%time in HDA) and quality (magnitude of heart rate deceleration)^{23,24}. Building upon this work, Aim 1 will test the hypothesis that **ASIBs** show **typical HDA initially** compared to **TDs**. Then from 1-3m, as ANS commands an increasingly larger role in regulating attention, the development of HDA for **ASIBs** will attenuate. In contrast, **PTs** will show **delayed HDA maturation** from 1-3m (adjusted age) due to broad ANS dysfunction²⁵.



Aim 2. Determine the moment-to-moment influence of autonomic regulation of attention on interactive behavior across the first year (6m, 9m, 12m). Autonomic regulation of attention, measured via heart activity and contextualized within moment-by-moment behavior, has extraordinary potential to transform early identification of ASD by revealing specific mechanistic relations between ANS, attention, and interactive behavior. Our preliminary data suggests: (1) temporal coordination between HDA and interactive behavior in **TDs** and (2) reduced HDA-behavior coordination in **ASIBs**. Aim 2 will test the hypothesis that **ASIBs** show weakening predictive associations between HDA and interactive behavior from 6-12 months, as early symptoms of ASD emerge²⁶. In contrast, we expect **PTs** to show initially weak associations that strengthen over time (with ANS and behavioral maturation)²⁵.



Aim 3. Predict ASD symptoms at age 3 years from infant autonomic and attentional features.

ASD biomarker discovery requires quantitative modeling techniques that adequately capture the complexity of biological processes involved in such a multifaceted, heterogeneous neurodevelopmental disorder^{27,28}. Aim 3 will leverage machine learning techniques to develop a biomarker predicting the later emergence of ASD symptoms based on early (1-12 months) features of the autonomic and attentional systems (variables from Aims 1-2). We hypothesize that features related to infant autonomic regulation of attention will predict ASD symptoms.

SIGNIFICANCE

Earlier Intervention Necessitates Earlier Identification

Autism spectrum disorder (ASD) is a lifelong condition that affects ~2.3% of the population and carries a lifelong financial cost of \$1.4-2.4 million per individual, thus representing a pressing public health challenge^{29,30}. Early intervention plays a pivotal role in the lives of children and families affected by ASD – improving functional outcomes for the individual, enhancing the quality of life for the family, and reducing a lifelong financial cost^{31,32}. Unfortunately, with ASD diagnosis and therefore intervention onset occurring after age 4-6 years^{33,34}, we are missing out on critical periods of early intervention that could capitalize on peak neuroplasticity within the first 2 years of life³⁵⁻³⁷. **This is a critical barrier in the field**—earlier identification of ASD within the first year of life can *catalyze the onset of very early intervention*, leading to greater *improvements in long-term outcomes*. Addressing this critical barrier by identifying the earliest point of developmental divergence in ASD is a high priority.

The search for biological and behavioral markers of ASD has resulted in increased understanding of altered neurodevelopmental pathways, and **atypical attention** is among the most promising biomarkers that persists across the lifespan³⁸⁻⁴². Research on infant attention suggests that infants with ASD exhibit reduced neonatal attention^{6,43}, diminished social monitoring^{44,45}, disrupted engagement of social sustained attention⁴⁶, and a developmental decline in eye-looking⁷, all prior to diagnosis. **Still, this evidence is limited to behavioral studies of attention, and few extend to earlier than 2-3 months of age.** To advance our understanding of disrupted attention in ASD, there is a need to examine specific biological substrates associated with this developmentally persistent phenotypic feature. The timing and context-specificity of attention differences measured at multiple levels (e.g., behavioral, physiological) in early development may reveal key neurodevelopmental mechanisms in ASD. This project takes the critical next step in examining **biological underpinnings of disrupted attention in ASD** by testing whether reduced autonomic regulation of attention (“heart defined attention”) is a **key early-emerging feature associated with ASD**. **Significantly**, this work will:

1. **Advance current etiological theories of ASD** that link, with heightened specificity, an established, lifelong cardinal feature of ASD – atypical attention – to autonomic regulation of attention in the first months of life.
2. **Pave the way for new developmental models of ASD** wherein disrupted attention has proximal effects on interactive behavior that interferes with learning and has distal, cascading effects on long-term outcomes.
3. **Lead to the development of new models for earlier intervention** that enrich early experiences and enhance learning opportunities *from the first months of life*, ultimately reducing lifelong costs and improving lifelong outcomes for individuals with ASD.

There is an urgent need to test specific mechanistic hypotheses about precisely when attention patterns diverge in ASD, the neurobiological systems implicated in this disruption, and the in-the-moment influence of attention on interactive behaviors in infancy, prior to the emergence of symptoms.

The First Three Months: A Critical Period of Neurodevelopment and Autonomic Maturation

The brain undergoes faster structural and functional maturation, as well as significant cross-domain integration, from birth to 3 months of age than at any other postnatal time⁴⁷. From birth, the autonomic nervous system (ANS) plays a vital role in the adaptive competence of the newborn. ANS has two major component systems: sympathetic (“fight or flight”) and parasympathetic (“rest and digest”). In response to ever-changing environmental contexts, these systems enact physiological changes that regulate basic functions, such as heart rate and respiration. In early prenatal development, physiological functions are driven primarily by sympathetic influence. Higher cortical processes begin to integrate autonomic control in the late prenatal and early postnatal period leading to increasing influence of the parasympathetic system. As ANS matures, parasympathetic tone serves as the foundation for several higher-level behaviors that support calm states of alertness and **sustained attention** to people and objects, ultimately providing countless opportunities for social and language learning. In this way, ANS is considered a “developmental resource” in the neonatal and early infant period that mediates infant-environment interactions¹¹. While some neurodevelopmental models of ASD suggest atypical emergence of cortically-driven endogenous attention systems⁴⁸, **the developmental course of parasympathetic influence on attention in ASD has never been investigated**, representing a clear scientific gap to be filled^{9,48,49,11}. Current evidence that forms the **scientific basis of this study** includes: 1a) *very early* disrupted attention, measured behaviorally, in ASD, 1b) an ANS-regulated marker of attention that is well-studied in TDs, but not ASD, 2) a developmental profile of PT infants who experience very early ANS dysregulation, but not ASD (in most cases), and 3) some evidence of a dynamic, temporally coordinated, ANS-attention-behavioral system in TDs, but less in ASD. This strong scientific foundation (described in further detail below) **calls for further research** on the “physiological phenomena” of attention⁴⁹ and presents **three specific gaps in prior research** that prevent a full understanding of the role of autonomic regulation in atypical attention in ASD: **(1)** The maturational course of

autonomically regulated attention and baseline autonomic functioning as it emerges *within the first three months of life* for infants at elevated genetic likelihood for ASD remains unknown. **(2)** The developmental course of (and relations between) autonomically regulated attention, baseline autonomic functioning, and ASD symptomology in infants who are *born with* baseline ANS dysfunction (PTs) has not been delineated. **(3)** The dynamic role of HDA in supporting *interactive behavior*, moment-by-moment, is entirely unexplored. These gaps have prevented our ability to test a full model of ASD wherein early autonomic dysregulation of attention *within the first months of life* has cascading effects on interactive behavior and the emergence of ASD symptoms. Examining this novel mechanistic hypothesis while capitalizing on an especially critical period of rapid neurodevelopment (1-3mos) will advance scientific knowledge of ASD pathogenesis, ultimately paving the way for novel paradigms of very early interventions.

1. Autonomic Regulation of Attention: A Critical Opportunity for ASD Research. Autonomic regulation and attention are critical components of an adaptive engagement system that selects relevant features of the environment and creates rich opportunities for learning and social interaction¹⁰. ANS serves a regulatory function in attention by modulating heart rate responses during periods of attention and information processing. In a healthy ANS-mediated attention system, a neonate overrides metabolic demands to generate a physiological response (heart rate deceleration) that facilitates and enhances attention and stimulus intake⁵⁰. This parasympathetically-driven physiological response, termed here **heart defined attention (HDA)**,⁵¹ is characterized by 3 discrete attention phases (Fig 1)

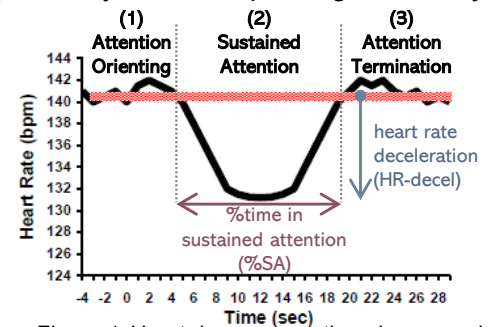


Figure 1. Heart defined attention phases and proposed DVs: %SA, HR-decel

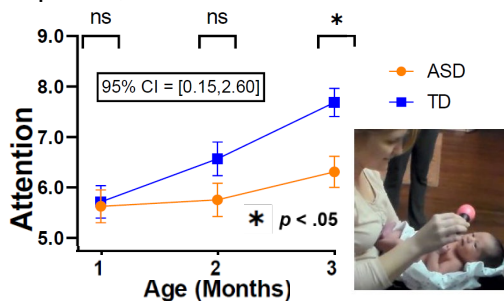


Figure 2. Infant attention scored from NICU Network Neurobehavioral (NNNS) exam. Higher score indicates better performance.

that are associated with cognitive and learning processes⁵². Some argue that HDA indices are more precise measures of information processing than looking time alone^{53–55}. Limited studies of infants later diagnosed with ASD show atypical behavioral measures of attention^{7,43,56} in very early infancy (see PI's work, Fig 2)⁵⁷ as well as atypical general autonomic functioning in later infancy^{58,59}, pointing to the possibility of autonomically-driven dysregulation of attention. **Our pilot work**, and that of others, show differences in HDA quantity (%SA: %time in heart-defined sustained attention) and quality (**HR-decel**: magnitude of heart rate deceleration; Fig 1) in infant siblings of children with autism (**ASIBs**)^{24,49} (see Fig 4 below).

2. Preterm (PT) Infants as a Model of Global ANS Dysfunction. Significant ANS maturation occurs in the second half of pregnancy, and preterm birth (born <37w gestation) significantly disrupts ANS development and the overall functioning of the PT newborn⁶⁰. For PTs, reduced ANS activity fails to normalize at term age equivalent^{25,61} and may alter long-term temporal programming of ANS maturation⁶². While the vast majority of PTs experience prolonged global ANS dysfunction, *only a small minority of PTs develop ASD* (7% in PTs vs. 20% in ASIBs)^{16,17}. Thus, inclusion of **PTs as a model for global ANS dysfunction** in this study allows for heightened precision and specificity in delineating the link between *attention-specific* ANS dysfunction (i.e., HDA) and ASD symptoms. **Systematic, longitudinal examination of autonomic measures of attention (HDA) in infants with elevated liability for ASD (ASIBs) contrasted to PT and TD control groups** is necessary to test our mechanistic hypothesis that disrupted autonomic regulation *specific to* attention (vs. global ANS dysfunction observed in PTs and measured in this study) carries critical consequences for learning and development⁶³ and is associated with ASD symptoms. Measurement of general, baseline autonomic function in addition to attention-specific autonomic regulation, as proposed in this study, will clearly show how autonomic regulation of attention differs across **ASIBs, PTs**, and **TDs**.

3. Modeling Autonomic, Attentional, and Behavioral Dynamics in ASD. A healthy, regulated autonomic and attention system creates rich opportunities for learning and social interaction¹⁰. After the third month of life, a newly emerging, cortically driven organization of behavior results in anticipation and preparation for interactions with people and objects. The dynamic coordination of autonomic arousal and attention affords calm states of alertness, an optimal context for learning⁶⁴. Quantification of the dynamics of this system⁶⁵ can reveal how autonomic regulation of attention is associated with, and may even predict, interactive behaviors as they unfold, moment-by-moment, in infancy. Autonomic arousal can index "anticipatory readiness" for dynamic changes in attention and behavior^{66,67}. Emerging research in ASD suggests unique patterns of ongoing physiological activity and attention during social tasks^{68,69}, however, **there is a significant lack of data on temporal dynamics of HDA and ongoing behaviors⁷⁰ in ASIBs** in infancy as symptoms of ASD begin to unfold. Such ANS-driven

dynamics could facilitate ongoing social engagement, which may help to guide ongoing interactive behavior, and ultimately increase opportunities for learning. **The proposed study will fill this gap** with an analysis of temporal dynamics of infant gaze, heart rate changes, and behavior during naturalistic interactions between age 6-12 months across ASIBs, PTs, and TDs. Critically, the direct link from **physiology** (heart-defined attention) to **in-the-moment interactive behavior** will enable translation of findings to pilot behavioral interventions and explore biofeedback tools^{19,71,72} to directly target the quality and quantity of infant HDA during interactions.

Model of ANS-Regulated Attention and ASD

Based on current literature and our preliminary data, we propose a model of ASD wherein initial global ANS (primarily sympathetic) function supporting neonatal arousal and reactivity is intact at birth. Yet, subsequently emerging parasympathetic regulation of attention is disrupted. This disruption emerges over the first year of life and interferes with infant sustained attention and ensuing behavior, leading to decreased learning, especially social learning, opportunities, and increased ASD symptoms. **Expected Outcomes by Group** (see Fig 3): **ASIBs**, who have *no/low* global ANS dysfunction at birth (Fig 3a), will show emerging ANS dysregulation of *attention* over the first year (attenuated HDA from 1-3m, weakening HDA-behavior associations from 6-12m; Fig 3b) leading to elevated ASD symptoms at age 3 (Fig 3c). **PTs**, who *do* have global ANS dysfunction at birth (Fig 3a), should show gradual improvement in HDA (1-3m) and HDA-behavior associations (6-12m), ultimately resulting in low ASD symptoms at age 3. The rate of ASD in PTs is higher than that of the general population, and we hypothesize that a small subset of PTs with elevated ASD symptoms at outcome will show *both* disrupted global ANS *and* ANS-dysregulated attention in early infancy. **TDs** (*no* global ANS dysfunction) will show *no/low* ANS dysregulation of attention and low ASD symptoms at age 3.

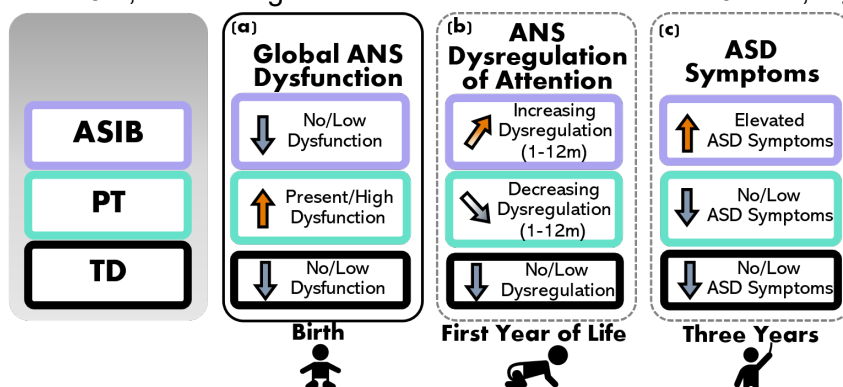
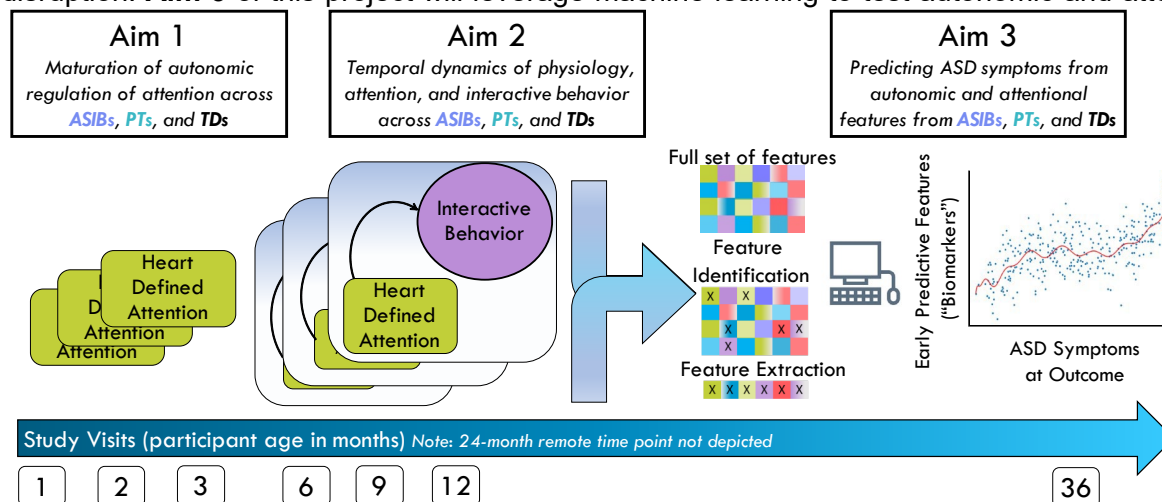


Figure 3. Driving framework for expected outcomes to be tested (b) and (c) given initial global ANS function/dysfunction at birth (a). ANS dysregulation of attention (b) will be measured early (1-3 months) via heart defined sustained attention (%time in sustained attention, HR deceleration). ANS dysregulation of attention will be measured later (6-12 months) via HDA-behavior associations.

Translation of Findings to Non-invasive Biomarkers of the ASD Phenotype

Early intervention plays a pivotal role in the lives of children and families affected by ASD – improving functional outcomes for the child, enhancing quality of life for the family, and reducing lifelong financial burden⁷³. Very early onset of intervention enhances child gains, which has prompted the PI and others to develop and pilot interventions for at-risk infants, some *as young as 6 months*^{31,74–77}. The nascent, but rapid, progress in the development of very early interventions demands innovative studies to identify early markers of social functioning. Offering intervention to all infants with genetic or congenital risk is likely not feasible. But a **valid, cost-effective, non-invasive biomarker** can help identify those who have the greatest likelihood of developing social-communication challenges and ASD, and thus are in greatest need. In a landmark study, Jones and Klin (2013)⁷ identified altered trajectories of social attention (2-6 months) as the earliest known predictor of ASD. However, this exceptional study did not evaluate potential biological mechanisms underlying such an early disruption. **Aim 3** of this project will leverage machine learning to test autonomic and attentional features from



1-12 months (Aims 1-2) as viable biomarkers of ASD symptoms. Notably, **Aim 3 is not contingent** on significant results from Aims 1-2 as statistical significance is not a requisite for including these variables in feature selection

procedures. See **study overview** above.

The proposed research will have a significant impact on the broader understanding of ASD by utilizing an innovative, developmentally informed framework: autonomic regulation of attention⁷⁸. We will generate linkages from ANS and attention to moment-by-moment infant behavior and utilize advanced analytic methods to predict ASD symptoms years later. The prior research and our preliminary data (see *Prelim. Data* below) provide support for the overall study premise to identify disrupted autonomic regulation of attention as an early indicator of ASD.

The potential for translation of findings to actionable interventions is high; promising studies of behavioral interventions for PT infants show positive maturational shifts of autonomic regulation⁷⁹, suggesting malleability of this early-developing system. Developmentally informed interventions that target this rapidly maturing autonomic system of attention have the potential to make significant, sustained changes in attention, cognitive, and social outcomes for infants with an elevated likelihood for ASD.

INNOVATION

This proposal seeks to shift current research paradigms by studying novel indices of attention across three unique infant groups using advanced modeling of dynamic, moment-by-moment, behavior in the first year of life.

Attention as an Autonomically-Regulated Process and Predictor of ASD. Atypical attention has been among the most promising early markers of ASD^{38,80}, but the biological underpinnings of disrupted infant attention in ASD is currently unknown. This proposal will utilize an innovative, physiological measure of attention (HDA) in addition to a baseline autonomic marker (respiratory sinus arrhythmia, RSA) to test the hypothesis that disrupted attention is linked specifically to altered ANS-regulated attention and is associated with later-emerging ASD symptoms. Repeated measurement of RSA and of HDA across interactive contexts from the first months of life is an innovative approach that will provide an enriched dataset more powerful than previous studies. This innovation will allow us to understand *mechanisms* and *roles* of ANS and attention in the unfolding of ASD.

Preterm Infants: Controlling for Global ANS Dysfunction. PT infants are born with an exceptionally immature autonomic regulatory system that may not fully develop until well after birth^{13,81–83}. Yet, the vast majority of PTs ultimately experience typical outcomes (~80-92% of PTs do not have ASD or developmental delays by preschool age^{84,85}). Thus, PT infants serve as a unique control group that experience early autonomic dysfunction and mostly typical, non-ASD outcomes. Measures of HDA and baseline RSA within this innovative three-group research design will allow us to delineate: 1) how global ANS dysfunction (in PTs) differs from autonomic dysfunction specific to attention (in ASIBs) and 2) how maturation and improvements in ANS over the first year of life for PTs may support interactive behavior and learning, potentially buffering against ASD-specific outcomes.

Moment-by-Moment Measures of a Dynamic, Interactive System. Dynamic biological rhythms are central to the development of the infant⁹. Physiological and social maturation are dynamic processes, and variability within this system during critical neurodevelopmental windows may predict later functioning^{9,86}. This proposal examines novel questions of how physiological correlates of attention (i.e., HDA) relate to infant interactive behavior. By measuring HDA during live interactions with people and objects, we will explore the dynamics of attention, physiology, and behavior in an integrated way that has never been done before. Aim 2 will capture this integration, generating a rich dataset that can inform the development of interventions.

APPROACH

Key empirical support for the proposed project include the PI and investigative team's preliminary data and previous research on disrupted attention in infants who later develop ASD^{6,44,45,48,87–89}. However, prior research has been largely limited to behavioral measurement of attention in contrived experimental contexts (e.g., eye tracking) in later infancy (after 3-6 months). Moreover, very few studies have directly compared ANS or attention in ASIBs and PTs. And no studies have examined the temporal dynamics of autonomic regulation of attention and interactive behavior in these groups. These issues limit advancement of etiological theories of ASD and implications for treatment. This proposal details specific plans to address the gaps in prior research.

Research Team. The project PI (Bradshaw) has >14 years of experience conducting research in largescale, NIH-funded, interdisciplinary, autism centers (Yale Child Study Center, UCSB, UCSD, Emory), resulting in 40 publications. As an early-stage investigator (ESI), the PI is currently **leading** two NIH-funded studies (K23, R21), one Foundation award (James S. McDonnell Foundation), and two internally funded pilot studies. In the proposed project, the PI will be supported by five Co-Is and one psychologist (below), four of whom are senior investigators with an established track record of grants management (R01s) and successful mentorship of and collaboration with ESIs. In addition to regular individual meetings between the PI and Co-Is, **regular meetings with the full investigative team** (PI+Co-Is) will occur at key points throughout the project (≥3x/year) to troubleshoot issues and think through larger project questions and plans from multiple perspectives. **Jane Roberts** (expertise in

autonomic dysfunction in ASD/Fragile X; primary mentor for PI's K23) has significant grants management experience and publications with the PI⁹⁰. **John Richards** (co-mentor on PI's K23, collaborator on internal pilot grant) developed the theoretical model of heart-defined attention^{52,91} and has led an R01 for >40 years focused on this model of infant attention in TD and PT infants. **Robin Dail** (expert in ANS in PT infants, collaborator on internal pilot grant) is leading a multisite R01 with PT infants in NICUs across North/South Carolina. Her expertise is critical for understanding ANS maturation in the context of PT birth and heightened medical morbidities. The PI's collaborative conceptual work with **Jana Iverson**⁸ (expertise in TD and ASIB infant motor/communication development) on developmental cascades in ASD has led to the novel questions proposed in this study. **Christian O'Reilly** (expertise in computational modeling, early biomarkers, and signal processing) has worked with the PI using machine learning to predict ASD likelihood from heart activity (preliminary data for Aim 3). Finally, a licensed **Clinical Psychologist** with expertise in developmental psychology and autism (Dr. Emily Neger) will assist the PI with clinical training, supervision, reliability monitoring, and expert clinical judgements. The unique, complementary areas of expertise among the investigative team, along with diversity across stage of career and discipline, will enhance diverse perspectives, enrich expertise, foster ingenuity, and result in successful achievement of scientific aims.

PRELIMINARY DATA

ASD-Specific Differences in Behavioral Measures of Early Infant Attention. In the neonatal period, attention to the environment occurs during calm states of alertness, facilitated by optimal levels of autonomic regulation. Our recent set of studies (P50MH100029)^{6,57} investigated *behavioral* measures of **attention** in 1-3-month-old ASIBs who later developed ASD. We identified specific timepoints, between 2-3 months of age, when maturation of attention in ASD (n=18) and ASIBs (n=41) lagged compared to TDs (n=39; *Fig 2*, above). These preliminary data highlight **feasibility** in **recruitment** and **measurement** of attention in neonates and **quantifiable differences** in behavioral measures of attention across groups. The current proposal will expand on these data by increasing sample size (n=75 ASIBs), adding a second comparison group (n=75 PTs), and examining a specific biological substrate of disrupted attention (heart-defined attention; HDA) as well as general autonomic integrity (baseline RSA). In addition, all data collection will occur in the home, which will greatly increase participant accessibility and enrollment (especially for communities of color and rural populations).

Infant Autonomic Regulation of Attention is Associated with ASD Symptoms. Preliminary data related to Aim 1 include trajectories of HDA during naturalistic parent-infant interactions from 1-4 months of age across n=10 ASIBs, n=20 PTs, and n=30 TDs (K23MH120476; R21DC017252; PI: Bradshaw). These trajectories are characterized by a significant increase in heart rate (HR) deceleration during sustained attention ($F=10.24$, $p < .001$, *Fig 4a*) and point to a dampened HR deceleration that is developmentally transient for PTs, but persistent for ASIBs^{23, 94,92,60}. Moreover, HR deceleration during sustained attention to people at 3 months is associated with early-emerging symptoms of ASD²⁶ (social-communication skills at 9mo⁹²; $r=0.50$; $p<.05$; *Fig 4b*). Our team's

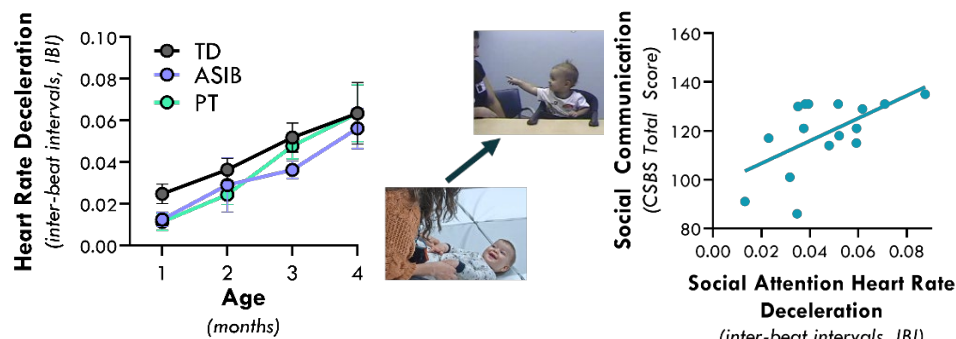


Figure 4a. Trajectories of HDA deceleration during social interactions.

Figure 4b. Association between HDA deceleration during social interaction at 3-4mos and 9mo communication.

ASD symptoms at age 3 (Aim 3).

Feasibility of Time-Locking HDA Phases to Interactive Behavior. Our data show that HDA-related HR deceleration during a social interaction at 3 months is associated with social-communication outcomes at 9 months⁹² (see above, *Fig 4b*). One interpretation of this preliminary finding is that more sustained attention during social interaction reflects increased social engagement and learning, leading to better social-communication skills 6 months later. Yet these data also point to questions about whether and how increased autonomic regulation of attention may facilitate interactive behavior (and therefore learning) **in the moment**. Aim 2 will address this question. Utilizing full electrocardiogram (ECG) waveform recordings (1024Hz) and frame-by-frame

past research also suggests dampened baseline RSA is present very early in PTs⁹³, but only later in ASD⁹⁰ (similar to other work⁵⁸). Building on these preliminary findings, we propose to test the hypothesis that autonomic dysregulation of attention is present early in ASIBs (Aim 1), disrupts temporal dynamics of physiology, attention, and interactive behavior (Aim 2), and is associated with elevated

video coding of gaze fixation, ongoing heart-defined phases of attention (HDA) are identified epoched based on the onset of interactive behavior (also coded frame-by-frame from video). Our preliminary data (K23MH120476) demonstrate **feasibility in capturing infant interactive behavior in 6-12-month-olds^{26,94} and synchronizing with HDA time-series data²³.**

In a preliminary analysis of N=12 infants, we identified the onset of interactive behavior (object exploration) and calculated the proportion of time (i.e., relative frequency) of HDA phase in a time-series of 5 seconds prior to each behavior onset (*Fig 5*). Findings point to between-group differences (TD vs ASIB) in autonomic regulation of attention immediately preceding behavior. As expected, TD infants showed a consistent pattern of HDA change within the 2 seconds immediately preceding the behavior, characterized by an increase in attention (increased **attention orienting**, green and **sustained attention**, red). By contrast, **ASIBs did not show a clear pattern of HDA change** immediately preceding the behavior. Instead, attention patterns remained stable up to 5 seconds prior to the behavior, characterized by less overall attention (more **inattention**, grey; less **attention orienting** and **sustained attention**). The proposed project will build upon this preliminary work in a rigorous examination of **moment-to-moment dynamics** of HDA and interactive behavior across the first year.

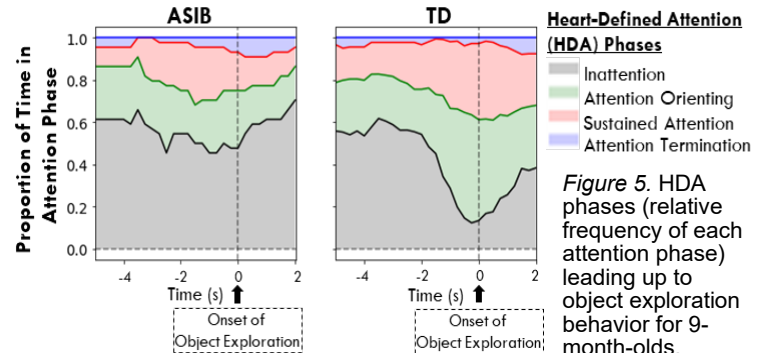


Figure 5. HDA phases (relative frequency of each attention phase) leading up to object exploration behavior for 9-month-olds.

Prediction of ASD Symptoms using Measures of ANS and Attention. In collaboration with Co-I (O'Reilly), ECG data from 3-6-month-old ASIBs (N=7) and TDs (N=16) during interactions were collected and cleaned. Several features were extracted (e.g., mean inter-beat-interval (IBI), standard deviation, root-mean-square differences between adjacent IBI, discrete functional analysis, sample entropy) and an ablation study was used for optimal feature selection. A multilayer perceptron was used to classify ASD likelihood (ASIB vs. TD) with 86% accuracy (precision: 65%, sensitivity: 65%, specificity: 94%). These preliminary results compare advantageously with a previous study reporting a 75.3% accuracy⁹⁵ and **demonstrate scientific and technical feasibility** in feature extraction and application of machine learning models to predict ASD likelihood. In this proposal, we will develop a more nuanced analysis predicting ASD symptoms on continuous scales using newly developed deep neural implementation of mixed-effect linear models⁹⁶. The proposed project will use autonomic and attentional features collected for Aims 1 and 2 to better understand how HDA can be leveraged in the formulation of biomarkers in ASD. **Our preliminary data suggest sufficient power** for applying these methods to a sample size that is over ten times larger than that of this pilot study (N=200; n=75 ASIBs). Further, large effect sizes (detectable on relatively small samples) are critical for developing effective ASD biomarkers.

Regression Over Continuous Scales Rather than Binary Diagnoses Classification. ASD symptoms will be captured using Autism Diagnostic Observation Schedule (ADOS-2) calibrated severity scores (CSS)⁹⁷. Our preliminary data (PI Bradshaw⁵⁷ and Co-I O'Reilly) suggest sufficient variability in CSS scores across groups of 3-year-olds with and without ASD to determine meaningful associations between early infant biomarkers and ASD symptoms. In the current proposal, we will further build on this direction using ANS-related predictors.

OVERALL EXPERIMENTAL DESIGN

Participants. A total of 200 infants will be included in this study: 75 ASIBs, 75 PTs, and 50 TDs. **Note that our analyses are not powered to detect associations between predictors and binary ASD classification. Rather, for Aim 3, our sample size is powered to detect associations with a continuous measure of ASD symptoms at age 3** (see *Power Analysis* below). Our power analyses, published studies,⁵⁷ and preliminary data suggest that this proposal is sufficiently powered to detect associations of infant biomarkers and ASD symptoms^{6,57,92}. All participants will be followed longitudinally from 1 month to 3 years, at which point **stable** ASD symptoms can be quantified using ADOS CSS^{98,99}. In accordance with NIH guidelines for consideration of **relevant biological variables**, we acknowledge that ASD prevalence is higher in males; however, males and females will be recruited equally, and sex included as a covariate in all analyses.

Inclusion Criteria: 1) **ASIB:** born fullterm (≥ 37 w), have a full biological sibling with ASD (confirmation of ASD in the older sibling: record review, direct clinical measures (SCQ¹⁰⁰, CBCL¹⁰¹, SRS-2¹⁰²), expert clinical judgement by licensed psychologist with expertise in ASD); 2) **TD:** born fullterm, have a full biological TD sibling (confirmed with SCQ, CBCL, SRS, expert clinical judgment), no history of ASD in 1st-3rd degree relatives; 3) **PT:** gestational age (GA) at birth between 24_{0/7}-32_{6/7} weeks, have a sibling, no family history of ASD in 1st-3rd degree relatives. **GA Considerations:** PTs will be evenly stratified into 3 GA bins to ensure relatively even distribution

of risk for ANS dysfunction: (24-26w; 27-29w; 30-32w). GA at birth will be included as a covariate in all analyses, similar to our previous work^{57,103}. **Exclusion criteria** for all participants: failed newborn hearing screen, known major hearing or visual impairments, non-febrile seizure disorders, known genetic/chromosomal syndromes (e.g., Down Syndrome), congenital heart condition known to effect heart rate, severe brain hemorrhage and periventricular leukomalacia (PVL). PTs will provide medical records for accurate eligibility assessment. To maximize recruitment efforts of PTs, we are not excluding twins (but multiple birth will be considered in analyses). TDs or PTs with an ASD diagnosis at outcome will be included in analyses (while considering GA and genetic likelihood in models) to increase generalizability.

Feasibility of Recruitment/Enrollment. The current project capitalizes on significant, established recruitment infrastructure, which has led to successful recruitment of ASIBs and PTs at USC. Resources include: the UofSC Neurodevelopment Registry (N>1,100; PIs: Roberts and Bradshaw), the USC Infants and Children Registry (N>150; PI: Richards, Bradshaw), multiple community partnerships (NICUs at the two largest hospital systems in the state, SC SPARK, CAN Center at USC, DDSN, Pediatric practices, autism services providers). USC investigators have **direct outreach access to all children with ASD in the state of South Carolina (total N>8,000; n=6,700<8yrs)** through the Department of Disabilities and Special Needs (DDSN). This outreach has been exceedingly successful for recruitment in the past (see *Letters of Support*). Our longitudinal studies (K23MH120476, R21DC017252) reflect consistent, successful recruitment and retention of ASIBs, PTs, TDs, **1-3 months of age with >95% retention**. The PI's lab consistently receives **30-50 referrals per week** from interested research participants in the local community and the average enrollment rate for ASIB studies at USC is **12-18 participants/month**. For 1-3-month-old ASIBs specifically, active recruitment in the PI's lab and at USC results in **enrollment of 4-6/month**. This track record demonstrates feasibility regarding enrollment for the proposed project (target average:10-15 enrollments/month across all groups). Our outstanding recruitment records are bolstered by established connections to supportive networks in South Carolina (ASD service providers, schools, pediatricians, hospital NICUs) and state organizations serving ASD and PT populations have pledged tremendous recruitment support (see *Letters of Support*). **To maximize enrollment, we will leverage in-home data collection** for study visits. The investigative team has ~20 years of experience with in-home data collection (TDs, PTs, ASIBs age 1wk-3yrs; N>450) and our data evidence that 99% of families (446/450+) prefer in-home visits when given the choice of home vs. lab. This approach will significantly increase participant recruitment/retention rates and **support access for families with limited resources or located in rural areas**.

Attrition. Resources for participant incentives and study visits include over-enrollment to account for expected attrition (rate of 5% based on current longitudinal studies of ASIBs, PTs, and TDs enrolling at 1-4 months). This includes an additional N=16 participants to account for attrition (~5% lost to follow-up, n=10). Thus, power analyses (described below) are based on the final expected sample of N=200.

Operational Considerations. The current project will capitalize on existing infrastructure and resources from the PIs ongoing NIH awards (K23MH120476, R21DC017252), foundation grants (James S. McDonnell Foundation), institutional awards, and institutional funds. The PI's lab is staffed with 1 research assistant professor, 1 postdoc, 4.5 research specialists, 1 psychologist, 1 fulltime recruitment coordinator, and 4 graduate and 20 undergraduate students. Two current RAs are trained in most proposed data collection and processing procedures and will be dedicated to this project. Two more RAs will be hired specifically for this project (see *Budget Justification*). The PI and lab psychologist are licensed clinical psychologists and at least two research staff are trained and reliable in all clinical measures.

EXPERIMENTAL PARADIGMS AND DATA COLLECTION

Data Collection Overview. A key component of this longitudinal study is dense sampling of infant behavior to capture change during a rapid period of neurodevelopment. Accordingly, study visits will occur at months 1, 2, 3, 6, 9, and 12, with an additional 24-month questionnaire-only check-in (which will also support retention), and a 36-month evaluation of ASD symptoms and overall development. **Adjusted age will be used for all PT visits.** At visit months 1-3, autonomic regulation of attention will be measured with simultaneous video and heart rate recording during administration of the well-established standardized neonatal measure NNNS-II (NICU Network Neurobehavioral Scales-II; see *Early Infant Attention Task* below)¹⁰⁴. At visit months 6-12, trained research staff will video record two standardized infant interactions. **Research staff will be masked to participant group; parents will be reminded at each visit to refrain from discussing any family or medical history with staff.** All research staff will be trained to reliability (>80% agreement) in administration and scoring (where applicable) of the clinical and experimental measures; reliability will be monitored monthly by the project psychologist. A comprehensive evaluation of ASD symptoms and overall development will be administered at the 36-month time

point. While study aims are not designed (nor powered) to use ASD diagnosis as a binary outcome, a clinical best estimate (CBE) procedure will be used to assign an overall diagnosis at 36 months in order to comprehensively characterize our sample. **ADOS calibrated severity scores** (RRB, Social Affect, Total) will be used as a metric of ASD symptoms for Aim 3, similar to our previous work⁵⁷. In addition, each study visit will begin with a 5-min baseline ECG recording while the infant is laying or sitting still watching a video of fractals with classical music (to encourage restfulness/ reduce movement). As in previous work^{58,59}, **baseline autonomic function** will be indexed with respiratory sinus arrhythmia (RSA; a measure of cardiorespiratory control that reflects general autonomic integrity¹⁰⁵; CardioBatch¹⁰⁶).

Early Infant Attention Task from the NNNS-II (Aim 1). Attention will be measured from 1-3m during a standardized, readily translatable measure (NNNS-II) that is grounded in a developmental, neurobehavioral framework and aligns with study aims and hypotheses. The NNNS-II is a 15-20-minute direct evaluation of neurological functioning and behavioral adaptation of at-risk (e.g., medically compromised, preterm) and healthy neonates¹⁰⁷. The PI has >6 years experience administering the NNNS, is a certified NNNS-II examiner and trainer, and has published on attention using the NNNS in TDs and ASIBs^{6,57}. Of interest to the current study are the NNNS-II Attention items, which examine the infant's ability to block out external (and internal) stimuli and attend to visual and auditory stimuli. Six attention items include inanimate (ball and rattle) and animate (face and voice) stimuli that are presented visually with sound (rattle; face/voice), visually without sound (ball; face), and auditory only (rattle/voice out of sight). Administration is done with the examiner holding the infant in their lap in a flat or semi-elevated position. To obtain optimal performance, examiners implement handling strategies to keep the infant in a calm, alert state (handling strategies are recorded). Attention item administration = 3-6min.

Infant Interaction Task (Aim 2). Infant HDA and interactive behavior will be recorded during two semi-structured, table-based infant interactions at 6, 9, and 12 months (see Fig 7 and our work using these protocols in 6-12-month-olds^{26,94}). To capture both social and object interactive behaviors, two interactions will be completed. The **Social** Interaction occurs with an examiner and caregiver (ecologically valid context for infants) and is designed to elicit social interaction behaviors, similar to common social-communication measures for infants (e.g., CSBS¹⁰⁸). The PI is a close collaborator of the CSBS author (Amy Wetherby)^{26,75,109} and our research staff are trained and supervised in the CSBS-BB by Dr. Wetherby's research staff. The **Object** Interaction (no caregivers) is designed to elicit independently initiated object exploration behavior. Our research team has successfully administered this experimental task to quantify object behaviors⁹⁴. Social and Object interactions are equated on duration and number of objects presented to the infant (n=5; Fig 7).

Autism Symptoms (Aim 3). ASD symptoms at 36m will be captured using ADOS-2 calibrated severity scores (CSS): Social Affect, Restricted and Repetitive Behaviors (RRBs), and Total⁹⁷. ADOS examiners will be trained to research reliability and supervised by the PI and project psychologist. When the administering examiner is unintentionally unmasked on a study visit, a masked examiner will score the ADOS from video to ensure scientific rigor. CSSs range from 1-10 and represent an excellent measure of ASD symptoms, relatively independent from IQ and language abilities^{97,110}.

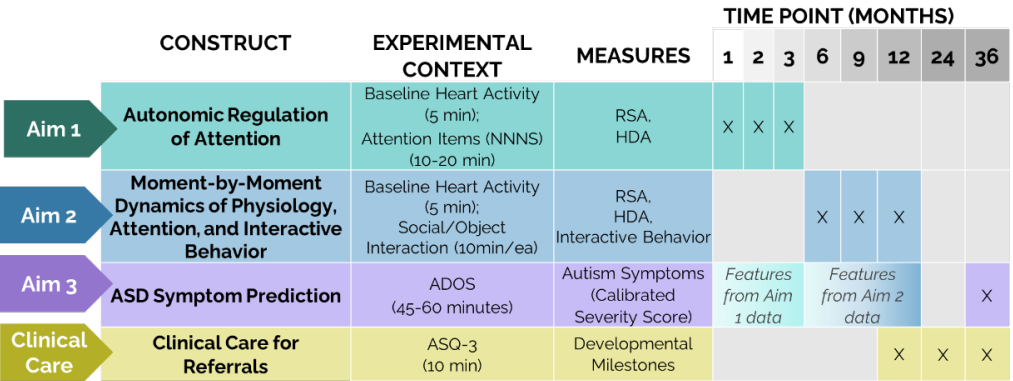


Figure 6. Data collection timeline and primary dependent variables. Note that some important measures are not depicted, including covariates (e.g., NICU-related morbidities, GA, motor skills).

	Social Interaction	Nonsocial Interaction
Protocol Description	Semi-structured interaction between infant, examiner, parent, and one object at a time. Examiner provides presses to elicit infant social interaction and communication.	Semi-structured interaction between infant and one object at a time. Distinct object shapes and functions provide various opportunities to elicit infant exploratory behavior.
Experimental Setup		
Task Length	10 min	10 min
Activities Presented	wind-up toy · squeaky toy · balloon · koosh ball · bubbles	rattle/sensory teether · soft blocks · ring stacker · noise maker · pop-up
Action	Earlier Actions - Eye contact, Vocalization	- Grasp, Mouth
Examples	Later Actions - Point, Word Approximations	- Stack, Press

Figure 7. Infant Interaction Protocol

Data Processing

Electrocardiogram (ECG) Recordings. ECG recordings will be collected at each time point between 1-12 months via Actiheart¹¹¹ wireless heart rate monitors, sampled at 1024Hz. The research team has extensive experience using these monitors for successful, continuous recording of infant ECG. Video recordings of all experimental paradigms will be collected for synchronization of ECG recordings with behavioral events (behaviorally coded later from video). Specialized software programs, CardioPeakSegmenter and CardioEditPlus^{106,112}, will be used for R peak detection, segmentation, and artifact removal.

Baseline Autonomic Function will be indexed using overall RSA from a 5-min baseline recording¹⁰⁶.

Synchronization of ECG and Video Recordings. Actiheart wireless heart rate monitors are incredibly user-friendly and allow unrestricted movement for infants and caregivers. The start time of the ECG recording (based on a remote computer time) is used as the synchronization signal and is captured (video recorded) at the beginning of the experiment (i.e., the start time and run time of the ECG recording is captured on the video that will later be coded for infant behavior). Our custom data processing pipeline then crops the ECG recording to align ECG and Video start times ($\text{time}_0(\text{ECG}) = \text{time}_0(\text{Video})$), allowing precise alignment of behavioral coding (described below) and ECG. This method has been used successfully in the PI's lab for the past 4 years^{23,92}.

Behavioral Coding of Infant Gaze. Coding of infant looking is needed to identify heart defined phases of attention. Thus, frame-by-frame video coding of infant looking using DataVyu software will be completed by trained and reliable undergraduate RAs masked to group. The PI has 16 years of experience in behavioral coding and the PI's lab has established procedures and manuals for coding infant looking during interactions^{23,92}. Coders will identify the onset and offset of infant looks to pre-identified target stimuli (object, caregiver, examiner) during NNNS attention items (1-3m) and infant interactions (6-12m). For all interactions, only one object at a time will be presented. Our established training/reliability procedures ensure high research rigor.

Behavioral Coding of Infant Interactive Behavior. The occurrence of infant interactive behaviors will be coded from video by masked RAs (Aim 2). Dr. Iverson (Co-I) is a renowned expert in capturing infant social¹¹³⁻¹¹⁶ and object¹¹⁷⁻¹¹⁹ interactive behavior in PTs^{120,121} and ASIBs^{114,117} and will assist with this process. Due to rapid development in social and object interaction skills from 6-12 months, behaviors will be categorized as "earlier" and "later" emerging (*Fig 7* above). This will allow our measures to account for maturation in interactive behaviors and avoid floor and ceiling effects at each time point. For example, we expect no infants show "later" social acts at 6m (words and gestures) but many infants show "earlier" social acts at 6m and 12m (social smiles).

HDA. The PI and Co-Is O'Reilly and Richards developed customized Matlab and Python scripts to synchronize heart rate (HR) with infant looking to calculate onsets and offset of HDA phases (sustained attention defined as 5 successive beats with HR slower than pre-look HR median^{22,23,92}). Summary variables calculated for analysis will include duration of sustained attention (%SA) and heart rate deceleration during sustained attention; these have been associated with enhanced learning¹²² and will serve as the primary DVs (Aims 1-2).

Covariates and Demographic Data. Sociodemographic data (race, ethnicity, income, education, home address) and medical history (pregnancy, birth, and early postnatal period) will be collected at the first visit. For PTs, morbidity scores¹²³ will be derived from medical records and in-depth NICU questionnaire (APGAR scores, overall time in NICU, presence/duration of medical supports (e.g., ventilator), neurological damage (IVH), infections, and surgeries). Due to the potential impact of NICU-related stimulation and stress on cardiorespiratory functions, these variables will be considered as covariates in analyses. Co-I Dail has expertise in evaluating and selecting relevant variables. Annual questionnaires about parental concerns, developmental milestones (ASQ-3¹²⁴), and any early intervention services will be collected. Given the impact of maternal depression on early social development, mothers will complete a depression measure (EPDS¹²⁵) at months 1, 3, and 6. The PI and project psychologist have the clinical expertise to evaluate and address clinical depression and will refer mothers for appropriate care. Motor questionnaires¹²⁶ will be completed at each visit to control for impact of motor delays on attention and interactive behavior. Outcome cognitive and motor skills will serve as covariates in all analyses.

Data Management. Management of ECG, video, questionnaires, and behavioral assessments will be led by the Project Manager. Undergraduates will be trained in cleaning and editing of raw ECG files (CardioEdit, CardioBatch), video files will be recorded to SD cards and immediately transferred to our secure server. REDCap will be used for parent questionnaires and to store behavioral data (using double data entry/validation). Original forms will be scanned, stored on a secure server, and shredded to protect participant confidentiality. Experimental data will be processed and linked to demographic/behavioral data for statistical analysis.

SPECIFIC AIM MEASURES AND ANALYSES

Aim 1. Compare the maturation of autonomic regulation of attention in the early infant period (1-3m) across ASIBs, PTs, and TDs.

Overview and Hypotheses. Behavioral measures of attention at 2-3 months reveal significant associations with ASD/ASD symptoms^{6,7,57}, but biological underpinnings have not been identified. Behavioral measures of looking coordinated with heart rate deceleration reflect focused attention, brain arousal system activation, and enhanced learning^{78,122,127}. Thus, HDA in early infancy reflects physiological regulation required to attend to, and learn from, the environment. **Aim 1 will directly test the hypothesis that the maturation of autonomic regulation of attention in young infants from 1-3 months is delayed for PTs and disrupted for ASIBs.**

Measures. Baseline autonomic function will be measured with RSA extracted from a 5-min baseline recording. Autonomic regulation of attention will be measured during the six Attention items of the NNNS-II (see *Data Collection Overview* above). Attention to each stimulus will be coded from video and temporally synchronized with ECG data to calculate: %time in heart-defined sustained attention (%SA = time in SA / task duration) and heart rate deceleration in sustained attention (HR-decel = maxHR–minHR during each SA phase).

Analytic Plan. We will utilize latent growth curve models (LGCM) to characterize trajectories of HDA (%SA; HRdec) as well as baseline autonomic function (RSA) from 1-3 months. LGCMs can be considered comparable to multilevel models and, in many cases, they are numerically identical, however there are several specific advantages to LGCM approaches¹²⁸. For a single manifest variable measured at a minimum of three time points, an LGCM can be fit to estimate two latent factors, i.e., intercept and slope. This approach is established within a structural equation model (SEM) framework, which allows us to take advantage of the flexibility and broadness of SEM models with the following added benefits: 1) ability to assess global model fit^{128,129}, 2) easy handling of missing data within and between visits using the full information maximum likelihood strategy¹³⁰, 3) inclusion of latent predictor(s)/outcome(s) in model^{128,129,131}. Recent research has confirmed that this technique can be applied to small sample sizes with missing data¹³⁰. We will first plot individual trajectories of HDA (%SA, HRdec) to explore differential patterns. The model fit will be determined using chi square tests and fit indices, such as the root mean square error of approximation and the comparative fit index. Mean values of growth factors (intercept, slope) will be estimated and compared across groups (TD, PT, ASIB). We will also examine the variance of the growth factors to test whether individuals share the same starting points and rate of change over time. Covariance between growth factors will provide information about individuals' growth rates related to their starting point. Effects of baseline or outcome characteristics (e.g., gestational age at birth, NICU-related morbidity, biological sex, SES, cognitive outcomes) will be controlled for in all analyses. This is critical as a substantial minority of preterm infants also show cognitive delays¹³². **Power Analysis.** Estimated power for growth curve models was calculated following a simulation-based approach (1000 simulation replicates)³⁴. Power rates were computed by assuming 3 measurement occasions and linear growth. Population values used for the simulation were obtained by fitting the linear latent class models (LCM) to our pilot data. Given 200 subjects, power rates to detect individual variability in intercepts and slopes were 0.99 and 0.99, respectively. To detect group differences in trajectories, we computed the power to detect a significant difference in terms of mean slopes between two groups (n=75, n=50) and estimated power was 0.81. Finally, given N=200, we can detect a medium effect ($R^2=0.25$) when predicting continuous outcomes (ASD severity) with 86% power.

Expected Outcomes and Interpretation. Due to global ANS dysfunction at birth, **PTs** will show reduced baseline RSA and autonomic regulation of attention (less %SA and less HRdec) from 1-3 months compared to **TDs** ($PT_{1-3mo} < TD_{1-3mo}$). Due to *intact* global ANS functioning for **ASIBs** at birth, baseline RSA for **ASIBs** will be comparable to **TDs** from 1-3 months. **ASIB** HDA, on the other hand, will be comparable to **TDs** at 1 month ($ASIB_{1mo} = TD_{1mo}$), but as attention and heart rate become increasingly coordinated over time, **ASIBs** will diverge from **TDs** by 3 months ($ASIB_{3mo} < TD_{3mo}$), similar to our behavioral work^{6,57}. Due to the hypothesized attention-specific ANS deficit for **ASIBs**, HDA will be reduced for **ASIBs** by 3 months ($ASIB_{3mo} < PT_{3mo} < TD_{3mo}$).

Aim 2. Determine the moment-to-moment influence of autonomic regulation of attention on interactive behavior across the first year (6m, 9m, 12m).

Overview and Hypotheses. **ASIBs**^{26,117,133,134} and **PTs**^{135–137} follow unique developmental trajectories of interactive behavior. We have three hypotheses regarding the predictive association between HDA and in-the-moment interactive behaviors: 1) In alignment with our pilot data, **TDs** will show strong, consistent patterns of HDA that predict interactive behaviors, 2) **PTs**' initially immature ANS will cause initially weak predictive associations that improve over time as ANS matures, 3) **ASIBs** will show disrupted associations that weaken (worsen) over time as ASD symptoms emerge. In addition, baseline ANS function trajectories will be compared.

Measures. Interactive behavior will be recorded at 6, 9, and 12 months using a standardized protocol (see above, *Infant Interaction Task*, Fig 7). The onset of each interactive behavior will be recorded and synchronized with time series HDA to examine, with fine temporal resolution, the influence of HDA on interactive behavior. Because motor

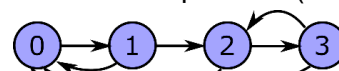


Figure 8. Possible transitions between HDA phases.

differences/delays may influence the frequency/function of interactive behavior¹³⁸, visit-specific motor questionnaires¹²⁶ will be included as a covariate.

Analytic Plan. We will use HDA times series locked to behavior onsets to study the dynamics of the autonomic attentional system leading up to interactive behavior. We will compare modeling capability of the following alternative approaches: **(1) Continuous-time Markov chain (CTMC):** Our previous work has employed Markov chains to understand dynamics of social attention in toddlers with ASD¹³⁹. A CTMC describes a continuous stochastic process in which the process changes state according to an exponential random variable, moving to a different state with probabilities specified for a Markovian process. We will use this approach to model the process of changing from one HDA phase to the next when moving backward in time, starting from a given action. The structure of the model is fixed given four HDA phases (not looking, orienting, sustained attention, termination) and includes one time constant per state and seven transition probabilities (see Fig 8), for a total of 11 parameters. These parameters will be estimated per subject and the distribution of these parameters will be compared between groups. Subject-specific parameters will be estimated in a Bayesian framework using the Markov Chain Monte Carlo (MCMC) approach. **(2) Simple historical mean:** We will compute the inter-beat-interval (IBI) historical mean and corresponding confidence intervals leading up to a behavior. These trajectories will be used to assess group differences in the historical mean and in its time derivatives (speed, acceleration, jerk) to assess group differences in autonomic-attention-action system dynamics. Cluster-based permutation tests will be used to determine periods of time showing statistical differences between groups while accounting for multiple comparisons. This approach is considered as a simpler approach to offer a benchmark for the other, more complex, approach, and previous studies show that simpler approaches often yield better outcomes¹⁴⁰. **Model Comparison.** We will evaluate the effect size obtained with these approaches to determine which one best captures the dynamics of HDA leading to marked behaviors and extract relevant features (e.g., fitted modeling parameters, historical means over temporal windows, rate of decrease in predictability) as potential biomarkers for ASD symptoms (see Aim 3). **Power Analysis.** Between-group comparisons for the different parameters estimated (i.e., CTMC parameters, IBI means/derivatives over time windows) will be entered into growth models to evaluate this longitudinal data. We therefore expect the same statistical power as Aim 1: range of .81-.99 to detect a medium effect for N=200.

Expected Outcomes and Interpretation: We hypothesize that: (1) baseline autonomic function (RSA) from 6-9 months will be reduced in **PTs** compared to **ASIBs** and **TDs**, but show gradual improvement over time; (2) **ASIB** RSA will be largely similar to **TDs** (similar to previous work showing later-emerging RSA disruption in ASD⁵⁸). In terms of HDA-behavior coordination, **PTs** (global ANS dysfunction at birth) will show initial reduced HDA-behavior coordination, and therefore weak HDA-behavior associations, compared to **TDs**. But the effect size of this group difference will decrease over time ($PT_{6mo} < TD_{6mo} \rightarrow PT_{12mo} \sim TD_{12mo}$). **ASIBs** (hypothesized developmental cascade wherein attention-specific ANS deficits are present early, by 3 months, and contribute to disrupted emergence of interactive behavior⁸) will show increasingly weak associations of HDA with interactive behaviors compared to **TDs** ($ASIB_{6mo} < TD_{6mo} \rightarrow ASIB_{12mo} \ll TD_{12mo}$). **ASIBs'** autonomic-attention-behavior system dynamics may be initially comparable to **PTs** ($ASIB_{6mo} = PT_{6mo} < TD_{6mo}$) due to prolonged ANS dysfunction for **PTs**. But by 12 months, ASIBs will significantly diverge ($ASIB_{12mo} < PT_{12mo} \sim TD_{12mo}$).

Aim 3. Predict ASD symptoms at age 3 years from infant autonomic and attentional features.

Overview and Hypotheses. Univariate statistical methods for biomarker discovery may be inadequate for capturing the complexity of biological processes involved in a multifaceted, heterogeneous neurodevelopmental disorder²⁷. Machine learning (ML) provides unique tools for examining relationships binding multiple interacting variables²⁸. Many biomarker studies of ASD (blood, metabolic) have implications for elucidating pathophysiology and improving diagnostic accuracy^{141–144}, but the lack of conceptual or functional links from **biomarker to behavior** prevents direct translation to behavioral interventions. This aim will incorporate biobehavioral measures of baseline autonomic function (RSA) and autonomic regulation of attention (HDA) to predict ASD symptoms. We hypothesize features collected from 1-12 months will predict ASD symptoms at age 3 years.

Measures. We will use a ML approach considering as features the variables and parameters estimated for Aims 1-2 (regardless of whether the impact of the variables has reach statistical significance for the specific analyses proposed in those aims), as well as simple additional statistics (e.g., overall time in HDA phases) and features established in our preliminary studies (i.e., statistics of heart rate variability, discrete functional analysis, sample entropy from pre-processed ECG with empirical mode decomposition and discrete wavelet transform). As in our previous work⁵⁷, ASD symptoms (36m) will be measured using the ADOS-2 Calibrated Severity Scores.

Analytic Plan. To optimize the use of our sample while precluding overfitting and ensuring reproducibility on new data, performance regarding ASD symptom prediction will be cross-validated using a leave-one-subject-out approach. These cross-validated predictions will be compared against the traditional regression approach (i.e.,

with no test-train split) to assess the degree of overfitting that would result from not splitting the dataset. If overfitting is negligible, this model will be considered final. Else, we will identify the subjects for which the cross-validation prediction is significantly different from the non-cross-validated model, reject the subjects, and re-test. Previous studies using ML in ASD with children and infants concentrated on diagnosis classification. They used various classification methods (support vector machine, k-nearest neighbors, naive Bayes, random forest) and obtained accuracy roughly in the 0.75-0.85 range^{145–147}. These studies vary widely with respect to the properties of their sample (e.g., large differences in sample size), their constructs (analysis of electroencephalogram versus spoken language), their objectives (diagnosis in infants versus in children), and their methods in general. As preliminary results, we showed that ANS properties assessed through ECG can be used for prediction by obtaining an 86% accuracy for the classification of ASD risk (ASIB vs TD) using a multilayer perceptron. In this proposal, we will use similar **deep learning models in the context of regression**⁹⁶ **to predict continuous phenotypic outcomes** (autism symptomology; ADOS CSS scores). Continuous outcomes will be assessed through the goodness of fit of regression models (e.g., coefficient of determination, R^2). **Power Analysis.** Sufficient power in our sample is supported by our preliminary demonstration of high predictivity of ECG-derived features for capturing ASD likelihood (classification accuracy=.86 with only $N=23$). Clinical significance, rather than mere statistical significance, requires large effect sizes (e.g., large R^2). As opposed to p-values, effect sizes do not depend on power and sample size. Reliability of R^2 estimates will be determined using confidence intervals computed with bootstrapping. For a sample size of 200 infants, a single predictor (derived from the combination of many features), and an $R^2=0.5$, the expected 95% confidence interval (CI) goes from 0.40 to 0.60 (for $R^2=0.6$, 95% CI = .51-.68). Thus, this study is well-sampled for reasonably narrow CIs on at least medium effect sizes.

Aim 3 Expected Outcomes and Interpretation: We hypothesize that early autonomic-attentional features will predict ASD symptoms that do not emerge until age 2-3 years. Further, through ablation study, we will examine which features from which time points (e.g., 1-3mos vs. 6-12mos) contribute most to predictability. We expect predictability to be highest for attention-specific features and to increase with age.

RIGOR AND REPRODUCIBILITY

Rigor and reproducibility of this project is apparent in our prospective longitudinal design with masked examiners, monitoring of inter-rater reliability/agreement (>80%) for experimental and clinical measures, consideration of potential confounds (GA, medical morbidities, sex), sample size selected based on a priori power analyses, and an investigative team with expertise in grant management/physiology (Roberts, Richards), ANS/prematurity (Dail), infant behavior (Iverson), and signal processing, machine learning, computational neuroscience (O'Reilly).

Potential Problems, Alternative Strategies, & Benchmarks for Success. This project will be initiated nearly upon funding as we have existing recruitment infrastructure and a team trained in most experimental and clinical measures. Activities will occur as follows: recruitment Y1-2, data collection/processing Y1-5, data analyses Y2-5, publications Y3-5 (see *Human Subjects-Study Timeline*). **Recruitment and attrition** are potential concerns for any longitudinal study. Considering past success of the PI and Co-Is in neonatal PT and ASIB recruitment, a plan to see participants in their home statewide, a large research staff, and significant resources dedicated to recruitment, we do not anticipate any difficulty with meeting enrollment targets. We will access institutional funds to support recruitment/retention, dedicate a fulltime recruitment coordinator solely to this project, utilize multiple state-specific autism databases, and employ retention strategies (frequent contact with families through social media and clinical feedback). It is highly possible that **examiners may be unmasked to participant group** during home visits. We mitigate this by: attempting to schedule visits when siblings are out of the home, reminding families to keep examiners masked to family history, collecting data on when examiners are unmasked (to explore in analytic models), process data using masked coders/psychologists (ECG data, attention data, behavioral data, ADOS scores). Our statistical design is **robust to within and between-visit missing data**, we will use state-of-the-art ECG artifact corrections using wavelet and empirical mode decomposition, and we rely on refined longitudinal modeling (i.e., growth curves). In order to ensure robust findings, we will cross-validate our results by employing additional analytic strategies (e.g., mixed effects regression, general linear models) when appropriate. **GA in PTs:** While recent evidence suggests that ANS dysfunction relates more to morbidity than GA¹²³, we will stratify PT recruitment by three evenly divided birth GA bins (all <32w) to ensure sufficient ANS immaturity. **Significant phenotypic heterogeneity** is expected in ASIB and PT groups. Therefore, comprehensive characterization of ASD and developmental profiles at 36m for all participants will allow for analyses to extend beyond between-group comparisons (Aims 1-2) to include key covariates or stratifiers that can parse out heterogeneity and reveal complex, meaningful interactions. **Aim 3 is not dependent on Aims 1-2** as significant results are not a requisite for including these variables in feature selection procedures (i.e., failure for a test to reach statistical significance does not prove the absence of useful predictive information) and, in fact, may elucidate novel findings above and beyond Aims 1-2.

PHS Human Subjects and Clinical Trials Information

OMB Number: 0925-0001

Expiration Date: 09/30/2024

Use of Human Specimens and/or Data

Does any of the proposed research in the application involve human specimens and/or data *

☒ Yes

☐ No

Provide an explanation for any use of human specimens and/or data not considered to be human subjects research.

Are Human Subjects Involved

☒ Yes

☐ No

Is the Project Exempt from Federal regulations?

☐ Yes

☒ No

Exemption Number

☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

☐ 6

☐ 7

☐ 8

Other Requested Information

Human Subject Studies

Study#	Study Title	Clinical Trial?
<u>1</u>	The Role of Autonomic Regulation of Attention in the Emergence of ASD	No

Section 1 - Basic Information (Study 1)

1.1. Study Title *

The Role of Autonomic Regulation of Attention in the Emergence of ASD

1.2. Is this study exempt from Federal Regulations *

☐ Yes ☒ No

1.3. Exemption Number

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8

1.4. Clinical Trial Questionnaire *

1.4.a. Does the study involve human participants?

☒ Yes ☐ No

1.4.b. Are the participants prospectively assigned to an intervention?

☐ Yes ☒ No

1.4.c. Is the study designed to evaluate the effect of the intervention on the participants?

☐ Yes ☒ No

1.4.d. Is the effect that will be evaluated a health-related biomedical or behavioral outcome?

☐ Yes ☒ No

1.5. Provide the ClinicalTrials.gov Identifier (e.g. NCT87654321) for this trial, if applicable

Section 2 - Study Population Characteristics (Study 1)

2.1. Conditions or Focus of Study

- Infant siblings of children with autism spectrum disorder
- Infants born preterm
- Infant siblings of typically developing children

2.2. Eligibility Criteria

This study aims to examine autonomic regulation of attention in infant siblings of children with autism spectrum disorder (ASIBs; n = 75) in comparison to preterm infants (PT; n = 75) and infant siblings of typically developing children (TD; n = 50). Participating infants will be assessed across development at ages 1, 2, 3, 6, 9, 12, 24, and 36 months.

ASIBs are eligible if they are full biological siblings of children diagnosed with ASD. ASD diagnosis of older siblings will be verified through a clinical best estimate diagnosis based on record review, documentation of direct clinical measures (i.e., Autism Diagnostic Observation Schedule, ADOS-2) by a licensed medical doctor or psychologist, and autism-specific questionnaires completed at study entry (Social Responsiveness Questionnaire (SRS-2) and Social Communication Questionnaire (SCQ)). If direct clinical measures are not evident, the study team will verify the presence of ASD using the ADOS-2 and developmental measures, followed by a clinical best estimate consensus.

TD infants are eligible if they have a full biological typically developing sibling (confirmed with SCQ, SRS-2, and expert clinical judgment) and no family history of ASD in 1st-3rd degree relatives.

PT infants are born preterm with a gestational age (GA) between 24 weeks, 0 days - 32 weeks, 6 days. PTs will have a sibling and no family history of ASD in 1st-3rd degree relatives. PTs will be evenly stratified into 3 GA bins (n=25/bin) to ensure even distribution of risk for autonomic nervous system dysfunction: (24-26w; 27-29w; 30-32w).

Exclusion criteria for all participants include: failed newborn hearing screen, known major hearing or visual impairments, non-febrile seizure disorders, known genetic/chromosomal syndromes (e.g., Down Syndrome), congenital heart condition known to effect heart rate, and severe brain hemorrhage and periventricular leukomalacia (PVL). PTs will provide medical records for accurate eligibility assessment. To maximize recruitment efforts of PTs, we are not excluding twins (but multiple birth will be considered in analyses).

2.3. Age Limits	Min Age: 1 Days	Max Age: 42 Months
2.3.a. Inclusion of Individuals Across the Lifespan	HumanSubjects_InclusionAcrossLifespan_BradshawR01.pdf	
2.4. Inclusion of Women and Minorities	HumanSubjects_InclusionWomenMinorities_BradshawR01.pdf	
2.5. Recruitment and Retention Plan	HumanSubjects_RecruitmentRetention_BradshawR01.pdf	
2.6. Recruitment Status	Not yet recruiting	
2.7. Study Timeline	HumanSubjects_Timeline_BradshawR01.pdf	
2.8. Enrollment of First Participant	08/01/2023	Anticipated

INCLUSION OF INDIVIDUALS ACROSS THE LIFESPAN

This study aims to examine the role of infant autonomic regulation of attention in the emergence of ASD. To meet this aim, 200 infants will be recruited across three groups: infant siblings of children with ASD (ASIBs, $n = 75$), preterm infants (PTs, $n = 75$), and infant siblings of typically developing children (TDs, $n = 50$). Participants will be assessed at eight time points (1, 2, 3, 6, 9, 12, 24, and 36 months) across development. Repeated assessments within the first three years of life allow for in-depth and nuanced assessment of attention and autonomic regulation development throughout infancy, thus allowing for the investigation of early ASD biomarkers prior to the age at which ascertainment of a stable ASD diagnosis is achievable (3 years). Using this approach will enable sophisticated longitudinal modeling while staying within the five-year time span of this project and will increase our understanding of the neurobiological underpinnings of attention disruption associated with ASD symptoms. Further, identification of earlier biomarkers of ASD has the potential to inform early interventions and targeted treatments that can improve long-term outcomes.

The PI and Co-Is have decades of experience conducting research with infants and children, especially those at an elevated likelihood for ASD, and thus have finely-tuned assessment procedures in place to keep children and their families happy, comfortable, and engaged in the proposed research measures. Further, all research staff who will be interacting with participating families and administering study protocols will have a background in child development, complete knowledge of the research protocol, and will be trained and closely supervised by the PI.

These procedures will be adapted or expanded upon as needed for the proposed study. As such, data collection procedures will be family-friendly and developmentally appropriate with survey questionnaire and interview completion, as well as direct assessment, scheduled to be both efficient and flexible based on each family's need. The length of direct assessment generally increases as the child ages, from 15-20 minutes in the first year to 3 hours at age 3 years. The 36-month timepoint can be completed in two study visits. Each visit will allow ample time for breaks, naps, feeding, etc., to ensure participant comfort and reduce participant fatigue. Data will be collected at each family's home for the family's convenience. The investigative team's long history of conducting in-home study visits suggests very high feasibility and family acceptability of this approach. All assessments will be scheduled to be efficient but flexible per family preference by offering accommodability in scheduling (e.g., evenings and weekends).

INCLUSION OF WOMEN AND MINORITIES

INCLUSION OF WOMEN

A total of 200 infants will be included in this study, distributed across three groups: infant siblings of children with autism spectrum disorder (ASIBs; $n = 75$), preterm infants (PT; $n = 75$), and infant siblings of typically developing children (TD; $n = 50$). In accordance with NIH guidelines for consideration of relevant biological variables, we will recruit an approximately equal number of males and females for each group. Additionally, while not the primary participant in this project, the biological mothers of each child will participate; thus, 200 women will participate in this study.

INCLUSION OF MINORITIES

Autism spectrum disorder (ASD) occurs in all known ethnic and racial groups, and therefore, our aim is for subject recruitment to be representative of the ethnic and racial distribution of South Carolina, within which recruitment and assessment are expected to occur. The 2020 U.S. Census Bureau Report estimates the demographics of South Carolinians as follows: 63.3% White, 25.0% Black/African American, 1.8% Asian, 0.5% American Indian and Alaska Native, 0.1% Native Hawaiian and Other Pacific Islander, 5.8% more than one race, and 3.5% Other. Local recruitment efforts will reflect state census figures and are the basis for the projections noted in the Inclusion Enrollment Report. We stringently aim to recruit a participant sample across groups that is representative of the local population, and as such, we aim for no more than 60% of our sample to be White. However, we recognize that barriers exist in recruiting under-represented minorities for research and will interface with psychologists, physicians, and agencies in the community to address these barriers. These efforts will be bolstered by our full-time Recruitment and Outreach Coordinator. The PI and Co-I Roberts have established relationships with a variety of organizations within South Carolina which will broaden outreach efforts. Furthermore, by offering in-home visits and flexibility in scheduling, we aim to minimize potential socioeconomic disparities in recruitment. In-home data collection minimizes the cost and burden associated with travel and increases accessibility for families living in rural areas. Moreover, we will offer flexibility in scheduling times, including weekends and evenings, to accommodate work schedules. Strategies to recruit participants from minority groups have been successful in the PI's previous studies (current enrollment is 40% non-white) and we anticipate ascertaining participation across racial and socioeconomic groups is highly feasible. However, we will require English proficiency of all participants due to funding limitations related to multilingual staff and challenges in translating all instruments for non-English speakers. Families of any ethnicity who speak English will not be excluded from the sample.

RECRUITMENT AND RETENTION PLAN

This study proposes 200 infants distributed across three groups: infant siblings of children with autism spectrum disorder (ASIBs; $n = 75$), preterm infants (PT; $n = 75$), and infant siblings of typically developing children (TD; $n = 50$). Participants will be assessed at 1, 2, 3, 6, 9, 12, 24, and 36 months of age.

Given that study visits will begin at 1 month of age, we have a narrow window during which to recruit participants, necessitating strong collaborations with community organizations, doctors' offices, state agencies, and clinicians who work with families and expectant mothers with children with ASD and typically developing children and with mothers of preterm infants. Participant recruitment and community outreach will be supported and operated by the full-time Recruitment and Outreach Coordinator who will help plan stakeholder engagement activities, liaise with organizational partners, and organize communication between the investigative team and community partners. The Recruitment and Outreach Coordinator will work with research staff and trained graduate and undergraduate students to interface with community contacts, screen potential participants, and schedule initial assessments.

Recruitment efforts will capitalize on existing infrastructure that bolsters existing collaborations and community networks. Participants will be recruited locally and across South Carolina through:

- 1) Community Provider Outreach Efforts: The PI and Co-Is have established collaborations with numerous community organizations, clinics, medical centers, parent support groups, and childcare centers, many of which have provided letters of support for recruitment efforts. Organizations that have agreed to support ASIB recruitment efforts include the South Carolina Department of Disabilities and Special Needs (DDSN), Prisma Health Children's Hospital Developmental Pediatrics, Prisma Health Children's Hospital NICU, Medical University of South Carolina Developmental Pediatrics, Palmetto Pediatrics & Adolescent Clinic, South Carolina Tri-County Pediatrics, BrightStart Early Intervention, Unumb Autism and Neurodevelopment Center, and the South Carolina Simons Powering Autism Research (SPARK) study.
- 2) The Director of Neonatology at Prisma Health and investigator at USC College of Nursing have provided support for PT recruitment efforts (see letters of support). The Prisma NICU admits ~200 within the eligible gestational age window of this study (<32 weeks). Further outreach efforts will include promoting the study at community and family events (e.g., Columbia's weekly farmer's market, the South Carolina State Fair, Autism Speaks events and walks). Significant funds have been allotted to design, print, and distribute research flyers to advertise to community partners.
- 3) Web-based advertisement: To bolster infant recruitment, we will advertise the study online on the PIs' respective websites and social media pages. We will also advertise online through the social media pages of community organizations and parent support groups supportive of our research, and as needed, will pay for online advertisements.
- 4) USC research registries: The USC has spearheaded the development of numerous research registry databases, including the Carolina Autism and Neurodevelopment (CAN) Center Research Registry and the University of South Carolina Infants and Children Research Registry. Both of these registries were designed by PI's Recruitment and Outreach Coordinator (Elizabeth Dixon) who has recruited over 1,100 families to the registry. Dixon is the named Recruitment and Outreach Coordinator on this proposal. The PI will have access to the contact information of potential participants registered in these databases, which enroll ~30 families monthly. Further, the PI can request periodic emails to families in these databases regarding research openings and how to self-refer.

The proposed research study is longitudinal, with participants being assessed at eight time points (seven of those in-person). Participating families are likely at an elevated risk of parental stress associated with the diagnosis and care of their child(ren) with ASD and/or the medical stress associated with preterm infants; thus, we will implement strategies that will not only ensure the collection of high-quality data but will enhance participants' willingness to participate and remain in the study. As the recruitment, retention, and assessment of our participants will require a great deal of effort and expertise, we will train a number of project staff to address these challenges. Additionally, we will facilitate the development of a personal relationship with our recruited families by assigning two research staff to conduct each assessment. To enhance continuity, one RA will be designated to all participant scheduling and at least one RA who has previously assessed the family will be scheduled to see the family at the next visit. This continuity will facilitate the personal relationships that are core to the successful retention of participants over the course of longitudinal studies.

The study will also capitalize on the PI's previous and existing protocols that support participant retention. As such, data collection procedures will be "family-friendly" with survey questionnaires, interview completion, and direct assessment scheduled to be both efficient and flexible based on each family's need. Data collection will occur in-home, which we hope will remove the potential burden associated with travel and assessment in an unfamiliar space. Furthermore, the PI's lab has established several free resources to support parents participating in research. These include a mindfulness-based stress reduction group and a respite care program that support families by: (1) providing an opportunity for parents to meet, connect, and share resources with each other, (2) delivering free respite care services at least one Saturday a month, and (3) creating a space for children to play and socialize with their peers and trained research staff.

The Recruitment and Outreach Coordinator (Elizabeth Dixon) will monitor recruitment and retention rates closely, as she has done the past year for our ongoing NIH-funded longitudinal study. Examples of efforts to bolster retention include (1) soliciting continued feedback on the informed consent process and research protocols through participation satisfaction surveys and (2) promoting community engagement by organizing information sessions, workshops, and outreach events for families and providers within the autism community.

Given the collective success of the PI and investigative team in recruiting ASIB, PT, and TD infants, and our established recruitment/retention infrastructure and statewide community connections, we do not anticipate issues meeting our recruitment goals. The PI's lab consistently receives 30-50 referrals per week from interested research participants and, when actively recruiting, the PI and Co-I (Roberts) have collective enrollment rates of 12-18 participants/month. This track record demonstrates the feasibility of successful enrollment for the proposed project. Further, Bradshaw and Roberts have over 10 years and 20 years, respectively, of experience conducting highly similar research protocols with great success. Based on the PIs' previous longitudinal studies utilizing similar designs (~5% attrition across all groups, ASIBs, PTs, TDs, for 3-year longitudinal studies when infants recruited before age 3 months), we anticipate minimal attrition.

STUDY TIMELINE

Progress and benchmarks for this project will be assessed weekly. In general, activities will occur as follows: recruitment Y1-2, data collection/processing Y1-5, data analyses Y2-5, and publications Y3-5. Specifically relevant to the work described in this study, we anticipate the following timeline.

- **Research Staff Training.** We anticipate having three already-trained research coordinators ready to begin the project upon its start date. In the event that this is not the case, staff will be hired immediately with training ensuing right away. Many training manuals for heart rate and behavioral data collection have already been developed. Current experimental protocols and training manuals will be modified to align with the experimental measures proposed in this study. A fourth research coordinator will be hired in Year 2 of the award to assist with data collection. This individual will begin training immediately. While training is likely an ongoing effort, the primary effort will take place in the first 1-2 years when staff are initially hired.
- **Subject Recruitment** will begin immediately, led by the Recruitment and Outreach Coordinator who is already working with the lab (Elizabeth Dixon). Undergraduate students will be recruited immediately to assist with recruitment tasks and potential participant screenings. Recruitment materials will be developed immediately and outreach to our many recruitment partners will occur as soon as we are notified of a high likelihood of funding. This will ensure a quick start to recruitment. Recruitment meetings with the PI, Recruitment and Outreach Coordinator, and the Project Manager will occur weekly for the first three years of the project.
- **Data Collection** will likely begin within two months of the start of the award as we plan to update existing experimental manuals and protocols and recruit participants immediately. The first two months will include training with pilot participants to ensure high fidelity of experimental protocol and clinical assessment administration. Throughout data collection (all 5 years of the project), reliability will be monitored by the project psychologist. Research staff will be masked to participant group during data collection. The occurrence of unintentional unmasking will be recorded for each study visit and monitored by the project psychologist.
- **Data Processing** will be an ongoing process with reliability standards established for data entry and verification as one of the first steps. Research staff will train undergraduate students in data cleaning and processing procedures, including data entry into REDCap, ECG data cleaning (removing noise and artifacts), and behavioral coding of attention and infant behavior. Undergraduate research assistants who clean and edit ECG data and who code infant behavior from video will be masked to participant group. Data processing reliability will be monitored monthly for the duration of the award. Research staff will be trained in running data through our customized data processing pipeline to output heart defined attention. Beginning in the last quarter of Year 1 of the award, descriptive statistics for dependent variables will be generated monthly for data quality checks. Quality checks will also include evaluation of the frequency and impact of unintentional unmasking of examiners during study visits. When unmasking occurs, a masked examiner will score assessments (e.g., ADOS) from video. These monthly data checks will occur monthly throughout the duration of the award.
- **Data Analysis** will begin with monthly descriptive of ongoing data collection and data quality checks as an initial data cleaning step. Subsequently, data analysis will occur on an ongoing basis from Years 2-5 of the award. Data analysis will be led by the PI with assistance provided by the postdoctoral fellow to be hired in Year 4 of the project.
- **Manuscript preparation and submission** will begin with publications of early preliminary data in Year 3 of the award. Increasingly sophisticated analyses will be forthcoming as the postdoctoral fellow is hired and longitudinal outcome data is acquired.

Activities	Year 1	Year 2	Year 3	Year 4	Year 5
Recruitment and Staff Training					
Data Collection and Processing					
Data Analyses					
Publications					

2.9. Inclusion Enrollment Reports

IER ID#	Enrollment Location Type	Enrollment Location
<u>Study 1, IER 1</u>	Domestic	South Carolina

Inclusion Enrollment Report 1

1. Inclusion Enrollment Report Title* : The Role of Autonomic Regulation of Attention: Inclusion Enrollment Report
2. Using an Existing Dataset or Resource* : ☐ Yes ☒ No
3. Enrollment Location Type* : ☒ Domestic ☐ Foreign
4. Enrollment Country(ies): USA: UNITED STATES
5. Enrollment Location(s): South Carolina
6. Comments:

Planned

Racial Categories	Ethnic Categories				Total
	Not Hispanic or Latino		Hispanic or Latino		
	Female	Male	Female	Male	
American Indian/ Alaska Native	2	2	0	0	4
Asian	4	3	0	0	7
Native Hawaiian or Other Pacific Islander	0	1	0	0	1
Black or African American	24	25	0	1	50
White	58	57	4	4	123
More than One Race	5	5	2	3	15
Total	93	93	6	8	200

Cumulative (Actual)

Racial Categories	Ethnic Categories									Total
	Not Hispanic or Latino			Hispanic or Latino			Unknown/Not Reported Ethnicity			
	Female	Male	Unknown/ Not Reported	Female	Male	Unknown/ Not Reported	Female	Male	Unknown/ Not Reported	
American Indian/ Alaska Native	0	0	0	0	0	0	0	0	0	0
Asian	0	0	0	0	0	0	0	0	0	0
Native Hawaiian or Other Pacific Islander	0	0	0	0	0	0	0	0	0	0
Black or African American	0	0	0	0	0	0	0	0	0	0
White	0	0	0	0	0	0	0	0	0	0
More than One Race	0	0	0	0	0	0	0	0	0	0
Unknown or Not Reported	0	0	0	0	0	0	0	0	0	0
Total	0	0	0	0	0	0	0	0	0	0

Section 3 - Protection and Monitoring Plans (Study 1)

3.1. Protection of Human Subjects

HumanSubjects_Protection_BradshawR01.pdf

3.2. Is this a multi-site study that will use the same protocol to conduct non-exempt human subjects research at more than one domestic site?

☐ Yes ☒ No ☐ N/A

Single IRB plan attachment

3.3. Data and Safety Monitoring Plan

3.4. Will a Data and Safety Monitoring Board be appointed for this study?

☐ Yes ☒ No

3.5. Overall structure of the study team

PROTECTION OF HUMAN SUBJECTS

1. Risks to Human Subjects

A. Human Subjects Involvement, Characteristics and Design

The proposed study is a five-year longitudinal study focused on examining autonomic regulation of attention in infant siblings of children with autism spectrum disorder (ASIBs; $n = 75$), preterm infants (PT; $n = 75$), and typically developing infants (TD, $n = 50$). More specifically, this study aims to determine specificity of autonomic regulation of attention as a key, early emerging feature of ASD and evaluate its predictive value and developmental consequences for interactive behaviors and ASD symptoms. To meet these aims, children will be assessed at eight timepoints across development from 1 month to 36 months of age, allowing for nuanced assessment of the development and consequences of autonomic regulation of attention in infants at an elevated likelihood of ASD compared to PT infants who experience global ANS dysfunction from birth, but largely typical outcomes. TD infants will be included as a second comparison group for which to study normative development of ANS and autonomic regulation of attention.

As previously detailed, infants will be recruited throughout South Carolina through (1) local and statewide parent support groups, community events, and service delivery units (i.e., private and public schools and childcare centers); (2) web-based advertisement, including a project website and Facebook account that will outline the study and solicit contact information from interested parties; and (3) the University of South Carolina's research registries. Recruitment efforts will be bolstered through a fulltime Recruitment and Outreach Coordinator and existing collaborations with community partners (see letters of support). At the time of recruitment, all procedures, as well as the potential risks and benefits, will be explained to participating families.

ASIBs are eligible if they are full biological siblings of children diagnosed with ASD. ASD diagnosis of older siblings will be verified through a clinical best estimate diagnosis based on record review, documentation of direct clinical measures (i.e., Autism Diagnostic Observation Schedule, ADOS-2) by a licensed medical doctor or psychologist, and autism-specific questionnaires completed at study entry (Social Responsiveness Questionnaire (SRS-2) and Social Communication Questionnaire (SCQ)). If direct clinical measures are not evident, the study team will verify the presence of ASD using the ADOS-2 and developmental measures, followed by a clinical best estimate consensus.

TD infants are eligible if they have a full biological typically developing sibling (confirmed with SCQ, SRS-2, and expert clinical judgment) and no family history of ASD in 1st-3rd degree relatives.

PT infants are born preterm with a gestational age (GA) between 24 weeks, 0 days - 32 weeks, 6 days. PTs will have a sibling and no family history of ASD in 1st-3rd degree relatives. PTs will be evenly stratified into 3 GA bins ($n=25/\text{bin}$) to ensure even distribution of risk for autonomic nervous system dysfunction: (24-26w; 27-29w; 30-32w).

Exclusion criteria for all participants include: failed newborn hearing screen, known major hearing or visual impairments, non-febrile seizure disorders, known genetic/chromosomal syndromes (e.g., Down Syndrome), congenital heart condition known to effect heart rate, and severe brain hemorrhage and periventricular leukomalacia (PVL). PTs will provide medical records for accurate eligibility assessment. To maximize recruitment efforts of PTs, twins will not be excluded (but multiple birth will be considered in analyses).

B. Study Procedures, Materials, and Potential Risks

Several sources of research materials will be collected at study visits, including (1) developmental testing of the child; (2) behavioral ratings from examiners; (3) questionnaires from caregivers; (4) heart activity; and (5) video recordings. Study visits are video recorded for offline behavioral coding and reliability assessments. Demographic information (i.e., sex, race, and ethnicity) and identifiable information (i.e., name, address, and date of birth) will be collected on all participants. However, only core research staff on the study's Institutional Review Board (IRB) who have completed ethics and compliance training will have access to identifiable information. The collection of identifiable information is necessary to maintain contact with participants throughout this longitudinal study and to set up repeated assessments.

There is no more than minimal risk anticipated in this study. Although there is a potential risk for breach of confidentiality with the research data, the anticipated risk is minimal, and every effort will be made to safeguard participants' confidentiality. Additionally, there is minimal risk anticipated for the collection of behavioral data and biomarkers, although participants may experience mild fatigue or fussiness. The length of direct child assessment generally increases as the child ages, from 15-30 minutes (1-12 month time points) to 3 hours at the 36-month timepoint. Longer assessments can be completed across two study visits. All study visits will be scheduled at each family's convenience (e.g., weekends, evenings) and frequent breaks (e.g., naps, feedings) will be allotted as needed.

PT infants and infant siblings of children with ASD are at an elevated risk of developmental delays and ASD, but there is the chance that families will be unsuspecting of the occurrence of developmental delay or ASD. Similarly, there is a small chance that a "TD" infant will, unbeknownst to the parents, meet the profile of developmental delay or ASD. The PI and staff psychologist will review annual screeners and questionnaires (Ages and Stages Questionnaire-3 and parent concerns questionnaire) for all participants (ASIBs, PTs, TDs) to screen for possible developmental delays. Participants who score below the ASQ-3 cutoff in any area (Communication, Fine Motor, Gross Motor, Problem Solving, Personal-Social) will be contacted by the PI or psychologist to describe the results and refer to our 0-3 state agency (BabyNet) for further evaluation and possibly intervention. Participants who score in the ASQ-3 "monitoring zone" in any area will be contacted to describe the results and gather further information. Based on additional information provided by the parent, these participants will either be referred to BabyNet for further evaluation or re-screened in 6 months. Regardless of ASQ-3 results, all families who report developmental concerns on annual questionnaires or during any study visits will be contacted to determine if a referral is warranted.

2. Adequacy of Protection Against Risks

A. Informed Consent and Assent

In accordance with policy 45 CFR 46.408, informed consent will be obtained from at least one parent of all children prior to data collection. Trained research staff will review the consent form with parents, allowing time for the parents to review the consent and ask any questions. Parents who choose to proceed will sign a paper copy of the consent form, which will be retained in a secure office space in a locked cabinet. Parents will also receive a paper copy of the consent form for their records.

B. Protections Against Risk

The proposed research protocol will be approved by the IRB at the University of South Carolina prior to the attainment of data, and approval will be obtained for any subsequent modifications. Participant confidentiality will be safeguarded by assigning all subjects a unique identification number, not linked with any identifiable information, which will be associated with all raw data collected. Identifiable information will be kept separate from all raw data during transport and will be stored separately. All personal, identifying information will be stored in a locked filing cabinet in a locked office or on a secure server in a password-protected database. Only core research staff on the study's IRB who have completed research ethics and compliance training will have access to identifiable information.

Data collection for all eight timepoints will be overseen by the PI. Research Coordinators, who have at least a bachelor's degree in psychology, child development, or a related field, will work with the PI and project psychologist to be trained on data collection with sensitive populations. Data collection will occur in the family's home to maximize participant comfort, thus minimizing the potential of stress from travel and anxiety over unfamiliar surroundings. Furthermore, to mitigate the potential for fatigue and fussiness during data collection, all study visits will be scheduled to allot for frequent breaks during the assessment. Visits will also be scheduled to occur at each family's convenience to accommodate individual schedules, including work, nap times, and intervention services. We will also make every effort to minimize discomfort during the collection of biomarkers. For example, heart activity will be recorded using standard research electrocardiogram electrodes that are disposable and placed on intact skin. The PI and project psychologist will be available to discuss any questions or concerns that arise during any study visit.

C. Vulnerable Subjects

As noted above, the rationale for including infants between 1 and 36 months of age is to assess autonomic regulation of attention in early development to identify this measure as a key, early-emerging feature of ASD. The PI has over 14 years of experience conducting research with infants and young children with ASD. The proposed study follows highly similar procedures to the PIs' past and current studies on development,

attention, and physiology, which have occurred without incident. Given the success of previous study procedures and feedback, we anticipate no more than minimal risk for this study given the study team's experience with children.

3. Potential Benefits of the Proposed Research to Research Participants and Others

There are no direct benefits from this study for participants and their families. The primary anticipated benefit is close monitoring of infant development by licensed clinical psychologists. Parents in the PI's current, ongoing studies have expressed deep appreciation for this monitoring and feedback. Parents will receive diagnostic information and a report if their child develops ASD over the course of the study and parents may view prospective surveillance and access to a diagnostic report as beneficial. We expect this study to advance research on the etiology of ASD and our hope is that families will find it satisfying to contribute to increasing the scientific knowledge in this domain and the potential for advancement in early intervention options. The PI makes valiant efforts to ensure recent findings are available to the families that participate in their studies through their website, newsletter, and Facebook page, providing both links to publications and family-friendly summaries. Although participants are not expected to personally benefit from the results of this study, the projected potential benefits to the autism community outweigh the minimal risks associated with the study.

4. Importance of the Knowledge to be Gained

Increased knowledge about the earliest emergence of disrupted autonomic regulation of attention in ASIBs and its relation to ASD symptoms will increase knowledge of the etiology of ASD and help lead to targeted interventions and support for families that can optimize outcomes. The results from the proposed study are expected to yield high-impact results that promote a more comprehensive understanding of the role of autonomic function in the daily lives of children with ASD.

Section 4 - Protocol Synopsis (Study 1)**4.1. Study Design****4.1.a. Detailed Description****4.1.b. Primary Purpose****4.1.c. Interventions**

Type	Name	Description
------	------	-------------

4.1.d. Study Phase

Is this an NIH-defined Phase III Clinical Trial? ☐ Yes ☐ No

4.1.e. Intervention Model

4.1.f. Masking ☐ Yes ☐ No

☐ Participant ☐ Care Provider ☐ Investigator ☐ Outcomes Assessor

4.1.g. Allocation**4.2. Outcome Measures**

Type	Name	Time Frame	Brief Description
------	------	------------	-------------------

4.3. Statistical Design and Power**4.4. Subject Participation Duration**

4.5. Will the study use an FDA-regulated intervention? ☐ Yes ☐ No

4.5.a. If yes, describe the availability of Investigational Product (IP) and Investigational New Drug (IND)/ Investigational Device Exemption (IDE) status

4.6. Is this an applicable clinical trial under FDAAA? ☐ Yes ☐ No

4.7. Dissemination Plan

Delayed Onset Studies

Delayed Onset Study#	Study Title	Anticipated Clinical Trial?	Justification
The form does not have any delayed onset studies			

REFERENCES

1. Ness, S. L., Bangerter, A., Manyakov, N. V., Lewin, D., Boice, M., Skalkin, A., Jagannatha, S., Chatterjee, M., Dawson, G., Goodwin, M. S., Hendren, R., Leventhal, B., Shic, F., Frazier, J. A., Janvier, Y., King, B. H., Miller, J. S., Smith, C. J., Tobe, R. H. & Pandina, G. An Observational Study With the Janssen Autism Knowledge Engine (JAKE®) in Individuals With Autism Spectrum Disorder. *Front Neurosci* **0**, 111 (2019).
2. Murias, M., Major, S., Davlantis, K., Franz, L., Harris, A., Rardin, B., Sabatos-DeVito, M. & Dawson, G. Validation of eye-tracking measures of social attention as a potential biomarker for autism clinical trials. *Autism Research* **11**, 166–174 (2018).
3. Bradshaw, J., Shic, F., Holden, A. N., Horowitz, E. J., Barrett, A. C., German, T. C. & Vernon, T. W. The Use of Eye Tracking as a Biomarker of Treatment Outcome in a Pilot Randomized Clinical Trial for Young Children with Autism. *Autism Research* **12**, 779–793 (2019).
4. McPartland, J. C. Developing Clinically Practicable Biomarkers for Autism Spectrum Disorder. *J Autism Dev Disord* **47**, 2935–2937 (2017).
5. Di Giorgio, E., Frasnelli, E., Rosa Salva, O., Luisa Scattoni, M., Puopolo, M., Tosoni, D., Apicella, F., Gagliano, A., Guzzetta, A., Molteni, M., Persico, A., Pioggia, G., Valeri, G., Vicari, S., Simion, F. & Vallortigara, G. Difference in Visual Social Predispositions Between Newborns at Low- and High-risk for Autism. *Sci Rep* **6**, 26395 (2016).
6. Bradshaw, J., Klin, A., Evans, L., Klaiman, C., Saulnier, C. & McCracken, C. Development of attention from birth to 5 months in infants at risk for autism spectrum disorder. *Dev Psychopathol* **32**, 491–501 (2020).
7. Jones, W. & Klin, A. Attention to eyes is present but in decline in 2-6-month-old infants later diagnosed with autism. *Nature* **504**, 427–31 (2013).
8. Bradshaw, J., Schwichtenberg, A. J. & Iverson, J. M. Capturing the complexity of autism: Applying a developmental cascades framework. *Child Dev Perspect* **16**, 18–26 (2022).
9. Feldman, R. From biological rhythms to social rhythms: Physiological precursors of mother-infant synchrony. *Dev Psychol* **42**, 175–188 (2006).
10. Brazelton, T. B. & Nugent, J. *The Neonatal Behavioral Assessment Scale*. (Mac Keith Press, 2011).
11. Porges, S. W. & Furman, S. A. The early development of the autonomic nervous system provides a neural platform for social behaviour: A polyvagal perspective. *Infant Child Dev* **20**, 106–118 (2011).
12. Cheng, Y. C., Huang, Y. C. & Huang, W. L. Heart rate variability in individuals with autism spectrum disorders: A meta-analysis. *Neuroscience and Biobehavioral Reviews* **118**, 463–471 (2020).
13. Patural, H., Barthelemy, J. C., Pichot, V., Mazzocchi, C., Teyssier, G., Damon, G. & Roche, F. Birth prematurity determines prolonged autonomic nervous system immaturity. *Clinical Autonomic Research* **2004 14:6 14**, 391–395 (2004).
14. Yiallourou, S. R., Witcombe, N. B., Sands, S. A., Walker, A. M. & Horne, R. S. C. The development of autonomic cardiovascular control is altered by preterm birth. *Early Hum Dev* **89**, 145–152 (2013).
15. Haraldsdottir, K., Watson, A. M., Goss, K. N., Beshish, A. G., Pegelow, D. F., Palta, M., Tetri, L. H., Barton, G. P., Brix, M. D., Centanni, R. M. & Eldridge, M. W. Impaired autonomic function in adolescents born preterm. *Physiol Rep* **6**, e13620 (2018).
16. Agrawal, S., Rao, S. C., Bulsara, M. K. & Patole, S. K. Prevalence of Autism Spectrum Disorder in Preterm Infants: A Meta-analysis. *Pediatrics* **142**, (2018).
17. Ozonoff, S., Young, G. S., Carter, A., Messinger, D., Yirmiya, N., Zwaigenbaum, L., Bryson, S., Carver, L. J., Constantino, J. N., Dobkins, K., Hutman, T., Iverson, J. M., Landa, R., Rogers, S. J., Sigman, M. & Stone, W. L. Recurrence risk for autism spectrum disorders: a Baby Siblings Research Consortium study. *Pediatrics* **128**, e488-95 (2011).
18. Reynolds, G. D. & Richards, J. E. Infant heart rate: A developmental psychophysiological perspective. in *Developmental psychophysiology: Theory, systems, and methods* (eds. Schmidt, L. A. & Segalowitz, S. J.) 173–212 (Cambridge University Press, 2008).
19. Goodman, M. S., Castro, N., Sloan, M., Sharma, R., Widdowson, M., Herrera, E. & Pineda, J. A. A Neurovisceral Approach to Autism: Targeting Self-Regulation and Core Symptoms Using Neurofeedback and Biofeedback. *NeuroRegulation* **5**, 9–9 (2018).

20. Porges, S. W. Autonomic regulation and attention. in *Attention and Information Processing in Infants and Adults: Perspectives from Human and Animal Research* (eds. Campbell, B., Hayne, H. & Richardson, R.) 201–203 (Lawrence Erlbaum Associates, Publishers, 1992).
21. Lansink, J. M. & Richards, J. E. Heart Rate and Behavioral measures of Attention in Six-, Nine-, and Twelve-Month-Old Infants during Object Exploration. *Child Dev* **68**, 610–620 (1997).
22. Richards, J. E. & Casey, B. J. Heart Rate Variability During Attention Phases in Young Infants. *Psychophysiology* **28**, 43–53 (1991).
23. Bradshaw, J., Vinyard, A., Platt, E. & Abney, D. Physiological Measures of Attention and Arousal During Dyadic Interactions as Predictors of Social Communication and ASD Symptoms. in *Multi-level dyadic pathways influence infants' emotion regulation across prenatal and postnatal development* (International Congress for Infant Studies, 2022).
24. Tonnsen, B. L., Richards, J. E. & Roberts, J. E. Heart rate-defined sustained attention in infants at risk for autism. *J Neurodev Disord* **10**, 7 (2018).
25. Mulkey, S. B., Govindan, R. B., Hitchings, L., Al-Shargabi, T., Herrera, N., Swisher, C. B., Eze, A., Russo, S., Schlatterer, S. D., Jacobs, M. B., McCarter, R., Kline, A., Maxwell, G. L., Baker, R. & du Plessis, A. J. Autonomic nervous system maturation in the premature extrauterine milieu. *Pediatric Research* 2020 89:4 **89**, 863–868 (2020).
26. Bradshaw, J., McCracken, C., Pileggi, M., Brane, N., Delehanty, A., Day, T., Federico, A., Klaiman, C., Saulnier, C., Klin, A. & Wetherby, A. Early social communication development in infants with autism spectrum disorder. *Child Dev* **92**, 2224–2234 (2021).
27. Huynh-Thu, V. A., Saeys, Y., Wehenkel, L. & Geurts, P. Statistical interpretation of machine learning-based feature importance scores for biomarker discovery. *Bioinformatics* **28**, 1766–1774 (2012).
28. He, Z. & Yu, W. Stable feature selection for biomarker discovery. *Comput Biol Chem* **34**, 215–225 (2010).
29. Buescher, A. V. S., Cidav, Z., Knapp, M. & Mandell, D. S. Costs of Autism Spectrum Disorders in the United Kingdom and the United States. *JAMA Pediatr* **168**, 721 (2014).
30. Zuvekas, S. H., Grosse, S. D., Lavelle, T. A., Maenner, M. J., Dietz, P. & Ji, X. Healthcare Costs of Pediatric Autism Spectrum Disorder in the United States, 2003–2015. *J Autism Dev Disord* **51**, 2950–2958 (2021).
31. Bradshaw, J., Steiner, A. M., Gengoux, G. & Koegel, L. Feasibility and Effectiveness of Very Early Intervention for Infants At-Risk for Autism Spectrum Disorder: A Systematic Review. *J Autism Dev Disord* **45**, 778–794 (2015).
32. Cidav, Z., Munson, J., Estes, A., Dawson, G., Rogers, S. & Mandell, D. Cost Offset Associated With Early Start Denver Model for Children With Autism. *J Am Acad Child Adolesc Psychiatry* **56**, 777–783 (2017).
33. van 't Hof, M., Tisseur, C., van Berckeleer-Onnes, I., van Nieuwenhuyzen, A., Daniels, A. M., Deen, M., Hoek, H. W. & Ester, W. A. Age at autism spectrum disorder diagnosis: A systematic review and meta-analysis from 2012 to 2019. *Autism* **25**, 862–873 (2021).
34. Maenner, M. J., Shaw, K. A., Baio, J., Washington, A., Patrick, M., DiRienzo, M., Christensen, D. L., Wiggins, L. D., Pettygrove, S., Andrews, J. G., Lopez, M., Hudson, A., Baroud, T., Schwenk, Y., White, T., Rosenberg, C. R., Lee, L. C., Harrington, R. A., Huston, M., Hewitt, A., Esler, A., Hall-Lande, J., Poynter, J. N., Hallas-Muchow, L., Constantino, J. N., Fitzgerald, R. T., Zahorodny, W., Shenouda, J., Daniels, J. L., Warren, Z., Vehorn, A., Salinas, A., Durkin, M. S. & Dietz, P. M. Prevalence of autism spectrum disorder among children aged 8 Years-Autism and developmental disabilities monitoring network, 11 Sites, United States, 2016. *MMWR Surveillance Summaries* **69**, 1–12 (2020).
35. Clark, M. L. E., Vinen, Z., Barbaro, J. & Dissanayake, C. School Age Outcomes of Children Diagnosed Early and Later with Autism Spectrum Disorder. *J Autism Dev Disord* **48**, 92–102 (2018).
36. Vivanti, G. & Dissanayake, C. Outcome for Children Receiving the Early Start Denver Model Before and After 48 Months. *J Autism Dev Disord* **46**, 2441–2449 (2016).
37. Dawson, G. Early behavioral intervention, brain plasticity, and the prevention of autism spectrum disorder. *Dev Psychopathol* **20**, 775–803 (2008).

38. Kaliukhovich, D. A., Manyakov, N. V., Bangerter, A., Ness, S., Skalkin, A., Goodwin, M. S., Dawson, G., Hendren, R. L., Leventhal, B., Hudac, C. M., Bradshaw, J., Shic, F. & Pandina, G. Social attention to activities in children and adults with autism spectrum disorder: effects of context and age. *Molecular Autism* 2020 11:1 **11**, 1–14 (2020).
39. McPartland, J. C., Bernier, R. A., Jeste, S. S., Dawson, G., Nelson, C. A., Chawarska, K., Earl, R., Faja, S., Johnson, S. P., Sikich, L., Brandt, C. A., Dziura, J. D., Rozenblit, L., Hellemann, G., Levin, A. R., Murias, M., Naples, A. J., Platt, M. L., Sabatos-DeVito, M., Shic, F., Senturk, D., Sugar, C. A., Webb, S. J. & Trials, the A. B. C. for C. The Autism Biomarkers Consortium for Clinical Trials (ABC-CT): Scientific Context, Study Design, and Progress Toward Biomarker Qualification. *Front Integr Neurosci* **0**, 16 (2020).
40. Del Valle Rubido, M., Hollander, E., McCracken, J. T., Shic, F., Noeldeke, J., Boak, L., Khwaja, O., Sadikhov, S., Fontoura, P. & Umbricht, D. Exploring Social Biomarkers in High-Functioning Adults with Autism and Asperger's Versus Healthy Controls: A Cross-Sectional Analysis. *Journal of Autism and Developmental Disorders* 2020 50:12 **50**, 4412–4430 (2020).
41. Frye, R. E., Vassall, S., Kaur, G., Lewis, C., Karim, M. & Rossignol, D. Emerging biomarkers in autism spectrum disorder: a systematic review. *Ann Transl Med* **7**, 792–792 (2019).
42. Constantino, J. N., Charman, T. & Jones, E. J. H. Clinical and Translational Implications of an Emerging Developmental Substructure for Autism. doi.org/10.1146/annurev-clinpsy-081219-110503 **17**, 365–389 (2021).
43. Karmel, B. Z., Gardner, J. M., Meade, L. S., Cohen, I. L., London, E., Flory, M. J., Lennon, E. M., Miroshnichenko, I., Rabinowitz, S., Parab, S., Barone, A. & Harin, A. Early medical and behavioral characteristics of NICU infants later classified with ASD. *Pediatrics* **126**, 457–67 (2010).
44. Shic, F., Macari, S. & Chawarska, K. Speech Disturbs Face Scanning in 6-Month-Old Infants Who Develop Autism Spectrum Disorder. *Biol Psychiatry* **75**, 231–237 (2014).
45. Chawarska, K., Macari, S. & Shic, F. Decreased Spontaneous Attention to Social Scenes in 6-Month-Old Infants Later Diagnosed with Autism Spectrum Disorders. *Biol Psychiatry* **74**, 195–203 (2013).
46. Jones, E. J. H., Venema, K., Earl, R., Lowy, R., Barnes, K., Estes, A., Dawson, G. & Webb, S. J. Reduced engagement with social stimuli in 6-month-old infants with later autism spectrum disorder: a longitudinal prospective study of infants at high familial risk. *J Neurodev Disord* **8**, 7 (2016).
47. Gao, W., Alcauter, S., Elton, A., Hernandez-Castillo, C. R., Smith, J. K., Ramirez, J. & Lin, W. Functional Network Development During the First Year: Relative Sequence and Socioeconomic Correlations. *Cerebral Cortex* **25**, 2919–2928 (2015).
48. Klin, A., Shultz, S. & Jones, W. Social visual engagement in infants and toddlers with autism: Early developmental transitions and a model of pathogenesis. *Neurosci Biobehav Rev* **50**, 189–203 (2015).
49. Perdue, K. L., Edwards, L. A., Tager-Flusberg, H. & Nelson, C. A. Differing Developmental Trajectories in Heart Rate Responses to Speech Stimuli in Infants at High and Low Risk for Autism Spectrum Disorder. *J Autism Dev Disord* **47**, 2434–2442 (2017).
50. Lester, B. M., Boukydis, C. F. Z. & LaGasse, L. Cardiorespiratory Reactivity During the Brazelton Scale in Term and Preterm Infants. *J Pediatr Psychol* **21**, 771–783 (1996).
51. Richards, J. E., Reynolds, G. D. & Courage, M. L. The Neural Bases of Infant Attention. *Curr Dir Psychol Sci* **19**, 41–46 (2010).
52. Richards, J. E. The development of sustained attention in infants. in *Cognitive neuroscience of attention* (ed. Posner, M. I.) 342–356 (Guilford Press, 2004).
53. Xie, W. & Richards, J. E. Effects of interstimulus intervals on behavioral, heart rate, and event-related potential indices of infant engagement and sustained attention. *Psychophysiology* **53**, 1128–42 (2016).
54. Reynolds, G. D., Guy, M. W. & Zhang, D. Neural Correlates of Individual Differences in Infant Visual Attention and Recognition Memory. *Infancy* **16**, 368–391 (2011).
55. Colombo, J. The Development of Visual Attention in Infancy. *Annu Rev Psychol* **52**, 337–367 (2001).
56. Macari, S., Milgramm, A., Reed, J., Shic, F., Powell, K. K., Macris, D. & Chawarska, K. Context-Specific Dyadic Attention Vulnerabilities During the First Year in Infants Later Developing Autism Spectrum Disorder. *J Am Acad Child Adolesc Psychiatry* (2020). [doi:10.1016/j.jaac.2019.12.012](https://doi.org/10.1016/j.jaac.2019.12.012)

57. Bradshaw, J., Shi, D., Hendrix, C. L., Saulnier, C. & Klaiman, C. Neonatal neurobehavior in infants with autism spectrum disorder. *Dev Med Child Neurol* **64**, 600–607 (2022).
58. Sheinkopf, S. J., Levine, T. P., McCormick, C. E. B., Puggioni, G., Conradt, E., Lagasse, L. L. & Lester, B. M. Developmental trajectories of autonomic functioning in autism from birth to early childhood. *Biol Psychol* **142**, 13–18 (2019).
59. Hogan, A., Hunt, E., Smith, K., Black, C., Bangert, K., Klusek, J. & Roberts, J. Trajectories of Heart Activity Across Infancy to Early Childhood Differentially Predict Autism and Anxiety Symptoms in Fragile X Syndrome. *Front Psychiatry* **12**, (2021).
60. Mulkey, S. B. & Plessis, A. d. The Critical Role of the Central Autonomic Nervous System in Fetal-Neonatal Transition. *Semin Pediatr Neurol* **28**, 29–37 (2018).
61. Patural, H., Pichot, V., Jaziri, F., Teyssier, G., Gaspoz, J. M., Roche, F. & Barthelemy, J. C. Autonomic cardiac control of very preterm newborns: A prolonged dysfunction. *Early Hum Dev* **84**, 681–687 (2008).
62. Mulkey, S. B. & du Plessis, A. J. Autonomic nervous system development and its impact on neuropsychiatric outcome. *Pediatric Research* **2018** *85*:2 **85**, 120–126 (2018).
63. Hendry, A., Johnson, M. H. & Holmboe, K. Early Development of Visual Attention: Change, Stability, and Longitudinal Associations. *Annu Rev Dev Psychol* **1**, 251–275 (2019).
64. Gardner, J. M., Karmel, B. Z. & Flory, M. J. Arousal modulation of neonatal visual attention: Implications for development. in *Visual information processing* (eds. Soraci, S. & Murata-Soraci, K.) 125–153 (Praeger Publishers/Greenwood Publishing Group, 2003).
65. De Barbaro, K., Johnson, C. M., Forster, D. & Deak, G. O. Methodological considerations for investigating the microdynamics of social interaction development. *IEEE Trans Auton Ment Dev* **5**, 258–270 (2013).
66. Wass, S. V., Clackson, K. & Barbaro, K. de. Temporal dynamics of arousal and attention in 12-month-old infants. *Dev Psychobiol* **58**, 623–639 (2016).
67. Wass, S. V., Noreika, V., Georgieva, S., Clackson, K., Brightman, L., Nutbrown, R., Covarrubias, L. S. & Leong, V. Parental neural responsivity to infants' visual attention: How mature brains influence immature brains during social interaction. *PLoS Biol* **16**, e2006328 (2018).
68. Billeci, L., Tonacci, A., Narzisi, A., Manigrasso, Z., Varanini, M., Fulceri, F., Lattarulo, C., Calderoni, S. & Muratori, F. Heart Rate Variability During a Joint Attention Task in Toddlers With Autism Spectrum Disorders. *Front Physiol* **0**, 467 (2018).
69. Watson, L., Roberts, J., ... G. B.-J. of A. and & 2012, undefined. Behavioral and physiological responses to child-directed speech of children with autism spectrum disorders or typical development. *Springer* (2011). doi:10.1007/s10803-011-1401-z
70. de Barbaro, K., Clackson, K. & Wass, S. V. Infant attention is dynamically modulated with changing arousal levels. *Child Dev* **88**, 629–639 (2017).
71. Dormal, V., Vermeulen, N. & Mejias, S. Is heart rate variability biofeedback useful in children and adolescents? A systematic review. *Journal of Child Psychology and Psychiatry* (2021). doi:10.1111/JCPP.13463
72. Lehrer, P. M. & Gevirtz, R. Heart rate variability biofeedback: how and why does it work? *Front Psychol* **0**, 756 (2014).
73. Koegel, L. K., Koegel, R. L., Ashbaugh, K. & Bradshaw, J. The importance of early identification and intervention for children with or at risk for autism spectrum disorders. *Int J Speech Lang Pathol* **16**, 50–56 (2014).
74. Rogers, S., Vismara, L., Wagner, A. L., McCormick, C., Young, G. & Ozonoff, S. Autism Treatment in the First Year of Life: A Pilot Study of Infant Start, a Parent-Implemented Intervention for Symptomatic Infants. *J Autism Dev Disord* **44**, 2981–2995 (2014).
75. Bradshaw, J., Trumbull, A., Stapel-Wax, J., Gillespie, S., George, N., Saulnier, C., Klaiman, C., Woods, J., Call, N., Klin, A. & Wetherby, A. Factors associated with enrollment into a clinical trial of caregiver-implemented intervention for infants at risk for autism spectrum disorder. doi.org/10.1177/1362361320928829 **24**, 1874–1884 (2020).
76. Koegel, L., Singh, A., Koegel, R., Hollingsworth, J. & Bradshaw, J. Assessing and Improving Early Social Engagement in Infants. *J Posit Behav Interv* **16**, 69–80 (2014).

77. Yoder, P. J., Stone, W. L. & Edmunds, S. R. For Which Younger Siblings of Children with ASD Does Parent-Mediated Intervention Work? *Autism* **25**, 58 (2021).
78. Conte, S. & Richards, J. Attention in Early Development. *Oxford Research Encyclopedia of Psychology* (2021). doi:10.1093/ACREFORE/9780190236557.013.52
79. Feldman, R. & Eidelman, A. I. Skin-to-skin contact (Kangaroo Care) accelerates autonomic and neurobehavioural maturation in preterm infants. *Dev Med Child Neurol* **45**, 274–281 (2003).
80. Murias, M., Major, S., Davlantis, K., Franz, L., Harris, A., Rardin, B., Sabatos-DeVito, M. & Dawson, G. Validation of eye-tracking measures of social attention as a potential biomarker for autism clinical trials. *Autism Research* **11**, 166–174 (2018).
81. Schneider, U., Schleussner, E., Fiedler, A., Jaekel, S., Liehr, M., Haueisen, J. & Hoyer, D. Fetal heart rate variability reveals differential dynamics in the intrauterine development of the sympathetic and parasympathetic branches of the autonomic nervous system. *Physiol Meas* **30**, 215 (2009).
82. De Rogalski Landrot, I., Roche, F., Pichot, V., Teyssier, G., Gaspoz, J. M., Barthelemy, J. C. & Patural, H. Autonomic nervous system activity in premature and full-term infants from theoretical term to 7 years. *Autonomic Neuroscience* **136**, 105–109 (2007).
83. Mathewson, K. J., Lieshout, R. J. Van, Saigal, S., Morrison, K. M., Boyle, M. H. & Schmidt, L. A. Autonomic Functioning in Young Adults Born at Extremely Low Birth Weight: doi.org/10.1177/2333794X15589560 **2**, 2333794X1558956 (2015).
84. Pascal, A., Govaert, P., Oostra, A., Naulaers, G., Ortibus, E. & Van den Broeck, C. Neurodevelopmental outcome in very preterm and very-low-birthweight infants born over the past decade: a meta-analytic review. *Dev Med Child Neurol* **60**, 342–355 (2018).
85. Kerstjens, J. M., De Winter, A. F., Bocca-Tjeertes, I. F., Ten Vergert, E. M. J., Reijneveld, S. A. & Bos, A. F. Developmental Delay in Moderately Preterm-Born Children at School Entry. *J Pediatr* **159**, 92–98 (2011).
86. Smith, L. B. & Thelen, E. Development as a dynamic system. *Trends Cogn Sci* **7**, 343–348 (2003).
87. Elsabbagh, M., Fernandes, J., Jane Webb, S., Dawson, G., Charman, T., Johnson, M. H. & British Autism Study of Infant Siblings Team. Disengagement of Visual Attention in Infancy is Associated with Emerging Autism in Toddlerhood. *Biol Psychiatry* **74**, 189–194 (2013).
88. Ellison, J. T., Paterson, S. J., Wolff, J. J., Reznick, J. S., Sasson, N. J., Gu, H., Botteron, K. N., Dager, S. R., Estes, A. M., Evans, A. C., Gerig, G., Hazlett, H. C., Schultz, R. T., Styner, M., Zwaigenbaum, L. & Piven, J. White matter microstructure and atypical visual orienting in 7-month-olds at risk for autism. in *American Journal of Psychiatry* **170**, 899–908 (American Psychiatric Association, 2013).
89. Macari, S., Milgramm, A., Reed, J., Shic, F., Powell, K. K., Macris, D. & Chawarska, K. Context-Specific Dyadic Attention Vulnerabilities During the First Year in Infants Later Developing Autism Spectrum Disorder. *J Am Acad Child Adolesc Psychiatry* (2020). doi:10.1016/j.jaac.2019.12.012
90. Roberts, J. E., Bradshaw, J., Will, E., Hogan, A. L., McQuillin, S. & Hills, K. Emergence and rate of autism in fragile X syndrome across the first years of life. *Dev Psychopathol* **32**, 1335–1352 (2020).
91. Richards, J. E. Sustained visual attention in 8-week-old infants. *Infant Behav Dev* **12**, 425–436 (1989).
92. Bradshaw, J. & Abney, D. H. Infant physiological activity and the early emergence of social communication. *Dev Psychobiol* **63**, e22145 (2021).
93. JE, R. Baseline respiratory sinus arrhythmia and heart rate responses during sustained visual attention in preterm infants from 3 to 6 months of age. *Psychophysiology* **31**, 235–243 (1994).
94. Federico, A., Platt, E., Dea, A. & Bradshaw, J. Differences in Object Manipulation in Infants at Low and Elevated Likelihood for Autism Spectrum Disorder. in *International Society for Autism Research Annual Meeting* (2022).
95. Anandhi, B., Jerriitta, S., Anusuya, I. G. & Das, H. Time Domain Analysis of Heart Rate Variability Signals in Valence Recognition for Children with Autism Spectrum Disorder (ASD). *IRBM* (2021). doi:10.1016/J.IRBM.2021.06.002
96. Simchoni, G. & Rosset, S. Using Random Effects to Account for High-Cardinality Categorical Features and Repeated Measures in Deep Neural Networks. in *35th Conference on Neural Information Processing Systems (NeurIPS 2021)* (2021).

97. Gotham, K., Pickles, A. & Lord, C. Standardizing ADOS Scores for a Measure of Severity in Autism Spectrum Disorders. *J Autism Dev Disord* **39**, 693–705 (2009).
98. Shumway, S., Farmer, C., Thurm, A., Joseph, L., Black, D. & Golden, C. The ADOS Calibrated Severity Score: Relationship to Phenotypic Variables and Stability over Time. *Autism Research* **5**, 267–276 (2012).
99. Bieleninik, Ł., Posserud, M. B., Geretsegger, M., Thompson, G., Elefant, C. & Gold, C. Tracing the temporal stability of autism spectrum diagnosis and severity as measured by the Autism Diagnostic Observation Schedule: A systematic review and meta-analysis. *PLoS One* **12**, e0183160 (2017).
100. Rutter, M., Bailey, A. & Lord, C. *Social Communication Questionnaire (SCQ)*. (Western Psychological Services, 2003).
101. Achenbach, T. M. *Manual for the Child Behavior Checklist/2-3 and 1992 Profile*. (University of Vermont Department of Psychiatry, 1992).
102. Constantino, J. N. *Social Responsiveness Scale, Second Edition (SRS-2)*. (Western Psychological Services, 2012).
103. Ralphe, J. L., Silva, S. G., Dail, R. B. & Brandon, D. H. The Association Between Very Premature Infant Body Temperatures Over Time and Respiratory Care. *Biol Res Nurs* **23**, 331–340 (2021).
104. Provenzi, L., Olson, K., Giusti, L., Montirosso, R., Desantis, A. & Tronick, E. NICU Network Neurobehavioral Scale: 1-month normative data and variation from birth to 1 month. *Pediatr Res* **83**, 1104–1109 (2018).
105. Grossman, P. & Taylor, E. W. Toward understanding respiratory sinus arrhythmia: Relations to cardiac vagal tone, evolution and biobehavioral functions. *Biol Psychol* **74**, 263–285 (2007).
106. Brain-Body Center. CardioBatchPlus software. (2016).
107. Tronick, E. & Lester, B. M. *NICU Network Neurobehavioral Scale (NNS) Manual*. (Paul H. Brookes Publishing Company, 2005).
108. Wetherby, A. M. & Prizant, B. M. *Communication and symbolic behavior scales (CSBS)*. (Brookes Publishing, 2003).
109. Pileggi, M. L., Brane, N., Bradshaw, J., Delehanty, A., Day, T., McCracken, C., Stapel-Wax, J. & Wetherby, A. M. Early Observation of Red Flags in 12-Month-Old Infant Siblings Later Diagnosed With Autism Spectrum Disorder. **30**, 1846–1855 (2021).
110. Wiggins, L. D., Barger, B., Moody, E., Soke, G., Pandey, J. & Levy, S. Brief Report: The ADOS Calibrated Severity Score Best Measures Autism Diagnostic Symptom Severity in Pre-School Children. *J Autism Dev Disord* **49**, 2999–3006 (2019).
111. CamNTEch & Ltd. ActiHeart.
112. Brain-Body Center. CardioEdit software. *Brain-Body Center* (2007).
113. Iverson, J. M. & Goldin-Meadow, S. Gesture paves the way for language development. *Psychol Sci* **16**, 367–371 (2005).
114. Paradé, M. V. & Iverson, J. M. The Development of Coordinated Communication in Infants at Heightened Risk for Autism Spectrum Disorder. *J Autism Dev Disord* **45**, 2218–2234 (2015).
115. West, K. L. & Iverson, J. M. Communication changes when infants begin to walk. *Dev Sci* e13102 (2021). doi:10.1111/desc.13102
116. Iverson, J. M., Capirci, O. & Caselli, M. C. From communication to language in two modalities. *Cogn Dev* **9**, 23–43 (1994).
117. Koterba, E. A., Leezenbaum, N. B. & Iverson, J. M. Object exploration at 6 and 9 months in infants with and without risk for autism: doi.org/10.1177/1362361312464826 **18**, 97–105 (2012).
118. West, K. L. & Iverson, J. M. Language learning is hands-on: Exploring links between infants' object manipulation and verbal input. *Cogn Dev* **43**, 190–200 (2017).
119. Zuccarini, M., Guarini, A., Savini, S., Iverson, J. M., Aureli, T., Alessandrini, R., Faldella, G. & Sansavini, A. Object exploration in extremely preterm infants between 6 and 9 months and relation to cognitive and language development at 24 months. *Res Dev Disabil* **68**, 140–152 (2017).
120. Benassi, E., Savini, S., Iverson, J. M., Guarini, A., Caselli, M. C., Alessandrini, R., Faldella, G. & Sansavini, A. Early communicative behaviors and their relationship to motor skills in extremely preterm infants. *Res Dev Disabil* **48**, 132–144 (2016).

121. Zuccarini, M., Guarini, A., Savini, S., Iverson, J. M., Aureli, T., Alessandroni, R., Faldella, G. & Sansavini, A. Object exploration in extremely preterm infants between 6 and 9 months and relation to cognitive and language development at 24 months. *Res Dev Disabil* **68**, 140–152 (2017).
122. Richards, J. E. Effects of attention on infants' preference for briefly exposed visual stimuli in the paired-comparison recognition-memory paradigm. *Dev Psychol* **33**, 22–31 (1997).
123. Schlatterer, S. D., Govindan, R. B., Barnett, S. D., Al-Shargabi, T., Reich, D. A., Iyer, S., Hitchings, L., Larry Maxwell, G., Baker, R., du Plessis, A. J. & Mulkey, S. B. Autonomic development in preterm infants is associated with morbidity of prematurity. *Pediatric Research* **2021** 91:1 **91**, 171–177 (2021).
124. Schonhaut, L., Armijo, I., Schönstedt, M., Alvarez, J. & Cordero, M. Validity of the Ages and Stages Questionnaires in Term and Preterm Infants. *Pediatrics* **131**, e1468–e1474 (2013).
125. Cox, J. L., Holden, J. M. & Sagovsky, R. Detection of postnatal depression. Development of the 10-item Edinburgh Postnatal Depression Scale. *Br J Psychiatry* **150**, 782–6 (1987).
126. Libertus, K. & Landa, R. J. The Early Motor Questionnaire (EMQ): A parental report measure of early motor development. *Infant Behav Dev* **36**, 833–842 (2013).
127. Linnemeyer, S. A. & Porges, S. W. Recognition memory and cardiac vagal tone in 6-month-old infants. *Infant Behav Dev* **9**, 43–56 (1986).
128. Chou, C., Bentler, P. M. & Pentz, M. A. Comparisons of two statistical approaches to study growth curves: The multilevel model and the latent curve analysis. *Struct Equ Modeling* **5**, 247–266 (1998).
129. Preacher, K., Wichman, A., Briggs, N. & MacCallum, R. *Latent growth curve modeling*. (2008).
130. McNeish, D. Brief Research Report: Growth Models With Small Samples and Missing Data. *The Journal of Experimental Education* **86**, 690–701 (2018).
131. Duncan, T. E. & Duncan, S. C. An introduction to latent growth curve modeling. *Behav Ther* **35**, 333–363 (2004).
132. Cheong, J. L., Doyle, L. W., Burnett, A. C., Lee, K. J., Walsh, J. M., Potter, C. R., Treyvaud, K., Thompson, D. K., Olsen, J. E., Anderson, P. J. & Spittle, A. J. Association Between Moderate and Late Preterm Birth and Neurodevelopment and Social-Emotional Development at Age 2 Years. *JAMA Pediatr* **171**, e164805 (2017).
133. Libertus, K., Sheperd, K. A., Ross, S. W. & Landa, R. J. Limited Fine Motor and Grasping Skills in 6-Month-Old Infants at High Risk for Autism. *Child Dev* **85**, 2218–2231 (2014).
134. Srinivasan, S. M. & Bhat, A. N. Differences in means-end exploration between infants at risk for autism and typically developing infants in the first 15 months of life. *Dev Psychobiol* **61**, 203–215 (2019).
135. Soares, D. de A., von Hofsten, C. & Tudella, E. Development of exploratory behavior in late preterm infants. *Infant Behav Dev* **35**, 912–915 (2012).
136. Ruff, H. A., Saltarelli, L. M., Capozzoli, M. & Dubiner, K. The differentiation of activity in infants' exploration of objects. *Dev Psychol* **28**, 851–861 (1992).
137. Lobo, M. A., Kokkoni, E., Cunha, A. B. & Galloway, J. C. Infants Born Preterm Demonstrate Impaired Object Exploration Behaviors Throughout Infancy and Toddlerhood. *Phys Ther* **95**, 51–64 (2015).
138. Iverson, J. M. Developmental Variability and Developmental Cascades: Lessons From Motor and Language Development in Infancy. *Curr Dir Psychol Sci* **30**, 228–235 (2021).
139. Shic, F., Chawarska, K., Bradshaw, J. & Scassellati, B. Autism, eye-tracking, entropy. in *2008 7th IEEE International Conference on Development and Learning* 73–78 (IEEE, 2008). doi:10.1109/DEVLRN.2008.4640808
140. Green, K. C. & Armstrong, J. S. Simple versus complex forecasting: The evidence. *J Bus Res* **68**, 1678–1685 (2015).
141. Hewitson, L., Mathews, J. A., Devlin, M., Schutte, C., Lee, J. & German, D. C. Blood biomarker discovery for autism spectrum disorder: A proteomic analysis. *PLoS One* **16**, e0246581 (2021).
142. Ristori, M. V., Mortera, S. L., Marzano, V., Guerrera, S., Vernocchi, P., Ianaro, G., Gardini, S., Torre, G., Valeri, G., Vicari, S., Gasbarrini, A. & Putignani, L. Proteomics and Metabolomics Approaches towards a Functional Insight onto AUTISM Spectrum Disorders: Phenotype Stratification and Biomarker Discovery. *International Journal of Molecular Sciences* **2020**, Vol. 21, Page 6274 **21**, 6274 (2020).

143. Chaitra, N., Vijaya, P. A. & Deshpande, G. Diagnostic prediction of autism spectrum disorder using complex network measures in a machine learning framework. *Biomed Signal Process Control* **62**, 102099 (2020).
144. Oh, D. H., Kim, I. Bin, Kim, S. H. & Ahn, D. H. Predicting Autism Spectrum Disorder Using Blood-based Gene Expression Signatures and Machine Learning. *Clinical Psychopharmacology and Neuroscience* **15**, 47 (2017).
145. Nakai, Y., Takiguchi, T., Matsui, G., Yamaoka, N. & Takada, S. Detecting Abnormal Word Utterances in Children With Autism Spectrum Disorders: Machine-Learning-Based Voice Analysis Versus Speech Therapists. doi.org/10.1177/0031512517716855 **124**, 961–973 (2017).
146. Bosl, W., Tierney, A., Tager-Flusberg, H. & Nelson, C. EEG complexity as a biomarker for autism spectrum disorder risk. *BMC Medicine* **2011 9:1 9**, 1–16 (2011).
147. Maenner, M. J., Yeargin-Allsopp, M., Braun, K. V. N., Christensen, D. L. & Schieve, L. A. Development of a Machine Learning Algorithm for the Surveillance of Autism Spectrum Disorder. *PLoS One* **11**, e0168224 (2016).

CONSORTIUM/CONTRACTUAL ARRANGEMENTS

We will enter into a contractual agreement for this proposed R01 award with Dr. Jana Iverson at Boston University. Efforts for the Iverson lab are detailed in the *Budget Justification*. Dr. Iverson will serve as PI of the subcontract with funds and efforts as outlined in our respective budget justifications. Dr. Iverson will provide expertise and oversight for the data collection, processing, analysis, and dissemination related to the project. In Year 1, she will work with the research team to ensure that data acquisition methods (filming and experimental paradigm) are suitable to extract the data of interest. In Years 1-3, she will assist PI Bradshaw in developing and executing a frame-by-frame coding scheme of infant object exploratory and social behavior. In Years 4-5, she will work with the PI to analyze, interpret, and disseminate project findings.

Michelle G. Fry, J.D., Ph.D.

State Director

Janet Brock Priest

Associate State Director Operations

Lori Manos

Associate State Director

Policy

Constance Holloway

General Counsel

Harley T. Davis, Ph.D.

Chief Administrative Officer Tracey

Hunt

Chief Financial Officer

Greg Meetze

Chief Information Officer



3440 Harden Street Extension
Columbia, South Carolina 29203

803/898-9600

Toll Free: 888/DSN-INFO

Home Page: www.ddsn.sc.gov

COMMISSION

Stephanie M. Rawlinson

Chairman

Barry D. Malphrus

Vice Chairman

Robin B. Blackwood

Secretary

Gary Kocher, M.D.

Eddie L. Miller

David L. Thomas

Michelle Woodhead

Jessica Bradshaw, Ph.D.

Assistant Professor

Department of Psychology

University of South Carolina

1800 Gervais St.

Columbia, SC 29201

Dear Dr. Bradshaw:

As you are aware, the South Carolina Department of Disabilities and Special Needs (DDSN) has authority over all the state's services and programs for South Carolinians with severe lifelong disabilities, including Intellectual and Developmental Disabilities and Autism Spectrum Disorders (ASD). Our primary responsibilities include planning, development and provision of a full range of services for children and adults. DDSN coordinates services with other agencies and promotes and implements prevention activities to reduce the occurrence of both primary and secondary disabilities.

We have a long history of collaboration with the University of South Carolina (USC) research projects on ASD, and especially with you, Dr. Bradshaw, and these efforts have been exceedingly successful. We are thrilled to embark on a new collaboration with you and we are very enthusiastic about your grant application titled, "The Role of Autonomic Regulation of Attention in the Emergence of ASD." On behalf of the Department of Disabilities and Special Needs I would like to express my excitement and support for your grant application investigating the link between autonomic function, attention, and social communication, and ASD.

In support of your grant efforts, we will partner with you for recruitment of infants with older siblings diagnosed with autism. In the past three years, we have done several mailouts that involved identifying children who have been diagnosed with autism and mailing information letters about your project to these potentially eligible families. These efforts have resulted in over 6,700 letters sent with a response rate of approximately 6%. Currently, our database indicates over 3,200 children diagnosed with ASD aged 8 years and younger who would be targeted for such recruitment letters in South Carolina. Thus, we are optimistic that we can be instrumental in helping you meet your target enrollment number of 100 newborn infant siblings in two years (~50/year) for your study.

Our work with your department has historically been a mutually beneficial relationship; we remain avid supporters of ASD research at USC and pledge to work collaboratively with you in this new study.

Sincerely,

A handwritten signature in blue ink that reads "Lori C Manos".

Lori Manos, Associate State Director-Policy

Victor Iskersky, MD
Director of Neonatology Palmetto Health
Medical Director, Palmetto Health Richland NICU

November 4th, 2022

Dear Dr. Bradshaw,

As the Director of Neonatology at Palmetto Health and the NICU Medical Director, I am pleased to provide a letter of support for your proposal on the on the role autonomic regulation of attention in the emergence of ASD. This will be the third grant-funded research project that we have worked on together over the past three years, which speaks to our collegial and successful collaboration. Moreover, I have a strong, longstanding collaborative relationship with your co-investigator Dr. Robin Dail. This will be our second research project with the three of us and I look forward to continuing this productive collaboration.

Your scientific agenda is well aligned with my own clinical and scientific interests, which lie in optimizing health outcomes for infants born preterm. Preterm infants experienced disrupted autonomic system function (respiratory and cardiac problems) and of course are at significant risk for poor neurodevelopmental outcomes. Your research design will use preterm infants as a fascinating control group – these neonates will have autonomic dysfunction but no genetic liability for autism. This will greatly add to our understanding of the neurodevelopmental outcomes for infants preterm. Your decision to focus on preterm infants is extremely well-informed and will significantly contribute to the literature.

I am committed to supporting this project through active recruitment of preterm infants as well as medical expertise, as needed, in neonatal development. In particular, the volume of preterm infants born at our hospitals is sufficient to fulfill your recruitment goals of N=75 infants within the two years of this proposal. I will also facilitate coordination with local obstetricians and OB/GYN offices to support recruitment and outreach efforts.

The aims of your proposal are innovative and results will be influential in the fields of neonatology, psychology, and ASD. I strongly support the present application, and look forward to being actively involved in this project.

Sincerely,

A handwritten signature in black ink, appearing to read 'V. Iskersky', with a stylized flourish at the end.

Victor Iskersky, MD



Kayla Everhart, PhD, RN
Assistant Professor
College of Nursing
University of South Carolina

November 4th, 2022

Dear Dr. Bradshaw,

As an Assistant Professor in the College of Nursing and a former Neonatal Intensive Care Unit (NICU) nurse, I am pleased to hear of your proposal investigating the relationship between infants' autonomic regulation of attention and the emergence of ASD. Including preterm infants as a group in your study will greatly add to our understanding of the role of ANS in development as well as preterm developmental outcomes. As you are aware, preterm infants show more autonomic dysfunction and stress signs which may contribute to decreased functioning following NICU discharge.

Our interests in promoting treatments and interventions for infants born preterm closely align, and I am excited to continue our work together. We have been working together closely for the past year as I have worked with you to train in the NICU Network Neurobehavioral Scale (NNNS), which I understand will also be a measure in your proposed study. I also have a close relationship with your co-investigator, Dr. Robin Dail, as she was my mentor in graduate school. I have very much enjoyed working with you both on our current pilot study investigating thermal gradients and ASD. Through this pilot study, we have meet weekly and we have evaluated babies together in the NICU and in your lab.

I am committed to supporting this proposed R01 through active recruitment of preterm infants as well as medical expertise, as needed, in neonatal development. Specifically, I am launching my own study to Evaluate the use of a commercial blood warmer (3M Hotline) to decrease hypothermia in preterm infants, funded by the Gerber Foundation and will recruit 140 preterm infants from our local NICU at Prisma Richland Hospital. In addition, I am still an active staff nurse in this NICU and am in frequent contact with families. Through these activities, I will share your study with eligible families in the NICU and will continue to collaborate with your recruitment coordinator to share referrals. I am well-versed in not only sharing information about your studies but providing informed consent via our collaborative pilot study. Therefore, I am confident in my ability to recruit a sufficient number of infants for your new R01.

In sum, strongly support the present application, and look forward to being actively involved in this project.

Sincerely,

A handwritten signature in cursive script that reads "Kayla Everhart".

Kayla Everhart, PhD, RN
Assistant Professor
College of Nursing-University of South Carolina
803-777-5999



Child Development and
Behavioral Health Pediatrics

Dear Dr. Bradshaw,

As a school psychologist within the Child Development and Behavioral Health Clinic at Prisma Health-Midlands, I am delighted to hear of your new research proposal aimed at promoting earlier diagnosis and treatment for autism within the first year of life. Our interdisciplinary clinic is comprised of developmental pediatricians, psychologists, and social workers, and we work to provide comprehensive assessments to inform treatment plans and services for children with developmental disabilities in the Midlands. I am pleased to hear how this research will help to expand the diagnosis window and interventions for children with autism at an earlier age.

As you know, our group has been a big supporter of your work and advocate for your ongoing research studies at UofSC. I have enjoyed this collaboration and I am committed to supporting this project through active recruitment of children with ASD and infant siblings of children with ASD through our clinic. I will support recruitment for your study by continuing to screen for eligibility and make direct referrals for families who qualify. I will also share this research opportunity with members of my team as an additional resource for families. Our clinic sees approximately 200 children with autism each month and I do not anticipate any challenges in achieving your recruitment goal of 100 infant siblings of children with ASD.

In closing, I strongly support your proposal titled, "The Role of Autonomic Regulation of Attention in the Emergence of ASD" and I believe that my efforts in recruitment will strongly contribute to your success.

Sincerely,

A handwritten signature in blue ink that reads "Emily Lowell".

Emily Lowell, PhD, NCSP
Psychologist #1244
Assistant Professor of Clinical Pediatrics
Child Development and Behavioral Health
Prisma Health, Midlands

Prisma Health—Midlands
9 Medical Park Dr., Ste. 210
Columbia, SC 29203

P 803-434-6598
F 803-758-0142

PrismaHealth.org



Deborah M. Greenhouse, MD, FAAP
Sarah G. Grooms, MD, FAAP
Nada J. Hamzy, MD, FAAP
Barry W. Cabiness, MD, FAAP
Naren Shenoy, MD, FAAP
Catherine C. Miller, MD, FAAP
Michael G. Barker, MD, FAAP
Richard D. Shrouds, MD, FAAP
James B. Dewar, MD, FAAP
W. Bradley Painter, MD, FAAP
Emilie Z. Bartlett, MD, FAAP
Sonya Hatfield Head, MD, FAAP
Melissa L. Mills, MD, FAAP
Jennifer M. Soroos, MD, FAAP
Kiersten Lofton, MD, FAAP
Kristen Rollins, MD, FAAP
Karen Sullivan, RN, IBCLC

www.palmettopediatrics.com



Downtown Office:
140 Park Central Dr
Columbia, SC 29203
(803) 779-4001
(803) 252-9458(f)



Northeast Office:
601 Clemson Rd
Columbia, SC 29229
(803) 788-4886
(803) 788-5020 (f)



Hardscrabble Office:
300B Rice Meadow Way
Columbia, SC 29229
(803) 788-6360
(803) 788-6668 (f)



Irmo Office:
7448 Broad River Rd
Irmo, SC 29063
(803) 732-0140
(803) 732-4848 (f)



Lexington Office:
1970 Augusta Hwy
Lexington, SC 29072
(803) 358-2370
(803) 358-2376 (f)



Corporate Office:
SC Pediatric Alliance
PO Box 291599
Columbia, SC 29229
(803) 788-6146
(803) 462-0312 (f)

Palmetto Pediatric and Adolescent Clinic
140 Park Central Dr
Columbia, SC 29203
(803) 779-4001

Jessica Bradshaw, PhD
Associate Professor
University of South Carolina

November 4th, 2022

Dear Dr. Bradshaw,

I am writing to express my support for your research proposal titled, "The Role of Autonomic Regulation of Attention in the Emergence of ASD." I am so pleased to hear how this research may help facilitate targeted treatment to children with autism at a very early age, as this reflects a significant need in the community and within service provision. We have had quite a successful collaborative partnership with the four research studies your team is currently recruiting for through my practice and I am enthusiastic to add this new project to the list!

I am a primary care pediatrician at Palmetto Pediatric and Adolescent Clinic in Columbia, South Carolina, and a fellow of the American Academy of Pediatrics (AAP). I remain active in the AAP on both a state and national level and serve as chapter coordinator for the Pediatric Research in Office Settings (PROS) research network of the AAP. I am also a member and past chairperson of the Richland School District Two Medical Advisory Committee. I am committed to supporting your project through active recruitment of children with autism, infant siblings of children with autism, and preterm infants through our clinic. I will support recruitment for your study by continuing to utilize your interest forms in our infant paperwork packets, which allows you to contact families directly who express interest in your study. As you know, this has led to over 120 participant referrals in the 12 months we have been using this strategy. I will also share this research opportunity with families who may qualify during office visits and maintain close collaboration with you and your recruitment team regarding referrals.

In closing, I strongly support the present application and I believe that my efforts in recruitment will strongly contribute to your success.

Sincerely,

A handwritten signature in black ink that reads "Deborah Greenhouse MD". The signature is fluid and cursive, with the last name being particularly prominent.

Deborah Greenhouse, MD



Heidi J. Steflik, MD, MSCR
Assistant Professor, Department of Pediatrics
steflikh@musc.edu

Division of Neonatology
165 Ashley Avenue, MSC 917
Charleston SC 29425-9170
Tel 843-792-2112
Fax 843-792-8801
MUSCKids.org

Dear Dr. Bradshaw,

I am pleased to hear of your new NIH research proposal titled, "The Role of Autonomic Regulation of Attention in the Emergence of ASD." As you are aware, preterm infants present with poor ANS systems and are at a higher risk for poor neurodevelopmental outcomes. Your research design using preterm infants as a control group is unique, as these neonates will have autonomic dysfunction but no genetic liability for autism. This will greatly add to our understanding of the neurodevelopmental outcomes for preterm infants and promote earlier interventions for these children.

I am a neonatologist and Assistant Professor at the Medical University of South Carolina (MUSC) and I am a site PI for Dr. Robin Dail's (your co-investigator) ongoing study on thermal regulation in preterm infants. Given this established collaboration and my strong interest in supporting research, I am very happy to commit to supporting your project through active recruitment of preterm infants. In particular, the volume of preterm infants born at our hospitals is sufficient to fulfill your recruitment goals of N=75 infants within the first two years of this proposal. I will assist with recruitment by sharing this research study with our medical team and distributing flyers throughout our unit. I will also ensure that our staff is in contact with your Recruitment Coordinator so that this process runs smoothly. I understand that you are seeing participants in their homes, which will help with your enrollment numbers, especially with patients at MUSC.

The aims of your proposal are innovative, and results will be influential in the fields of neonatology and neurodevelopment more broadly. I strongly support the present application and look forward to staying in touch about your findings.

Sincerely,

A handwritten signature in blue ink that reads "Heidi Steflik".

Heidi J. Steflik, MD, MSCR

Assistant Professor of Pediatrics

Medical University of South Carolina



**DEVELOPMENTAL AND
BEHAVIORAL PEDIATRICS**

**135 RUTLEDGE AVENUE
P.O. BOX 250567 CHARLESTON, SC 29425**

**(843) 876-0504
FAX (843) 876-1518**

Jessica Bradshaw, Ph.D.

Associate Professor
Department of Psychology
College of Arts and Sciences

November 1st, 2022

Dear Dr. Bradshaw,

I am pleased to write this letter in support of recruitment for your NIH R01 "The Role of Autonomic Regulation of Attention in the Emergence of ASD." I feel confident your project, which aims to examine the developmental trajectory of autonomic nervous system dysfunction and attention in infant siblings of children with ASD, will make an important contribution to our knowledge regarding biomarkers for ASD symptoms. I am excited that your research may eventually help facilitate targeted treatment to the children who need them most, which may ultimately impact the developmental outcomes for children with ASD. This is an area of great importance to the autism community. I have enjoyed collaborating with your investigative team through individual grants and the SPARK study, and I am very supportive and engaged in the statewide research collaboration your team is spearheading through the CAN Center at UofSC (Carolina Autism and Neurodevelopment Research Center at UofSC).

My program of research focuses on the identification, treatment, and epidemiology of ASD. I have had success in the recruitment of participants for my research program. For example, in the past three years, our team at MUSC has recruited more than 1500 SC participants with ASD, as well as more than 2500 family members to take part in the nationwide SPARK autism genetics study. I also direct a busy autism diagnostic clinic in which we typically evaluate 10 children for new autism-related concerns per week, and this clinic will serve as a good referral source for infant siblings of children with autism as some mothers are pregnant with their second child by the time they bring their older child into our clinic for ASD-related concerns. Further, I have a great deal of experience in working with social media and community partners to promote research participation.

Recruitment is often the most challenging part of any research study and I look forward to helping your team achieve your recruitment goals on schedule. As such, I will provide you with support in the recruitment of infant siblings. Specifically, I will work with your recruitment coordinator to develop a protocol within my own busy autism diagnostic clinic, as well as the two other Developmental Evaluation Centers in SC, to share study

information with pregnant mothers of children with ASD. As you know, we have a very successful social media program in SC aimed at improving participant involvement in ASD research (*SC Autism Research Opportunities* on Facebook and Instagram), and I will work with your team to share targeted marketing campaigns based on the success that we have had in our own recruitment efforts. In addition, I will help facilitate the contact of clinical patients and prior research participants who have consented to be contacted about future research opportunities.

I strongly support the present application, and while I know that all referrals will not qualify or agree to participate, I believe that my efforts in recruitment will strongly contribute to your success and that we will continue to work productively together.

Sincerely,

A handwritten signature in black ink that reads "Laura Carpenter". The signature is written in a cursive, flowing style.

Laura Arnstein Carpenter, Ph.D.
Professor of Pediatrics and Psychiatry
SC LEND Training Director
Medical University of South Carolina



Jessica Bradshaw
Associate Professor
Department of Psychology
University of South Carolina
1800 Gervais St.
Columbia, SC 29201

May 10th, 2022

Dear Dr. Bradshaw:

The Simons Foundation Powering Autism Research for Knowledge (SPARK) is pleased to support your NIH R01 application titled “The Role of Autonomic Regulation of Attention in the Emergence of ASD”.

As you know, SPARK’s aim is to collect the largest ever cohort of individuals with ASD, from whom we collect medical, phenotypic, and genetic information to facilitate more rapid research to advance our understanding of the causes of ASD and to improve the lives of the individuals and families it affects.

SPARK is centrally coordinated, fully funded by the Simons Foundation Autism Research Initiative (SFARI) and governed by the Western Institutional Review Board® (WIRB® Protocol #20151664; PI: Wendy Chung). It includes collaboration with 31 university-affiliated research clinics in 26 states across the U.S. In addition, numerous national and local autism community organizations across the U.S. are partnering with SPARK to help recruit participants and spread the word about this landmark study. Working with SPARK central, this strong network of collaborators has enrolled 85,000 children and 20,000 adults with ASD into SPARK in just 5 years since it was launched. We expect to steadily increase the number of consented participants in the coming years.

The SPARK research match service has been designed to facilitate ASD research efforts by informing participants via email about studies for which they qualify and sending follow-up reminder email(s) after the initial invitation email to SPARK participants who have not yet responded. If they express interest, we would send their contact information directly to you to support your local recruitment efforts. Additionally, approved researchers can request phenotypic data on family members who are registered in SPARK, to link with the data gathered during studies like yours. Please note that launch of projects utilizing SPARK research match is contingent on full SPARK Participant Access Committee approval.

For your study, we will target recruitment efforts to families with children with ASD under the age of 13 years. This will support recruitment of 100 infant siblings of children with ASD for your study.

Sincerely,

Wendy K. Chung, MD PhD
Director of Clinical Research, SFARI

The Unumb Center
FOR NEURODEVELOPMENT



The Unumb Center for Neurodevelopment
1505 Blanding Street
Columbia, South Carolina 29201
803-929-0011

Jessica Bradshaw, PhD
Associate Professor
Department of Psychology
University of South Carolina

Dear Dr. Bradshaw,

I am writing on behalf of The Unumb Center for Neurodevelopment to offer our strong support of your project titled, "The Role of Autonomic Regulation of Attention in the Emergence of ASD." The Unumb Center is a not-for-profit organization offering individualized diagnostic and intervention services for individuals with autism. I am the Director of Clinical Services at The Unumb Center, where I oversee ABA, psychology, and pre-employment transition services and provide comprehensive psychological evaluations for individuals with developmental disabilities. The Unumb Center is committed to improving the lives of children and families with ASD, and it is clear that your proposed project is consistent with this mission.

We are committed to showing our support through outreach efforts and will announce this voluntary research opportunity to our families through newsletters, mailings, flyers, and direct conversations with clients. This is already evident through our recruitment efforts for your ongoing studies. Given the significant impact your study will have on families affected by ASD, we are fully committed to helping you reach your recruitment goal of enrolling 100 infant siblings of children with ASD. As an organization dedicated to early diagnosis and intervention, we are thrilled to have you as a collaborator at USC. We have already discussed ways in which your research may be immediately translatable to the community, and we eagerly await the unfolding of your research program.

We look forward to working with you on this important project, and many to follow.

Sincerely,

Kathleen Franke, Ph.D., BCBA
SC Licensed Psychologist #1572
Director of Clinical Services



Helping families today, reach goals for tomorrow

Beth Bunge
Bright Start Early Intervention
720 Gracern Road, Suite 450
Columbia, SC 29210

Jessica Bradshaw, Ph.D.
Assistant Professor
Department of Psychology
University of South Carolina
1800 Gervais St.
Columbia, SC 29201

November 5, 2022

Dear Dr. Bradshaw,

On behalf of Bright Start of South Carolina, I would like to express my enthusiasm and full support for your project titled, "The Role of Autonomic Regulation of Attention in the Emergence of ASD." As you are aware, there can be significant barriers for children to receive early autism evaluations and services, and I am excited to hear how this grant will support earlier identification for autism within the first year of life.

As the co-founder of Bright Start, a prominent early intervention provider serving almost 6000 children, teens and adults and families across South Carolina, I am committed to showing our support through outreach efforts and will help you reach your recruitment goal of enrolling 100 newborn infant siblings in two years (~50/year) for your study. To support these efforts, we will continue to include your interest forms in our paperwork packets with new families, allowing you to contact interested families directly about your study. We will also encourage our providers to contact your team directly if a family may benefit from your study opportunity.

The state of South Carolina has a tremendous need for research and policy focused on early identification and intervention and I am thrilled that your research focuses on these exact issues. As an organization dedicated to early intervention, we are excited to partner with you on this important project and look forward to future collaborations.

Sincerely,

Beth Bunge
Beth Bunge, Co-Founder
Bright Start

Charleston

4925 Lacross Rd. Suite 209
North Charleston, SC 29406
843.872.6335

Anderson

PO Box 568
Sandy Springs, SC 29677
864.540.8530

Columbia

720 Gracern Rd. Suite 450
Columbia, SC 29210
803.929.1112
Letters of Support

Aiken

PO Box 288
Aiken, SC 29802
Toll-free: 855.221.9265

Greenville

6 Pelham Ridge Drive
Greenville, SC 29615
864.627.5097

Rock Hill

1420 Ebenezer Road, Suite 103
Rock Hill, SC 29732
803.366.8434



**1775 Hazan CT
Mount Pleasant, SC 29464-9602
Phone: (843) 327-9996
Fax: (843) 388-6294**

**Jessica Bradshaw, Ph.D.
Assistant Professor
University of South Carolina
1800 Gervais Street
Columbia, SC 29201**

November 2, 2022

Dear Dr. Bradshaw,

I am excited to hear about your new project that will work to support earlier diagnosis and treatment for children with autism. As a supervisor and owner of Lowcountry Early Intervention, I am writing to express my support of your new proposal titled "The Role of Autonomic Regulation of Attention in the Emergence of ASD." My team currently serves approximately 70 children in the Lowcountry area and we regularly collaborate with other providers and practitioners who serve the autism community.

As you are aware, there can be significant barriers for children to receive an early autism evaluation, including insurance hurdles and long wait lists, among other factors. We are thrilled to hear that this research will help address these gaps, helping children gain access to services early on. We have collaborated with your team in the past regarding referrals and are thankful for how this research opportunity may provide earlier assessments for families.

We are committed to showing our support through outreach efforts and will help you reach your recruitment goal of enrolling 100 newborn infant siblings of children with autism into your study. To support this, we will share your study with our families and will continue to collaborate with your team regarding referrals. We will also continue to connect you with other partners and providers in the Lowcountry area who work directly with families and who can further extend this opportunity to the ASD community. As an organization dedicated to early intervention, we are excited to partner with you on this important project and look forward to future collaborations.

Sincerely,


Renee Johnson Cochrane

Eric Craft
Trevett's Print + Mail
6065 St. Andrews Rd.
Columbia, SC 29212
803-781-3150

Jessica Bradshaw, Ph.D.
Assistant Professor
Department of Psychology
University of South Carolina
1800 Gervais St.
Columbia, SC 29201

November 5th, 2022

Dear Dr. Bradshaw,

As a project manager and team lead with Trevett's Fulfillment Service, I am pleased to write a letter of support for your grant application titled, "The Role of Autonomic Regulation of Attention in the Emergence of ASD." Our company has over 30 years of experience providing printing, mailing, and fulfillment services to our customers in Columbia, SC, and we have enjoyed collaborating with your team in the past for your last 3 mailouts.

For these mailouts, we have printed your study's flyers and coordinated with the South Carolina Department of Disability and Special Needs to distribute these flyers to families in the SC DDSN database. In total, we have mailed approximately 5,200 letters to families over the last year. I am pleased to hear how successful these mailings have been in supporting recruitment needs for your project.

We recognize the value of your research and the importance of connecting your study with families across South Carolina and look forward to supporting your project in the future.

Sincerely,

A handwritten signature in black ink, appearing to read "Eric M. Craft", with a long, sweeping horizontal line extending to the right.

Eric M. Craft

RESOURCE SHARING PLAN

For this project, data will be collected from infant siblings of children with autism spectrum disorder (ASD), infants born preterm (24-32 weeks gestation), and infant siblings of typically developing children. Data collected as part of this project will be disseminated in an open and timely manner in line with the objective of PA-21-201 to elucidate the etiology, epidemiology, diagnosis, and optimal means of service delivery in relation to autism spectrum disorders. Thus, raw data collected in support of this project will be shared biannually through the NIMH Data Archive (NDA), in accordance with the mandate set forth PA-21-201, to fully comply with the NIH initiative to accelerate and advance autism research. Shared datasets will include demographic, clinical, behavioral, and electrocardiogram (ECG) data collected from infants and caregivers. During the informed consent process, research staff will engage in specific discussions and provide documentation that participants' data will be shared for research purposes through the NIH NDA repository and allow for participants to provide or deny consent for such data to be shared. When sharing de-identified data, every effort will be taken to safeguard the privacy of participants. The PI will also share data upon request, under her auspices, through data-sharing agreements with other researchers. Moreover, the data collected and analyzed throughout this project will be disseminated at local, national, and international conferences, as well as published in high-impact journals. All final peer-reviewed manuscripts will be submitted to PubMed Central, adhering to NIH Public Access Policy guidelines.

Trained research staff will ensure the collection of high-value data, and the full-time Project Manager will dedicate 40% of time to managing a quality-controlled database, which will bolster the feasibility of sharing accurate, comprehensive, and informative data. We have designed the budget to account for our data sharing efforts and have protocols in place to collect needed information.